Active Inference of Interpersonal Trust Within and Across Mental Disorders

Eckert, Anna-Lena[1,2*]; Pawlowski, Janik[1]; Rief, Winfried[2,3]; Zimmer, Raphaela I.[4]; Hermann, Andrea[4,5,6]; Endres, Dominik[1,2]; Kirchner, Lukas[2,3]

Background. Interpersonal problems are common in mental disorders like depression and social anxiety disorder. However, there is a lack of formal models to explain idiosyncrasies in patients' interpersonal functioning. Following modern computational accounts of perception and action, interpersonal experience and behavior results from implicit inference about hidden environmental properties.

Methods. We simulate interpersonal decision-making in a "trust game" using a POMDP generative model and active inference. Simulated agents decide to either keep or invest an initial budget to a trustee. If the agents invests, the trustee can cooperate with or exploit the agent's trust. Agents perform active inference of the context (cooperative vs. hostile) and action selection. By introducing systematic biases to the generative model, we simulate several subtypes of depression and social anxiety. We then collected data of N=25 patients with diagnosed depressive disorders and fit the generative model to the data.

Results. Agents with biased expectations or preferences showed idiosyncratic interpersonal inference. They more readily infer to be in a hostile context and avoid investing with the trustee. Biases also affected the total earned rewards, with socially anxious and depressed agents receiving higher average rewards in hostile contexts compared to "healthy" agents (i.e., agents with an accurate-to-optimistic generative model), but "healthy" top optimistic agents showing superior performance in volatile environments. Fitting the model to the data resulted in individual parameters for patients with low symptom scores compared to those with high symptom scores, with group differences particularly in transition dynamics B, outcome preferences C and prior beliefs D.

Discussion. Our simulations formalize complex interpersonal phenomena within Active Inference. Interpersonal biases in mental disorders can be formalized and investigated with respect to their functional role. The model has potential applications in psychopathological research, personalized diagnostics, and individualized treatment planning. Further empirical work is necessary to validate the empirical basis and clinical usefulness of formal models of interpersonal decision-making.

Keywords. Active Inference; depression; social anxiety disorder; interpersonal functioning; economic games; agent-based decision making

Word Count. Total: 9,871; Introduction: 1,826; Methods: 3,767; Results: 2,348; Discussion: 1,930.

Author affiliations.

[1] Theoretical Cognitive Science Group, Department of Psychology, Philipps-University Marburg, Marburg, Germany

[2] The Adaptive Mind Research Cluster of the Universities of Marburg, Giessen, Frankfurt and Darmstadt

[3] Clinical Psychology and Psychotherapy, Department of Psychology, Philipps-University Marburg, Marburg, Germany

[4] Department of Psychotherapy and Systems Neuroscience, Justus Liebig University Giessen, Giessen, Germany

[5] Center for Mind, Brain and Behavior, Universities of Marburg, Giessen and Darmstadt, Germany

[6] LOEWE Center DYNAMIC, Universities of Marburg, Giessen, Frankfurt, Darmstadt; Germany.

* Corresponding author, OrcID 0000-0002-3520-5714, eckertan@staff.uni-marburg.de

Introduction

Impairments in interpersonal experience and behavior are highly prevalent in mental disorders and can cause tremendous suffering and social disadvantages (social anxiety disorder: Alden & Taylor, 2004; eating disorders: Arcelus et al., 2013; schizophrenia: Hooley, 2010; major depression: Kupferberg et al., 2016; borderline personality disorder: Lazarus et al., 2014; post-traumatic stress disorder: Scoglio et al., 2022). Determining whether interpersonal impairments are the cause or consequence of a mental health problem is challenging. It is likely that longitudinal bidirectional relationships exist (Beeson et al., 2020; Kirchner et al., 2022). Impairments in interpersonal experience and behavior predict important clinical outcomes and play a role in the development and maintenance of mental disorder (Backs-Dermott et al., 2010; Bell et al., 2018; Kirchner et al., 2022; Quilty et al., 2013; Tokuda et al., 2008; Woods et al., 2021). Upon closer examination, interpersonal symptoms tend to share significant similarities across diagnostic categories. For instance, social withdrawal is common in depressed (Fernández-Theoduloz et al., 2019), and socially anxious individuals (Hudson & Rapee, 2000); but also in individuals with chronic pain (Harris, 2014), post-traumatic stress disorder (Maercker & Horn, 2013), or borderline personality disorder (Liebke et al., 2017). It follows that a breadth of different underlying causes may result in commonalities at the level of observable behavior (Gao et al., 2017). These commonalities are of particular interest in light of generally high comorbidity rates among mental disorders (Conway et al., 2022; Cuthbert & Insel, 2013; Dalgleish et al., 2020). However, meaningful differences in interpersonal experience and behavior can still be identified between- and within diagnostic categories. Joormann & Gotlib (2006) demonstrated differences between depressed and socially anxious individuals in terms of sensitivity to social stimuli. Also within the spectrum of depressive disorders, different "interpersonal subtypes" can be identified (e.g., "vindictive" vs. "socially avoidant", Simon et al., 2015).

There is a lack of formal theories or models that can effectively integrate and accurately operationalize the differences and commonalities in patients' interpersonal experiences and behaviors across- and within disorders. We have previously described the theoretical structure and functional form of such a model (Kirchner et al., 2022), where we conceptualized impairments in interpersonal experience and behavior as the result of biased implicit inferences about hidden properties ("states") of the environment. Individual biases can stem from, for example, divergent probability estimates about the environment, differences in evaluating interpersonal observations or the advantageousness attributed to these observations.

Promises of Computational Approaches in Mental Health

The promise of applying formal, normative models to mental health conditions is to identify distinct behavioral or neural phenotypes, or *computational assays* (Karvelis et al., 2023; Montague et al., 2012). Formal computational models of clinical phenotypes, coupled with empirical data, may open novel avenues for researching heterogeneity, developing personalized treatments and predicting treatment outcomes (Schwartenbeck & Friston, 2016). Bayesian accounts of behavior have gained substantial

momentum in psychiatry and clinical psychology, reframing mental disorders as caused by aberrant implicit inference (Pio-Lopez et al., 2022). Bayesian inference has been applied to a wide range of mental disorders, including schizophrenia (Fletcher & Frith, 2009; Haarsma et al., 2022) and autism (Palmer et al., 2017; Pellicano & Burr, 2012), panic disorder (Maisto et al., 2021), social anxiety disorder (Gerrans & Murray, 2020), post-traumatic stress disorder (Linson & Friston, 2019), anorexia nervosa (Barca & Pezzulo, 2020), and depression (Barrett et al., 2016; M. Berg et al., 2022). While conceptually attractive, many theoretical accounts suffer from two crucial limitations. First, they tend to be developed within the boundaries of a specific, categorical diagnosis. Research has shown repeatedly that traditional, categorical nosologies show limited predictive power, arbitrary boundaries between sub-clinical and clinical expressions of a trait or symptom, and high levels of comorbidity (Kotov et al., 2018). Developments such as the HiToP (Kotov et al., 2018) or RDoC (Cuthbert & Insel, 2013) are moving away from a description of mental disorders by a set of distinct diagnostic categories and towards dimensional perspectives on clinical symptoms and syndromes. Secondly, many studies lack rigorous mathematical operationalizations that allow for computer simulations and a thorough formal evaluation of theoretical assumptions considering data. The Active Inference framework may help address these shortcomings, providing flexible and tractable algorithms for inference (Parr et al., 2022).

Active Inference – A Framework to Model Mental Disorders

Active Inference is a theoretical account of perception, action and learning in complex systems (Friston et al., 2016; Heins et al., 2022; Parr et al., 2022). It is commonly used to model decision-making under uncertainty in cognitive neuroscience, but its application now extend to a wide range of fields and disciplines, including clinical psychology and psychiatry (Benrimoh et al., 2018; Montague et al., 2012; Smith et al., 2021). Complex behavior under Active Inference results from the imperative to maximize a lower bound on Bayesian model evidence, or minimize variational free energy (VFE). In other words, Active Inference ensures that incoming sensory information is well-explained by an internal, formal model of the world. To effectively navigate and interact with the (interpersonal) environment, agents need to infer its current *state* and decide what actions to pursue to resolve uncertainty about the environment and obtain preferred observations and outcomes. To accomplish this, it is crucial for the brain to entertain a generative model of its surroundings. A *generative model* is a formal conceptualization of the processes that generate sensory observations (Friston et al., 2016). The generative model is used to form predictions, such as expectations about an interpersonal environment, which are based on sensory information and prior beliefs. Deviations between sensory information and model predictions can be used to update the generative model. This iterative process ensures that predictions continue to minimize VFE over time. To achieve the optimal balance between trusting prior beliefs or the current sensory information, their respective precision needs to be estimated (Brown et al., 2013). The process of minimizing VFE is thought to underlie *perception*. Beyond perception, Active Inference offers a solution for *action selection* that dynamically interacts with VFE-based perceptual inference. Desirable states or outcomes (preferences) are part of the agent's generative model and can

be reframed as a type of prior prediction (Da Costa et al., 2020). Actions are selected in a way that makes these predictions more probable. In Active Inference, actions hence ensure that preferred outcomes or states are reached or maintained. Action selection requires minimizing the expected free energy (EFE) E over future states, given a sequence of actions. Policies that lead to preferred outcomes are more likely to be pursued by the agent. Agents are hence equipped to choose *proximal* actions that lead to the achievement of a *distal* goal state, i.e., planning-as-inference. Through this, Active Inference offers a computational strategy to solve the explore-exploit dilemma (Heins et al., 2022). In Active Inference, the state of the environment is usually modelled as unobservable, or hidden, to the agent. The agent has to infer the hidden state from sensory observations, which are generated by the hidden state in a probabilistic manner (Da Costa et al., 2020). An agent further needs a representation of how, probabilistically, their own actions alter environmental states. This ensures that only such actions are chosen that either resolve the agent's uncertainty (explore) or bring the agent closer to their preferred outcome (exploit). Partially observable Markov decision processes (POMDPs) have proved to be a convenient choice for generative models in Active Inference (Da Costa et al., 2020; Heins et al., 2022). POMDPs model the environment in terms of hidden states, which change dynamically over time as a function of the current state and the agent's actions.

To summarize, agents perpetually infer the real-world causes of incoming sensory observations. Agents will choose policies, or action sequences, that resolve uncertainty and maximize the likelihood of achieving preferred states. To do this, agents need to estimate how observations relate to hidden states, how hidden states evolve over time, and how their actions impact the evolution of hidden states.

Neuroeconomic Decision-making and Interpersonal Behavior in Mental Disorders.

Neuroeconomic games are complex and strategic, yet controlled experiments (Glimcher & Rustichini, 2004; Robson et al., 2020) that have been utilized to study interpersonal decision-making in both healthy participants and clinical populations (King-Casas & Chiu, 2012). In the present work, we focus on trust games, where an agent needs to decide whether to keep or invest a small monetary wage with a trustee. If the agent decides to invest, the trustee can decide to cooperate, usually resulting in an increased payout for the agent; or defect, resulting in a loss for the agent. Previous research suggests idiosyncrasies in gameplay behavior in mental disorders (King-Casas & Chiu, 2012; Robson et al., 2020). In particular, gameplay behavior in individuals with depression and social anxiety disorder is well documented (Ong et al., 2017; C. Sripada et al., 2013; C. S. Sripada et al., 2009; Zhang et al., 2012). Individuals with remitted major depressive disorder tend to defect less when in the role of the trustee compared to healthy controls, which has been related to a reduced ability to cope with the socio-emotional demands of the trust game (Ong et al., 2017; Zhang et al., 2012). In social anxiety, functional imaging data suggests decreased processing of the opponent's motivation (C. S. Sripada et al., 2009) and decreased neural responses to a cooperative trustee compared to controls (C. Sripada et al., 2013). Additionally, decision-making in neuroeconomic gameplay has been associated with important clinical variables. In an

exemplary study, individuals with a history of high-lethality suicide attempts were less sensitive to the costs of punishing their partner in a trust game compared to patients with a history of low-lethality suicide attempts (Szanto et al., 2014). Hence, interpersonal decision-making in gameplay allows us to model clinically relevant experiences and behavior (Hinterbuchinger et al., 2018; King-Casas & Chiu, 2012; Robson et al., 2020; Tzieropoulos, 2013; Wang et al., 2015).

Active Inference in Interpersonal Decision-making

Interpersonal contexts are highly complex and recursive. An individual needs to constantly monitor the dynamic actions of another individual, and predict future actions based on prior experiences with this individual or situation to choose appropriate and timely reactions. Moutoussis and colleagues (2014) equip simulated agents with variational Bayesian message passing and engage in a trust game. They demonstrate that the agents are able to iteratively update beliefs about an opponent and make decisions that maximize reward over time (Moutoussis et al., 2014). Similarly, Schwartenbeck et al. (2015) model behavior and neural activity in a neuroeconomic "limited offer" game using Active Inference. Participants had to decide whether they would accept a lower offer or wait for a higher one, at the risk of losing any offers if they waited for too long. Active Inference showed superior performance compared to classical utility maximization schemes in explaining participant behavior (Schwartenbeck et al., 2015). Finally, Cullen and colleagues (2018) simulated agents equipped with Active Inference in an artificial game-play environment and demonstrate that agents are capable of forming representations of their environments that help guide research into neurobiological underpinnings of mental disorder (Cullen et al., 2018).

Methods

Task

We use a classical economic game to simulate interpersonal decision-making in depression (see **Figure 1A**). In the *trust game* (J. Berg et al., 1995; King-Casas & Chiu, 2012), the agent's task is to play a multi-round investment-return game. In our simplified version, the agent receives a small amount of money in each trial, which they can either keep to themselves or invest their budget with a trustee. The artificial trustee can either keep or return half the multiplied wage invested by the agent. If the trustee decides to return half of the investment (*return*), the investment is multiplied by three, and split evenly between agent and trustee, leading to a surplus reward created by the joint venture (i.e., the agent invests 0.5 and receives 0.75). If, in contrast, the trustee decides to keep the agent's investment to themselves (*defect*), the agent loses all their budget to the trustee. In other words, if the agent invests, he can encounter both a return of investment (*win*) or a loss of their investment (in case the trustee *defects*). In the simulation, there were two different interpersonal contexts (**Figure 1B**). In the cooperative context, the trustee collaborates (i.e., shares the agent's investment) with a probability of $p = 0.8$. In the exploitative, or hostile context, the trustee tends to keep the agent's investment to themselves and shares with a probability of $p = 0.2$. We simulated two games: a cooperative-to-hostile game, where a cooperative phase (t=20) was followed by a hostile phase (t=20), and a hostile-to-cooperative game (t=20 cooperative trials, followed by t=20 hostile trials). To maximize rewards over time, the agent should frequently share with the trustee in cooperative contexts and keep their initial wage in the hostile context. For this, the agent requires an accurate representation of the current context, which they need to infer from observing the trustee's behavior over several rounds of the game. The agent hence needs to solve the dilemma of learning as much as possible about the trustee's intentions (*explore*) on the one hand and maximizing reward (*exploit*) on the other.

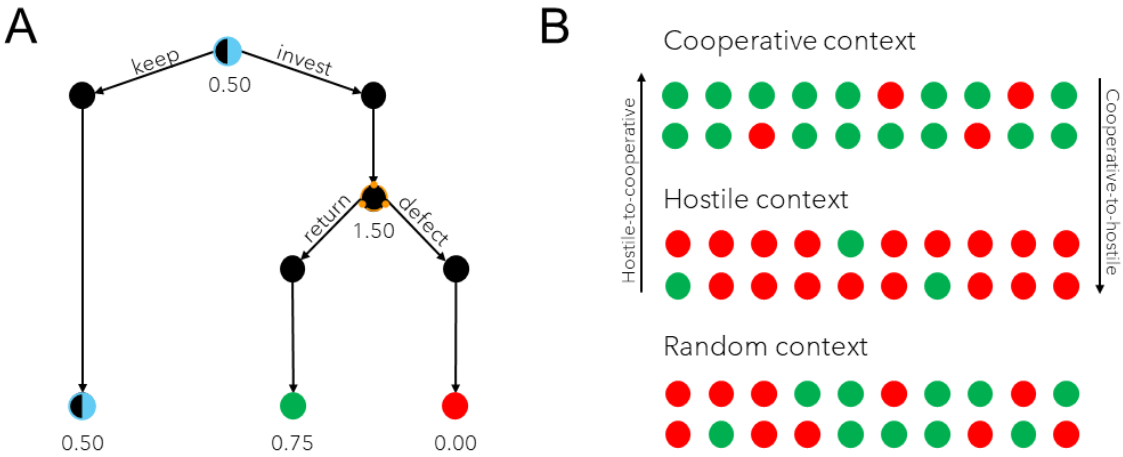


Figure 1. Trust Game and experimental design.

A. The agent receives a budget of 0.50 at the beginning of each trial. The agent must decide to keep or invest their budget. If the agent decides to invest their budget with the trustee, the budget is multiplied by factor 3. The trustee can decide to collaborate with the agent and return half of the tripled budget (0.75), resulting in a larger overall payoff for the agent and the trustee. If the trustee, decides to keep the agent's investment to themselves (defect), the agent does not receive anything.

B. There were three contexts: cooperative (trustee returns agent's investment in 80% of trials), hostile (trustee returns agent's investment in 20% of trials); *and* random (50% return of investment when sharing with the trustee). In the simulations, we focused on the cooperative- and hostile context and switches between these two, with two types of games: a hostile-to-cooperative game (t=20 cooperative followed by t=20 hostile trials) and a cooperative-to-hostile game (t=20 hostile trials followed by t=20 cooperative trials).

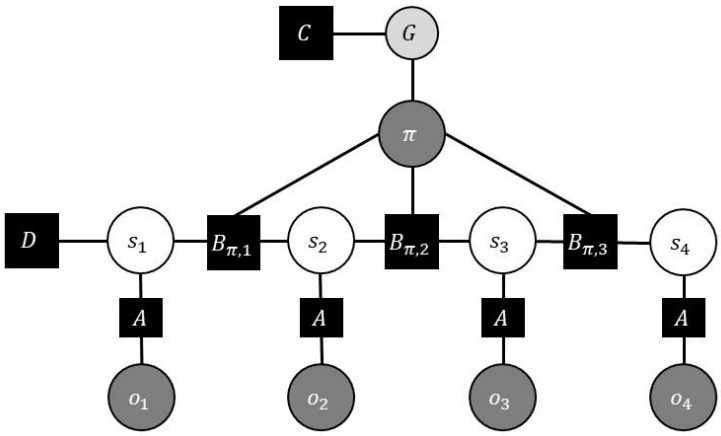


Figure 2. Generative model

Model parameters (A, B, C, D) are shown in boxes. A is the observation model, connecting observations o_t to hidden states s_t. B is the transition parameter, governing state transitions that depend on previous hidden states s_{t-1} and policies π. C represents prior beliefs (or prior preferences) over outcome observations o. D represents the agent's initial beliefs about the hidden states s. Finally, G is the expected free energy over policies, which depends on prior preferences C. Based on G, those policies with the highest probability of minimizing uncertainty over states s and maximize the probability of the preferred rewards are assigned the highest probability.

Model

The simulations were implemented in Python (v. 3.8.3) and the pymdp toolbox for Active Inference in discrete state spaces (Heins et al., 2022). The game's context can be cooperative ($p(share) = 0.8$), hostile ($p(share) = 0.2$) or random ($p(share) = 0.5$). The agent's task is to infer the current context and choose their actions accordingly. If, for example, the agent infers to be in a cooperative context, they should invest their money with the trustee for an increased outcome. If, on the other hand, they infer to be in a hostile context, they should prefer to keep their money. We here simulate 40 trials *t* of the task, with one context switch after *t*=20.

The agent's generative model was determined by four parameters A, B, C and D. For a graphical overview of the agent's generative model, see **Figure 2.** The A matrix, or likelihood, describes the agent's beliefs about the mapping between observations o_t (i.e., defection vs. cooperation) and the hidden state of the environment s_t (i.e., cooperative vs. hostile). For example, an agent might expect to observe a cooperative reaction in a cooperative context with a probability of $p = 0.8$. In this case, the agent's belief is accurate, since the probability of observing a "share" in the cooperative environment is set to $p = 0.8$ in the environment. However, the agent can hold inaccurate beliefs, which we assume to be a central mechanism underlying mental disorders.

The B**,** or transition matrix, comprises the agent's beliefs about the evolution of the hidden states over time, given the present state and their own actions. If, for example, the agent infers to be in a cooperative context, they may believe that this context is stable and expect to remain in a cooperative state in the next timestep with a probability of $p = 0.9$. The B matrix parameterizes all possible state transitions.

The C vector formalizes the agent's prior preferences over outcomes. In the trust game, a "win" after engaging in a joint venture with the trustee may be the preferred outcome, whereas a "loss" of their investment after the trustee defects is the least preferred outcome. In the following, "win" here refers to the multiplied return on investment the agent receives when the trustee shares the amount previously invested by the agent.

Finally, the D vector contains the agent's prior beliefs over hidden states, i.e., the agent's beliefs about states at the start of the game. An agent may for example believe that the cooperative, hostile or random state are equally likely, assigning a probability of $p(context = x) = 0.333$ to each context. We now introduce the first agent, a healthy-to-optimistic agent, with a detailed explanation of the underlying generative model.

Healthy agent. We first developed a healthy-to-optimistic agent with a generative model close to the probability structure of the generative process (i.e., the environment). Their observation model consisted of three matrices: (1) the reward observation model A_1, containing the agent's expectations about reward outcomes associated with each context, (2) the behavior observation model A_2, i.e. the agent's internal

mapping between the context and the expected behavior of the trustee, and (3) the choice observation model A_3, which maps the agent's executed action to the expectation about the sensed action.

The healthy agent's reward observation is here reported separately for observing a certain outcome given they have decided to share with the trustee:

$$A_1 = \underbrace{\begin{pmatrix} .9 \\ .15 \\ .5 \end{pmatrix}}_{win}, \underbrace{\begin{pmatrix} .1 \\ .85 \\ .5 \end{pmatrix}}_{loss} \begin{matrix} coop. \\ hostile \\ random \end{matrix}$$

For example, when sharing in the cooperative context (first row), the agent expects to observe a return of their investment (win) with a probability of $.9$, and to observe a $loss$ as a consequence of the trustee defecting with a probability of $.1$.

In contrast, when the agent decides to keep their budget, they deterministically observe their initial budget as outcome. Note that this relationship remains the same for all subsequent agents.

$$A_2 = \underbrace{\begin{pmatrix} 1. \\ 1. \\ 1. \end{pmatrix}}_{budget} \begin{matrix} coop. \\ hostile \\ random \end{matrix}$$

The agent's behavior observation model is equivalent to the agent's reward observation model above, with the exception that it connects context and trustee behavior. There are three possible trustee behaviors to be observed: return, defect, and unknown. The return and defect outcomes are distributed as follows for the healthy agent:

$$A_2 = \underbrace{\begin{pmatrix} .9 \\ .15 \\ .5 \end{pmatrix}}_{return}, \underbrace{\begin{pmatrix} .1 \\ .85 \\ .5 \end{pmatrix}}_{defect} \begin{matrix} coop. \\ hostile \\ random \end{matrix}$$

When the agent decides to *keep* their initial budget, the trustee's behavior is unknown.

$$A_2 = \underbrace{\begin{pmatrix} 1. \\ 1. \\ 1. \end{pmatrix}}_{unknown} \begin{matrix} coop. \\ hostile \\ random \end{matrix}$$

For all subsequent agents, their behavioral observation model A_2 is equivalent to their reward observation model A_1. For simplicity, we therefore only report all agent's reward observation model A_1.

Finally, the agent's choice observation model A_3 maps the agent's executed action to the expectation about the sensed action. Here, for all agents, this mapping is deterministic across all contexts, i.e. (simplified matrix):

$$A_3 = \underbrace{\begin{pmatrix}1.\\0.\end{pmatrix}}_{\substack{share\\(observed)}}, \underbrace{\begin{pmatrix}.0\\1.\end{pmatrix}}_{\substack{keep\\(observed)}} \Bigg\} \begin{matrix} share \\ keep \end{matrix}$$

Next, the healthy agent's transition model was given by

$$B_1 = \underbrace{\begin{pmatrix}0.9\\0.32\\0.5\end{pmatrix}}_{cooperative}, \underbrace{\begin{pmatrix}0.02\\0.6\\0.3\end{pmatrix}}_{hostile}, \underbrace{\begin{pmatrix}0.08\\0.08\\0.2\end{pmatrix}}_{random} \Bigg\} \begin{matrix} coop. \\ host. \\ rand. \end{matrix}$$

The rows in the matrix refer to the three contexts [*cooperative, hostile, random*] in the current timestep, the columns reference the next timestep. The first column of the matrix hence contains the agent's beliefs about transitioning from a cooperative context into a cooperative one in the next timestep (p=0.9), into a hostile one (p=0.02, middle column) or a random context (p=0.08, right column).

The agent's policy-dependent transition probabilities are set up in a way that the agent always transitions into the action state determined by the previous action. In other words, in the generative model, the agent transitions into the action state at $t+1$ given by the previous action at t. Context dynamics cannot be directly influenced by the agent's action. The transitions between policies is deterministic over time, i.e.

$$B_2 = \underbrace{\begin{pmatrix}1\\0\\0\end{pmatrix}}_{\substack{share\\t}}, \underbrace{\begin{pmatrix}0\\1\\0\end{pmatrix}}_{\substack{keep\\t}}, \underbrace{\begin{pmatrix}0\\0\\1\end{pmatrix}}_{\substack{start\\t}} \Bigg\} \begin{matrix} share\ (t+1) \\ keep\ (t+1) \\ start\ (t+1) \end{matrix}$$

B_2 stays the same for all subsequent agent models, which is why report only B_1 for subsequent agents.

The agent's preferences are formalized in the C vector, given by

$$C = softmax\begin{pmatrix}2.5\\-2.2\\1.\end{pmatrix}\Bigg\} \begin{matrix} win \\ loss \\ budget \end{matrix} \rightarrow \begin{pmatrix}0.81\\0.07\\0.18\end{pmatrix}$$

This translates to a strong relative preference for observing a "win" outcome (i.e. a return of their investment), a relative aversion to observe "loss" outcomes (i.e. losing their investment to a defecting trustee), and a slight preference for observing a "budget" outcome (i.e., keeping their initial budget) compared to the "loss" outcome. Finally, the agent holds prior beliefs over hidden states, which are comprised in the D matrix

$$D = \begin{pmatrix}0.6\\0.35\\0.05\end{pmatrix}\Bigg\} \begin{matrix} coop. \\ host. \\ rand. \end{matrix}$$

The agent believes the cooperative state to be more probable a priori. At the same time, a hostile state is possible, but less likely than a cooperative state. A random environment is very unlikely to the agent. These beliefs were chosen in line with the literature on optimism biases in healthy individuals, where

positive outcomes are assumed to be more likely than negative ones (Dieguez et al., 2015; Galinsky et al., 2008; Sharot, 2011).

Learning is implemented by updates to the model's parameters after making observations. Learning in active inference is assumed to happen at a slower timescale compared to hidden state inference, but follows the same imperative of minimizing variational free energy (Heins et al., 2022). In our simulations, model parameters A, B & D are updated using Dirichlet posteriors, which are computed from the Dirichlet prior over the parameter distribution, observations, and a learning rate (Heins et al., 2022). The learning rate is used to scale the magnitude of the update to the posterior over each parameter. The learning rates for the healthy agent are $lr_A = 0.1, lr_B = 3.0, lr_D = 1.0$. The observation model A is updated only slowly since it contains stable beliefs about what types of observations expects under a certain context. In contrast, B is updated with a large learning rate to achieve the flexible decision-making characteristic for healthy individuals in dynamic environments. In all simulations, C is not updated, underlying which is the assumption that agents hold relatively stable reward outcome preferences over the course of a game. Finally, D is updated with a learning rate of 1.0 to achieve updating with observations, while not deviating from prior beliefs too quickly. For further details on and derivations of update rules, see Heins et al., 2022 (appendix: learning.py).

The simulations reported in the next section are the result of deviations from this normative, or "healthy" agent's generative model and should be understood in relation to the parameter settings introduced above. For a tabular overview of all agents, see **Table 1**.

Between- and within-disorder simulations

Depression. The depressed agents' generative model comprises a range of pessimistic beliefs (Barkus & Badcock, 2019; Hames et al., 2013; Kupferberg et al., 2016), which are formalized predominantly within the B and D matrices. Research suggests that individuals with symptoms of depression respond more strongly to unpleasant stimulation compared to controls (Gradin et al., 2015; Haley & Strickland, 1986; Pulcu et al., 2015), which may be a consequence of more precise "pessimistic" prior beliefs (D) and more fatalistic transition probabilities (B), where positive contexts are assumed to turn aversive with a high probability, and negative contexts are believed to remain more stable. Furthermore, patients with depression show difficulties adapting flexibly to changing contexts in neuroeconomic games (Pulcu et al., 2015), which we formalize by introducing context switches throughout the game. Finally, there is some evidence for decreased reward sensitivity in depression (Radke et al., 2013; Scheele et al., 2013; Wang et al., 2014), which we operationalized in the depressed agent's generative model via a slightly more ambiguous preference over outcomes (C). Both agent's observation model A is equivalent to the healthy agent's parameter.

Reward-insensitive and pessimistic. The first agent is designed to model a loss-averse and pessimistic sub-type of depression. The generative model for this agent is hence given by

$$B = \left.\underbrace{\begin{pmatrix}0.2\\0.05\\0.15\end{pmatrix}}_{cooperative}, \underbrace{\begin{pmatrix}0.7\\0.9\\0.15\end{pmatrix}}_{hostile}, \underbrace{\begin{pmatrix}0.1\\0.05\\0.7\end{pmatrix}}_{random}\right\} \begin{matrix}coop.\\host.\\rand.\end{matrix}$$

$$C = softmax\begin{pmatrix}0.8\\-2.2\\1.\end{pmatrix}\left\}\begin{matrix}win\\loss\\budget\end{matrix}\right. \rightarrow \begin{pmatrix}0.44\\0.02\\0.53\end{pmatrix}$$

$$D = \begin{pmatrix}0.15\\0.8\\0.05\end{pmatrix}\left\}\begin{matrix}coop.\\host.\\rand.\end{matrix}\right.$$

The agent believes the cooperative context to develop towards a more hostile context with a higher probability (B, p(host.|coop)=0.7) compared to a cooperative (p(coop.|coop) = 0.2) or random (p(random|coop) = 0.1) context. Furthermore, observing a return of investment, or "win" outcome, has lost utility for this agent compared to the healthy agent (C, p(win)=.44); instead, they have a stronger preference for observing their initial budget (p(budget)=.53). Finally, the agent is more certain a-priori about being in a hostile context, as parameterized by their heightened prior (D) on hostile contexts (p(hostile)=.8).

The reward-insensitive and pessimistic agent shows only small updates to their observational model A, equivalent to the healthy-to-optimistic agent ($lr_A = 0.1$). However, to simulate rigidity in the face of dynamically evolving environments, this agent does not update their B parameter considering new observations. D is updated in a similar way to the healthy agent ($lr_D = 1.0$).

Fatalistic. In another simulation of a depressed agent (*fatalistic* in the following), we aim at modelling treatment-resistant depression, marked by rigid fatalistic beliefs where a low probability is assigned to transitioning into a cooperative context when in a hostile or random context. The cooperative context is assumed to be the least stable context over time. The agent exhibits a stronger preference for the "win" outcome and a greater loss aversion than the healthy agent. Further, a low prior probability is assigned to the cooperative context a-priori, with a higher a-priori belief in a random (or unpredictable) context compared to the reward-insensitive and pessimistic depressed agent.

$$B = \left.\underbrace{\begin{pmatrix}0.6\\0.05\\0.1\end{pmatrix}}_{cooperative}, \underbrace{\begin{pmatrix}0.3\\0.8\\0.6\end{pmatrix}}_{hostile}, \underbrace{\begin{pmatrix}0.1\\0.15\\0.3\end{pmatrix}}_{random}\right\} \begin{matrix}coop.\\host.\\rand.\end{matrix}$$

$$C = softmax\begin{pmatrix}2.5\\-2.5\\1.5\end{pmatrix}\left\}\begin{matrix}win\\loss\\keep\end{matrix}\right. \rightarrow \begin{pmatrix}0.72\\0.004\\0.26\end{pmatrix}$$

$$D = \begin{pmatrix}0.1\\0.6\\0.3\end{pmatrix}\left\}\begin{matrix}coop.\\host.\\rand.\end{matrix}\right.$$

In this agent, parameters A and D are updated equivalent to the healthy-to-optimistic agent ($lr_A = 0.1, lr_D = 1.0$). However, their learning rate over B is reduced compared to the healthy agent ($lr_B = 1.0$), modelling beliefs over the game's dynamics that are less changeable in light of new observations.

Social anxiety disorder. When modelling an agent with social anxiety, we assume perturbations to the preferred outcome matrix C, leading to a reduction in interpersonal exploration and increased avoidance behaviors. Specifically, an increased risk, ambiguity and loss aversion might be underlying social anxiety in the context of the trust game (Alden & Taylor, 2004; Hartley & Phelps, 2012). In empirical studies, individuals with social anxiety show increased risk aversion and avoidance behaviors (Charpentier et al., 2017; Lorian & Grisham, 2010). Furthermore, studies suggest a decreased neural response to cooperative partners as a consequence of a deficit in social learning over time (C. Sripada et al., 2013) and reduced mPFC activity, which may be related to reduced learning about and/ or exploration of the trustee's intentions (Sripada et al., 2013). Such avoidant behaviors could be caused by an increased loss aversion (C) and pessimistic prior beliefs about the social context (D) (Taylor & Wald, 2003).

Insecure-avoidant. The first socially anxious agent was designed to model an *insecure-avoidant* sub-type. We first introduce some ambiguity to the agent's observation model A:

$$A_1 = \underbrace{\begin{pmatrix} .6 \\ .15 \\ .4 \end{pmatrix}}_{win}, \underbrace{\begin{pmatrix} .4 \\ .85 \\ .6 \end{pmatrix}}_{loss} \begin{matrix} coop. \\ hostile \\ random \end{matrix}$$

The transition model is perturbed in a way that the agent is biased towards expecting an unpredictable, or random context in the next timestep when currently in a cooperative or random context. The hostile context is assumed to remain stable over time:

$$B = \underbrace{\begin{pmatrix} 0.2 \\ 0.1 \\ 0.2 \end{pmatrix}}_{coop.}, \underbrace{\begin{pmatrix} 0.3 \\ 0.4 \\ 0.3 \end{pmatrix}}_{hostile}, \underbrace{\begin{pmatrix} 0.5 \\ 0.5 \\ 0.5 \end{pmatrix}}_{random} \Bigg\} \begin{matrix} coop. \\ host. \\ rand. \end{matrix}$$

We furthermore manipulate the preference over outcomes C in the insecure-avoidant subtype such that

$$C = softmax \begin{pmatrix} 2.5 \\ -3.5 \\ 1.0 \end{pmatrix} \begin{matrix} win \\ loss \\ keep \end{matrix} \rightarrow \begin{pmatrix} 0.815 \\ 0.002 \\ 0.182 \end{pmatrix},$$

which leads to the agent's relative inability to reliably connect observations to the hidden context.

The insecure-avoidant agent assigns a very low preference to the loss outcome. Consequently, the agent's priority is to avoid observing losses. The agent further holds idiosyncratic prior beliefs:

$$D = \begin{pmatrix} 0.4 \\ 0.2 \\ 0.4 \end{pmatrix} \begin{matrix} coop. \\ host. \\ rand. \end{matrix}$$

This choice was made to operationalize the agent's indecisiveness with respect to interpersonal contexts, where most contexts are believed to be random and unpredictable a-priori.

Similar to the healthy-to-optimistic agent, the insecure-avoidant agent updates their observation model A only slowly, with a learning rate of $lr_A = 0.1$; and their prior over contexts with a learning rate of $lr_D = 1.0$. Their transition model B is not updated to model rigid, fatalistic beliefs of the game's dynamics that are not changed by new observations.

Trauma-and-defeat. We simulate an additional socially anxious agent. Like the insecure-avoidant type, the agent's A_1 parameter is characterized by observational ambiguity:

$$A_1 = \underbrace{\begin{pmatrix} .6 \\ .4 \\ .5 \end{pmatrix}}_{win}, \underbrace{\begin{pmatrix} .4 \\ .6 \\ .5 \end{pmatrix}}_{loss} \begin{matrix} coop. \\ host. \\ rand. \end{matrix}$$

The B matrix for the trauma-and-defeat agent is set so that the hostile and random contexts tend to remain stable, while the cooperative context is assumed to develop into a hostile context in the next timestep:

$$B_1 = \underbrace{\begin{pmatrix} 0.2 \\ 0.15 \\ 0.2 \end{pmatrix}}_{coop.}, \underbrace{\begin{pmatrix} 0.6 \\ 0.8 \\ 0.3 \end{pmatrix}}_{hostile}, \underbrace{\begin{pmatrix} 0.2 \\ 0.05 \\ 0.5 \end{pmatrix}}_{random} \begin{matrix} coop. \\ host. \\ rand. \end{matrix}$$

Their behavior and decision-making is governed by observational preferences of

$$C = softmax \begin{pmatrix} 1.2 \\ -2.2 \\ 1.3 \end{pmatrix} \begin{matrix} win \\ loss \\ keep \end{matrix} \rightarrow \begin{pmatrix} 0.46 \\ 0.01 \\ 0.51 \end{pmatrix}.$$

These parameters were chosen to simulate loss avoidance on the one hand, and a relative insensitivity to rewards on the other – note that here, the agent slightly favors keeping their initial wage over observing a win from the joint venture with the trustee.

This agent is further characterized by pessimistic prior beliefs (D), e.g., a consequence of traumatic previous interpersonal experiences:

$$D = \begin{pmatrix} 0.2 \\ 0.8 \\ 0.0 \end{pmatrix} \begin{matrix} coop. \\ host. \\ rand. \end{matrix}.$$

The agent's learning is characterized by similar updates applied to the A and D parameters as the healthy-to-optimistic agent ($lr_A = 0.1, lr_D = 1.0$). However, learning over the game's course is reduced compared to the healthy agent with $lr_B = 1.0$.

For each agent type, their average earned rewards are computed by simulating a timeseries with T=40 for 20 times per agent type, starting with the cooperative context and switching to the hostile context and vice versa.

Name	A_1	B	C $\begin{pmatrix} win \\ loss \\ keep \end{pmatrix}$	D $\begin{pmatrix} coop. \\ host. \\ neut. \end{pmatrix}$
Healthy	$\underbrace{\begin{pmatrix} .9 \\ .15 \\ .5 \end{pmatrix}}_{win}, \underbrace{\begin{pmatrix} .1 \\ .85 \\ .5 \end{pmatrix}}_{loss} \left.\right\} \begin{matrix} coop. \\ hostile \\ random \end{matrix}$	$\underbrace{\begin{pmatrix} 0.9 \\ 0.32 \\ 0.5 \end{pmatrix}}_{coop.}, \underbrace{\begin{pmatrix} 0.02 \\ 0.6 \\ 0.3 \end{pmatrix}}_{hostile}, \underbrace{\begin{pmatrix} 0.08 \\ 0.08 \\ 0.2 \end{pmatrix}}_{rand.} \left.\right\} \begin{matrix} coop. \\ host. \\ rand. \end{matrix}$	$sm\begin{pmatrix} 2.5 \\ -2.2 \\ 1. \end{pmatrix}$	$\begin{pmatrix} 0.6 \\ 0.35 \\ 0.05 \end{pmatrix}$
Depression *reward-insensitive and pessimistic*	$\underbrace{\begin{pmatrix} .9 \\ .15 \\ .5 \end{pmatrix}}_{win}, \underbrace{\begin{pmatrix} .1 \\ .85 \\ .5 \end{pmatrix}}_{loss} \left.\right\} \begin{matrix} coop. \\ hostile \\ random \end{matrix}$	$\underbrace{\begin{pmatrix} 0.2 \\ 0.05 \\ 0.15 \end{pmatrix}}_{coop.}, \underbrace{\begin{pmatrix} 0.7 \\ 0.9 \\ 0.15 \end{pmatrix}}_{hostile}, \underbrace{\begin{pmatrix} 0.1 \\ 0.05 \\ 0.7 \end{pmatrix}}_{rand.} \left.\right\} \begin{matrix} coop. \\ host. \\ rand. \end{matrix}$	$sm\begin{pmatrix} 0.8 \\ -2.2 \\ 1. \end{pmatrix}$	$\begin{pmatrix} 0.15 \\ 0.8 \\ 0.05 \end{pmatrix}$
Depression *fatalistic*	$\underbrace{\begin{pmatrix} .9 \\ .15 \\ .5 \end{pmatrix}}_{win}, \underbrace{\begin{pmatrix} .1 \\ .85 \\ .5 \end{pmatrix}}_{loss} \left.\right\} \begin{matrix} coop. \\ hostile \\ random \end{matrix}$	$\underbrace{\begin{pmatrix} 0.6 \\ 0.05 \\ 0.1 \end{pmatrix}}_{coop.}, \underbrace{\begin{pmatrix} 0.3 \\ 0.8 \\ 0.6 \end{pmatrix}}_{hostile}, \underbrace{\begin{pmatrix} 0.1 \\ 0.15 \\ 0.3 \end{pmatrix}}_{rand.} \left.\right\} \begin{matrix} coop. \\ host. \\ rand. \end{matrix}$	$sm\begin{pmatrix} 2.5 \\ -2.5 \\ 1.5 \end{pmatrix}$	$\begin{pmatrix} 0.1 \\ 0.6 \\ 0.3 \end{pmatrix}$
Social anxiety *insecure avoidant*	$\underbrace{\begin{pmatrix} .6 \\ .15 \\ .4 \end{pmatrix}}_{win}, \underbrace{\begin{pmatrix} .4 \\ .85 \\ .6 \end{pmatrix}}_{loss} \left.\right\} \begin{matrix} coop. \\ hostile \\ random \end{matrix}$	$\underbrace{\begin{pmatrix} 0.2 \\ 0.1 \\ 0.2 \end{pmatrix}}_{coop.}, \underbrace{\begin{pmatrix} 0.3 \\ 0.4 \\ 0.3 \end{pmatrix}}_{hostile}, \underbrace{\begin{pmatrix} 0.5 \\ 0.5 \\ 0.5 \end{pmatrix}}_{random} \left.\right\} \begin{matrix} coop. \\ host. \\ rand. \end{matrix}$	$sm\begin{pmatrix} 2.5 \\ -3.5 \\ 1.0 \end{pmatrix}$	$\begin{pmatrix} 0.4 \\ 0.2 \\ 0.4 \end{pmatrix}$
Social anxiety *trauma and defeat*	$\underbrace{\begin{pmatrix} .6 \\ .4 \\ .5 \end{pmatrix}}_{win}, \underbrace{\begin{pmatrix} .4 \\ .6 \\ .5 \end{pmatrix}}_{loss} \left.\right\} \begin{matrix} coop. \\ host. \\ rand. \end{matrix}$	$\underbrace{\begin{pmatrix} 0.2 \\ 0.15 \\ 0.2 \end{pmatrix}}_{coop.}, \underbrace{\begin{pmatrix} 0.6 \\ 0.8 \\ 0.3 \end{pmatrix}}_{hostile}, \underbrace{\begin{pmatrix} 0.2 \\ 0.05 \\ 0.5 \end{pmatrix}}_{random} \left.\right\} \begin{matrix} coop. \\ host. \\ rand. \end{matrix}$	$sm\begin{pmatrix} 1.2 \\ -2.2 \\ 1.3 \end{pmatrix}$	$\begin{pmatrix} 0.2 \\ 0.8 \\ 0.0 \end{pmatrix}$

Table 1. Generative model parameters for all simulations.

A_1: left column: probability of observing a return of investment ("win") by trustee for all contexts (rows); right column: probabilities of observing a "loss" (trustee defecting) across contexts (rows). B**:** transition probability from context (cooperative, hostile, random, three matrices) to cooperative (top row), hostile (middle row) and random contexts (bottom row). C**:** Preference over rewards, with win (top row), loss (middle row) and keeping the same (bottom row). sm: softmax. D: Prior over contexts, with cooperative (top), hostile (middle) and random (bottom) context priors.

Data collection.

We collected behavioral data from patients with a diagnosis of a depressive disorder (ICD-10 diagnoses F32-F33) who are undergoing treatment in the outpatient clinic at Justus-Liebig-Universität Gießen. Patients gave written informed consent prior to participation. All procedures were in line with the declaration of Helsinki and approved by the institutional review boards of the Philipps-Universität Marburg and Justus-Liebig-Universität Gießen (reference number #2022-10k). The data reported here were collected as part of a larger study (pre-registration: https://aspredicted.org/see_one.php) A total of N=23 patients participated in the trust game and were randomly assigned to one of three conditions. In the hostile-to-cooperative condition, participants played with a trustee who returns their investment with a relatively low probability (p=0.2) for t=30 trials, followed by a cooperative phase in which the trustee returns the patients' investment with a higher probability (p=0.8). In the cooperative-to-hostile condition, patients first interacted with cooperative trustee for t=30 trials, followed by t=30 trials of playing with more hostile trustees who return the patient's investment relatively rarely. In a third condition, the trustee's behavior was random, i.e. determined by coin-flip in each trial. Participants first received written instructions about the game. They were instructed that they will be playing with other participants who are currently online. The experimental paradigm was equivalent to the simulated task (**Figure 1**) with two exceptions. Firstly, participants were instructed that they are playing with a real trustee via an online platform, and jittered waiting times were introduced between trials and after a share action in line with the cover story. Secondly, participants played 60 trials of the trust game, each of which was with one of 10 possible trustees which were introduced with a code name. In the end, the participant received 50% of the monetary amount earned in the game as a financial compensation for their participation.

After completing the game, patients filled out several clinical and demographic surveys, including the Beck's Depression Inventory (BDI, Beck et al., 1988) which will be used in the current analysis to group patients into high- and low depressed individuals.

Optimization.

We fit the model to the behavioral data of participants playing the trust game using optimization, with the goal of obtaining a set of individual parameters A, B, C, D and ε for each participant. We introduced ε as a free parameter which determines the weighting between hedonic (reward-oriented) and epistemic (information-oriented) drives by scaling the update especially to parameter A after observations. This was done to obtain more behaviorally flexible strategies during the model fitting and initialization during the fitting. The parameters obtained through the optimization best capture actual participant behavior, yielding computational-level approximations of the participant's generative model while performing the task.

For the optimization, we defined a cost function

$$c = \sum_{i=0}^{I} q_i \cdot log\left(\frac{q_i}{as_i}\right) + (1 - q_i) \cdot log\left(\frac{1 - q_i}{1 - as_i}\right)$$

where q_i are the normalized behavioral responses in trial i ($q_i = 1$ indicates a share action, or investment by the agent; and $q_i = 0$ indicates that the agent keeps their budget):

$$q = \frac{\sum r_t = 1}{\sum T}$$

a_s are the probabilities of "share", or invest, choices obtained from running the active inference model with optimized parameters. Minimizing c is equivalent to finding parameters that yield simulated actions as close as possible to the observed actions r_t, which enter c in normalized form q. In other words, minimizing c identifies parameters of the generative model that result in the best possible match between simulated- and empirically observed actions.

The objective function was used to find the optimum of 15 parameter distributions in total, with 3 parameter values for A, six parameter values for B, three parameter values for C and two parameter values for D, the exact number of parameters required for the simulation. We furthermore introduced parameter ε, which was used to weight hedonic (reward-oriented) vs. epistemic drives by scaling the update on parameters A and B after a new observation. ε was another free parameter that was included in the optimization. The optimization was performed using the Powell's conjugate direction method of the module scipy-optimize (scipy version 1.11.4, Virtanen et al., 2020). Tolerance was set to $tol = 1e^{-8}$ with the maximum number of iterations set to 200. The optimization started with a random initialization for all parameters that were bounded to psychologically plausible ranges (in logit space, $A{:}\, p(\text{win}|\text{coop}) = (1.0, 4.0), p(\text{win}|\text{hostile}) = (-4.0, -1.0), p(\text{win}|\text{random}) = (-1.0, 1.0); B{:}\, p(c|c) = (0.0, 4.0), p(h|c) = (-2.0, 0.0), p(c|h) = (-2.0, 0.0), p(h|h) = (0.0, 4.0), p(c|r) = (-2.0, 0.0), p(h|r) = (-2.0, 0.0); C{:}\, p(\text{win}) = (1.0{,}4.0), p(\text{loss}) = (-4.0, -1.0), p(\text{budget}) = (-1.0, 1.0), D{:}\, p(\text{cooperative}) = (-2.0{,}2.0), p(\text{hostile}) = (-2.0{,}2.0), p(\text{random}) = (-2.0{,}2.0);\ \varepsilon = (0.0, 1.0)$). Learning rates were kept constant across all optimization runs and equivalent to the simulation (Learning Rate $LR(A) = 0.1, LR(B) = 3.0, LR(D) = 1.0$). The constraints on the parameters set by the nature of the game were maintained in the optimization (constraints on A; a zero probability to obtaining a return on investment when the agent chooses to keep their budget; $p(\text{win}|\text{keep} - \text{budget}) = 0$ etc). The optimization algorithm hence looked as follows:

1. Read participant dataset
2. Randomly initialize the required 14 parameters in logit space within pre-defined bounds
3. Softmax logit parameters and initialize the agent model with parameters and participant data
4. Compute the cost function
5. Reiterate and restrict the search for new parameter values towards minimizing the cost function
6. Repeat steps 3-5 until change in objective is below tolerance levels

The whole procedure was performed 10 times per participant to account for the stochastic variance in the minimization scheme, and the run with the lowest cost is chosen for each participant. The optimization was performed at the Marburg High Performance Computing Cluster 3 (MarC3). Results of the optimization were used to analyze group differences in generative model parameters between patients scoring low on the BDI (mild depressive symptoms) and patients scoring high on the BDI (moderate to severe depressive symptoms; median split and minimal/ maximum score split, defined by scores that lie one standard deviation above- or below sample median score). We furthermore investigated potential parameter differences between individuals who played a hostile-to-cooperative game vs. a cooperative-to-hostile game. Finally, we ran a series of simulations with the optimal parameters of the patients with the least- and most severe depressive symptomatology, respectively. For this, the parameters obtained from the optimization were inserted in the agent model and twenty runs each of a cooperative-to-hostile and a hostile-to-cooperative game were simulated.

Results

Simulations

Healthy interpersonal inference. **Figure 3** shows exemplary results of an agent with an accurate-to-optimistic generative model ("healthy" agent, appendix **Figure 28** for results averaged over N=20 runs). When starting in a cooperative context (**Figure 3A**, blue area), some initial defects on the trustee's side are processed adequately and the hostile context becomes more likely. During a long series of observing returns on their investment, the agent forms a precise posterior belief about being in a cooperative context (max. $p(\text{context} = \text{cooperative}) = 0.95$). After the context switches to hostile, this agent accurately infers to be in a hostile context (max. $p(\text{context} = \text{hostile} = 0.89$). Any individual returns on their investment leads to a large posterior belief to be in a cooperative context again, which is reminiscent of an optimism bias (Bracha & Brown, 2012; Sharot, 2011). After observing several defects, the agent adopts a more careful behavioral strategy and starts avoiding the investment. However, the agent's strong preference for *win* outcomes (return on investment) and their epistemic drive leads to recurrent probing of the trustee.

When faced with the negative context first, the agent forms precise posterior beliefs about being in a hostile context (max. $p(\text{context} = \text{hostile} = 0.96)$. Here, the agent starts to adopt an avoidant behavioral strategy after t=13 timesteps, during which 10 defects were observed. Note that the posterior belief $p(\text{context} = \text{hostile})$ becomes less certain with each "keep" action, resulting in a drive towards exploration. When the context switches to a cooperative one, the agent quickly returns to precise posterior beliefs about being in a cooperative context and interacting with the trustee.

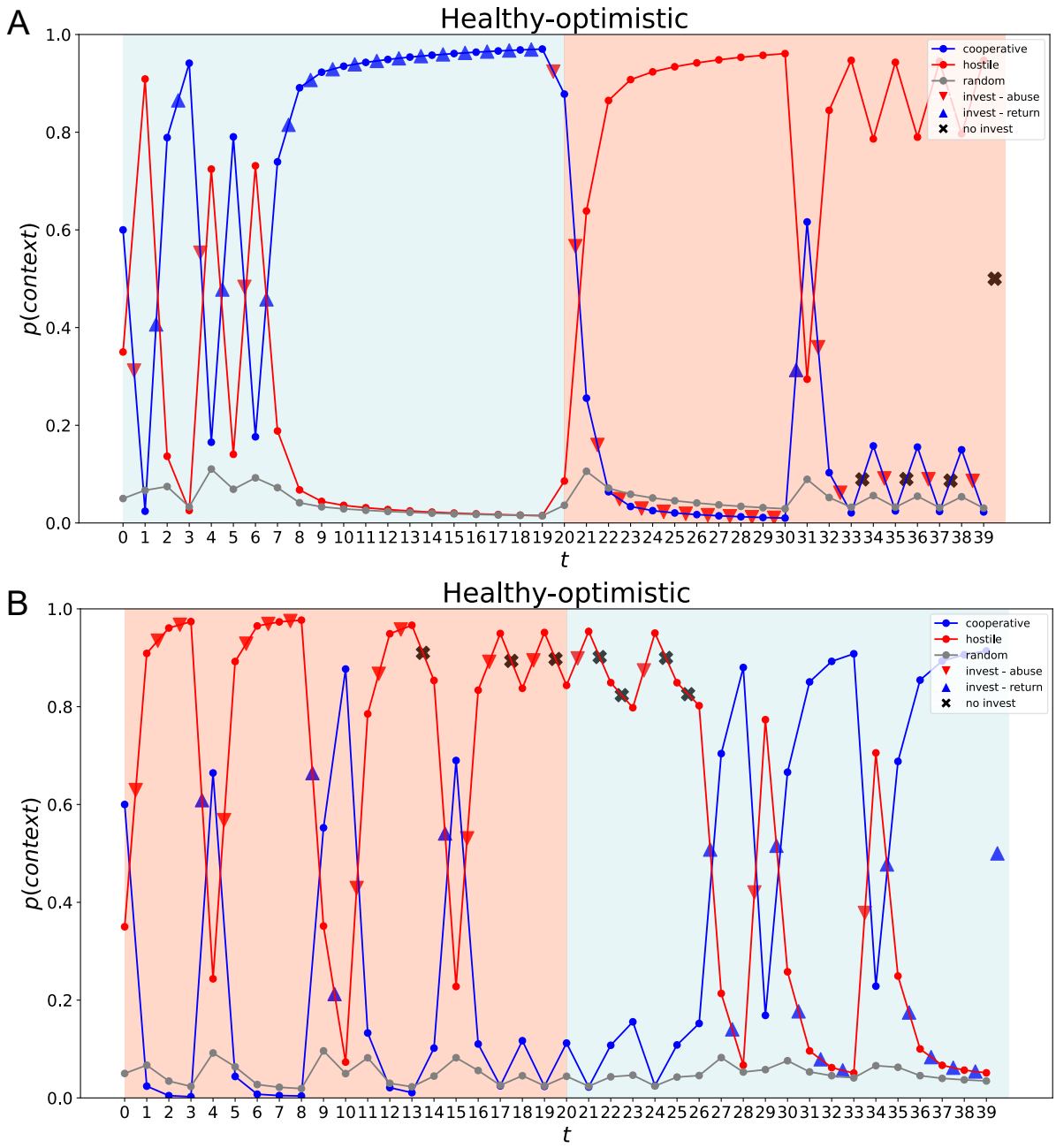


Figure 3. Simulation of a healthy-to-optimistic agent

A. The healthy agent (i.e., an agent with an optimistic-to-accurate generative model of the task) playing the trust game, first in a cooperative context (blue background, epochs 0-20), followed by a hostile context (red background, epochs 20-39). After observing returns from the trustee (blue triangle), the agent is confident that they are in a cooperative context. When the switch to a more hostile context happens, the agent's posterior belief switches and they believe to be in a hostile context, but their behavioral strategy switches relatively late in the time series.

B. Similar as **A**, but the game starts with a hostile context, followed by a cooperative one. While the agent quickly believes to be in a hostile context and starts to adapt their behavior accordingly, they continue to explore, which allows them to detect the switch to a more cooperative context after 20 timesteps.

Depression in our models results from a combination of processes, comprising pessimistic prior beliefs about context (D), pessimistic beliefs about the future (B), and decreased reward sensitivity (C). We simulated two sub-types of depression, simulating perturbations in the generative model that are in line with the literature on depression.

Depression: reward-insensitive and pessimistic. The first model of depression (*reward-insensitive and pessimistic*) holds a generative model which compared to healthy 1) assumes cooperative contexts to be more volatile compared to hostile contexts; 2) a relative insensitivity to rewards, and 3) a higher prior belief to be in a hostile context. When first observing the trustee return their investment, the agent does not accurately infer a cooperative context (max. $p(\text{context} = \text{cooperative}) = 0.68$). In the example of **Figure 4A-B** (averaged results suppl. **Figure 29**), the agent experiences one defect at t=3, after which their posterior belief to be in a cooperative context approaches zero. Subsequently, observing a series of cooperative interactions, the agent infers a cooperative context with a maximum posterior probability of $p(\text{context} = \text{cooperative}) = 0.51$. Few further defects (a total of t=3 in the cooperative context) lead to the agent adopting a rigid avoidant strategy from t=17 onwards. The context switch from a cooperative to a hostile context is hence not represented in the agent's posterior beliefs. When faced with a hostile context first (**4B**), the agent's poster belief to be in a hostile context approaches $p(context = hostile) = 0.96$ here and they stop investing with the trustee, while continuing to probe their behavior. After the context switch, the agent adopts an avoidant strategy after observing another defect. Thereafter, the agent stops exploring their trustee's behavior for the rest of the time series.

Depression: fatalistic. An agent with more rigid, fatalistic beliefs about the evolution of hidden states over time and pessimistic prior beliefs is shown in **Figure 5C-D** (averaged results supplementary **Figure 30**). When faced with the cooperative context first, the fatalistic agent comes to the precise posterior belief to be in the cooperative context (max. $p(\text{context} = \text{cooperative}) = 0.9$), and they continue to interact with the trustee. After observing defects in the hostile context, the agent increasingly avoids the interaction with the trustee. Their belief to be in a cooperative context drops close to $p(\text{context} = \text{cooperative})\approx 0.05$. When in the hostile context first, they accurately infer to be the context's nature. However, the agent continues to explore the trustee's behavior and correctly infers a cooperative context to be more likely after observing returns on their investment.

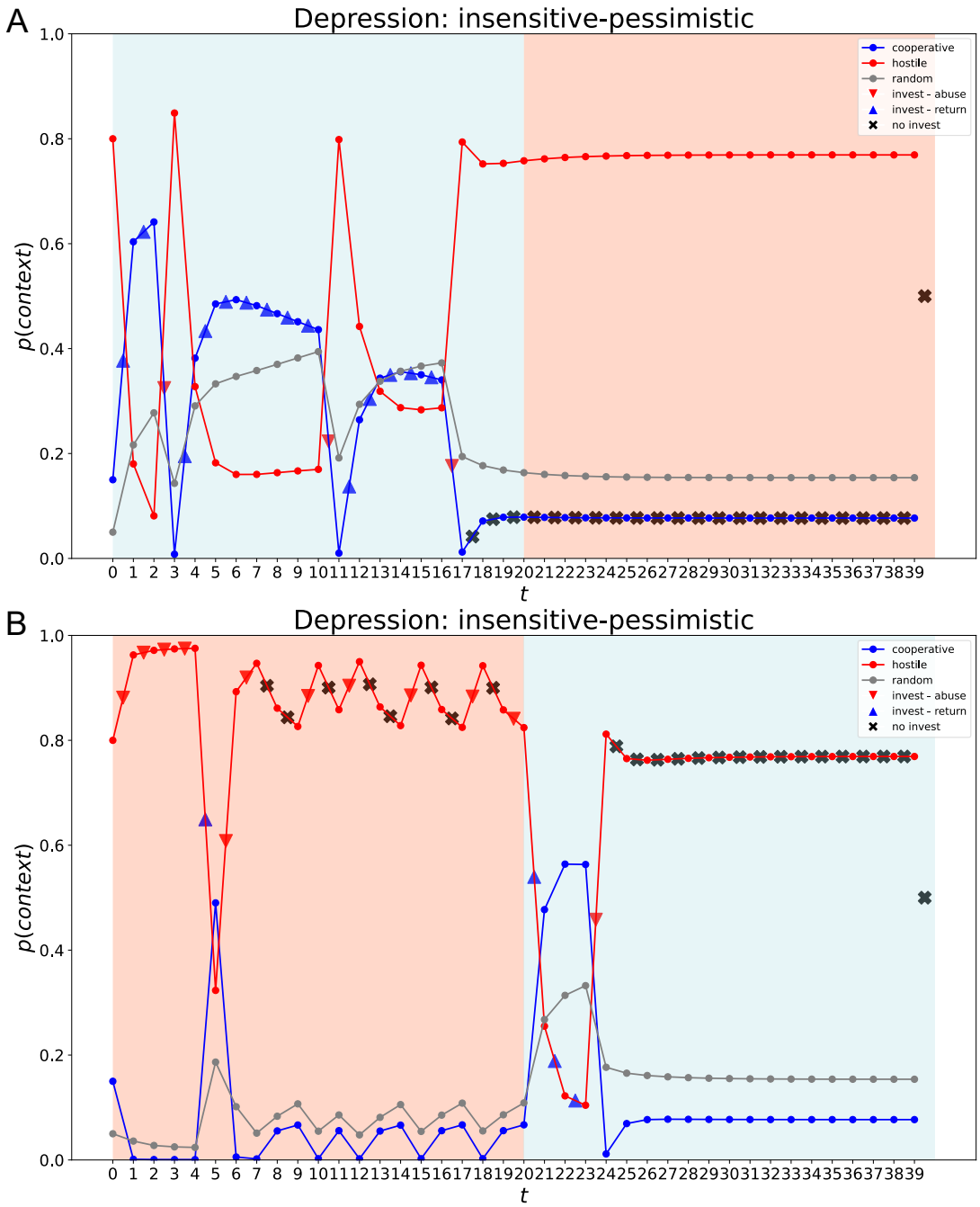


Figure 4. Simulation of depression, reward-insensitive and pessimistic.

A The insensitive-pessimistic (Type 1 depression) agent cannot identify the cooperative nature of the context as clearly as the healthy agent in Fig. 2. Individual defects by the trustee cause spikes in the posterior belief to be in a hostile context. The agent here furthermore avoids the interaction before the context switch, and this behavioral strategy remains rigid for the rest of the time series.

B When starting in a hostile context, the agent makes a few negative experience when investing with the trustee and as a consequence their posterior belief to be in a hostile context approaches p(context=hostile)=0.95. The agent shows a mixture of continued exploration of the trustee behavior and avoidance during the hostile context. After the context switches, rigid avoidance behavior is adopted after an individual defect by the trustee.

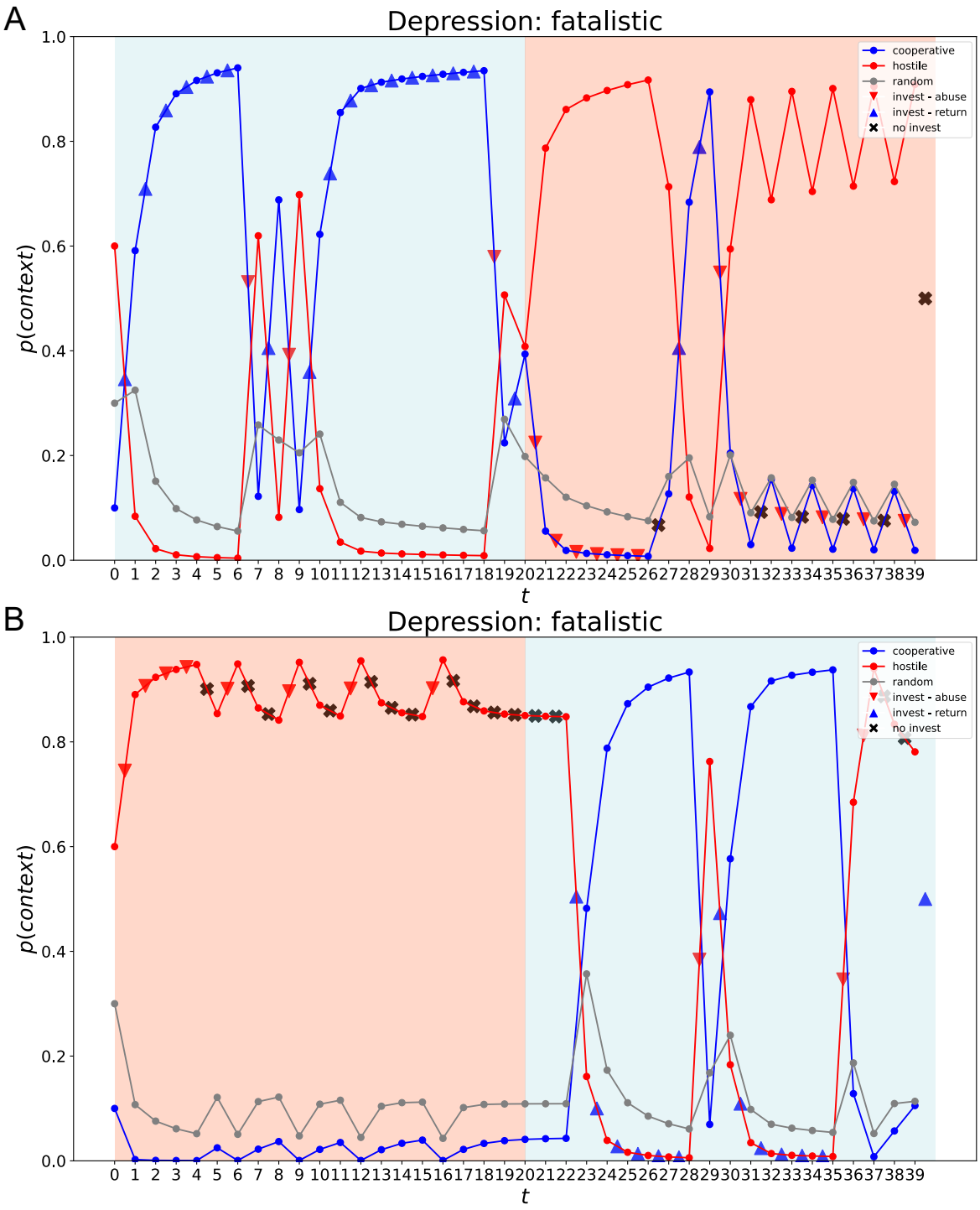


Figure 5. Simulation of depression, fatalistic.

A shows the fatalistic agent in a cooperative, then hostile context. During multiple cooperative interactions, the agent's belief to being in a cooperative context p(context=cooperative) accumulates in line with observing returns of their investment. Upon transitioning into a hostile context, the agent's beliefs rapidly shift to favor the hostile state.

B The fatalistic agent, when first exposed to a hostile context, correctly infers its nature and begins to avoid investing with the trustee. However, due to continued exploration, the agent registers the context switch. Individual defects in the cooperative context influence the agent's posterior beliefs drastically compared to the healthy-to-optimistic agent (**Fig. 2**)

Socially anxious interpersonal inference. The generative models of the socially anxious agents are characterized by more ambiguous observation models (A), fatalistic beliefs in the dynamics of the context over time (B), heightened loss aversion (C), and/ or more pessimistic prior beliefs (D) compared to healthy interpersonal inference.

Social anxiety: insecure-avoidant (Type 1). **Figure 6A-B** shows the behavior of the *insecure-avoidant* social anxiety agent in the trust game (averaged results supplementary **Figure 31**). This agent's observational model is marked by a more ambiguous mapping between observations and the corresponding context (A). The agent further believes that the context dynamics will develop towards random or unpredictable contexts (B) and an increased focus on loss avoidance (C). A random and a positive context are believed to be more likely a-priori than a hostile context (D). When first immersed in a cooperative context, the agent infers the random context to be the most likely cause of observing returns of investment (**Figure 6A**). When first exposed to the hostile context, the random- or hostile context are assumed to be the most likely cause of observing defects (**Figure 6B**). The agent quickly retreats to an avoidant behavioral strategy.

Social anxiety: trauma-and-defeat (Type 2). This agent is characterized by an ambiguous observation model (A) and the belief that cooperative contexts are volatile than hostile or random contexts, which are assumed to be more stable (B). The agent is further characterized by reward insensitivity (C) and a increased prior belief to be in a hostile interpersonal context (D) compared to the healthy-to-optimistic agent. This agent infers a hostile context to be the most likely when in a cooperative context first (**Figure 7A,** averages appendix **Figure 32**). The agent does not deviate much from their high prior belief to be in a hostile context, despite observing a series of returns of their investment. Observing one defect in the cooperative phase leads the agent to adopt avoidance behaviors that are maintained until the end of the game. When in a hostile context first (**Figure 7B**), the agent continues to believe to be in a hostile context and avoids investing with the trustee after observing a few defects for the rest of the time series.

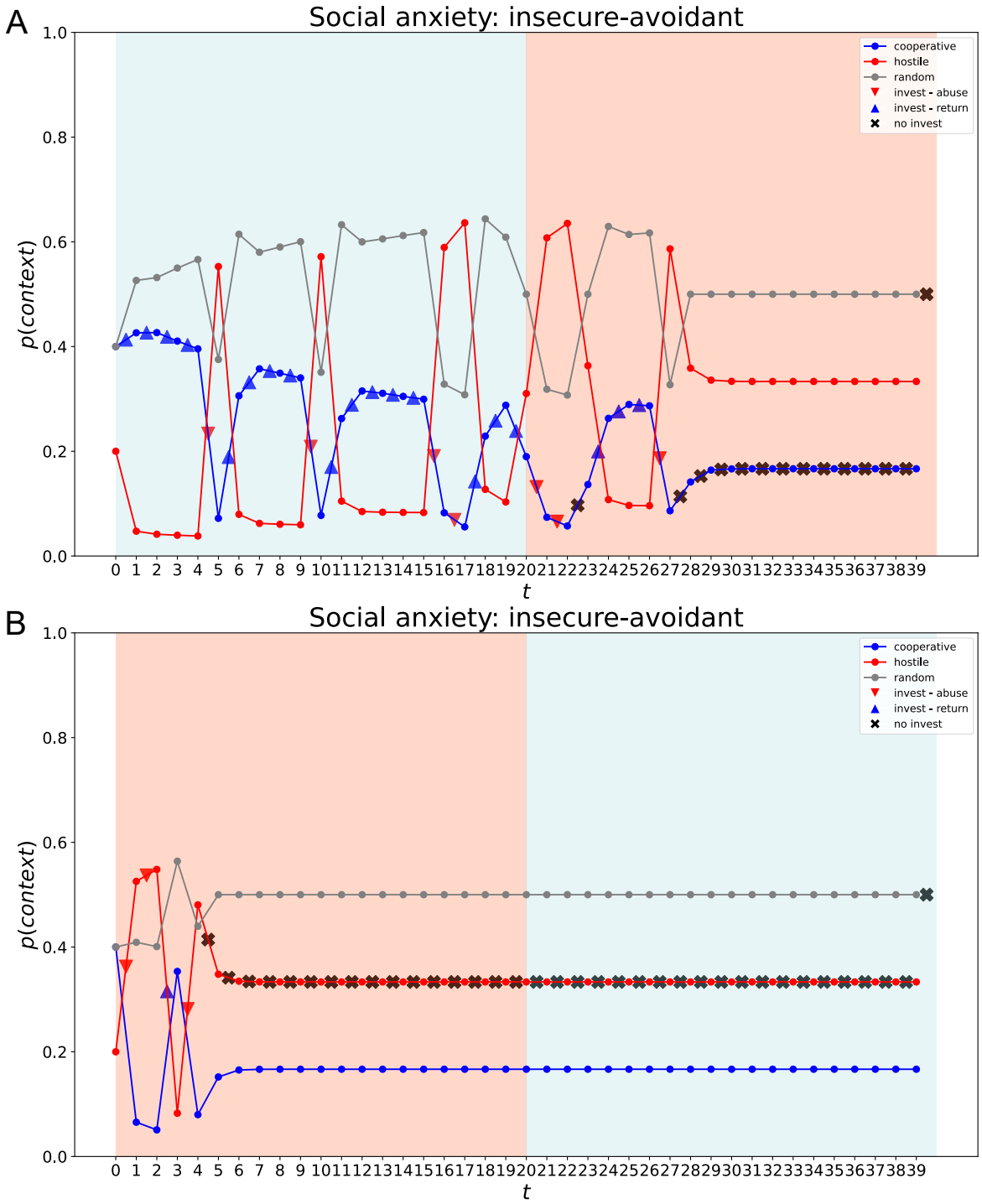


Figure 6. Social anxiety, insecure-avoidant

A shows a socially anxious agent in a cooperative context, followed by a switch hostile context. The agent infers a random context to be the most likely cause of observing returns of their investment. After the context switches, a hostile context is inferred as more likely than a random context and an avoidant behavioral strategy is adopted until the end of the game.

B Similarly, when first immersed in a hostile context, the agent first infers a random- or hostile context to be the most likely causes of observing defects. After observing defects for t=3 out of t=4 timesteps, the agent avoids any investment with the trustee for the rest of the game.

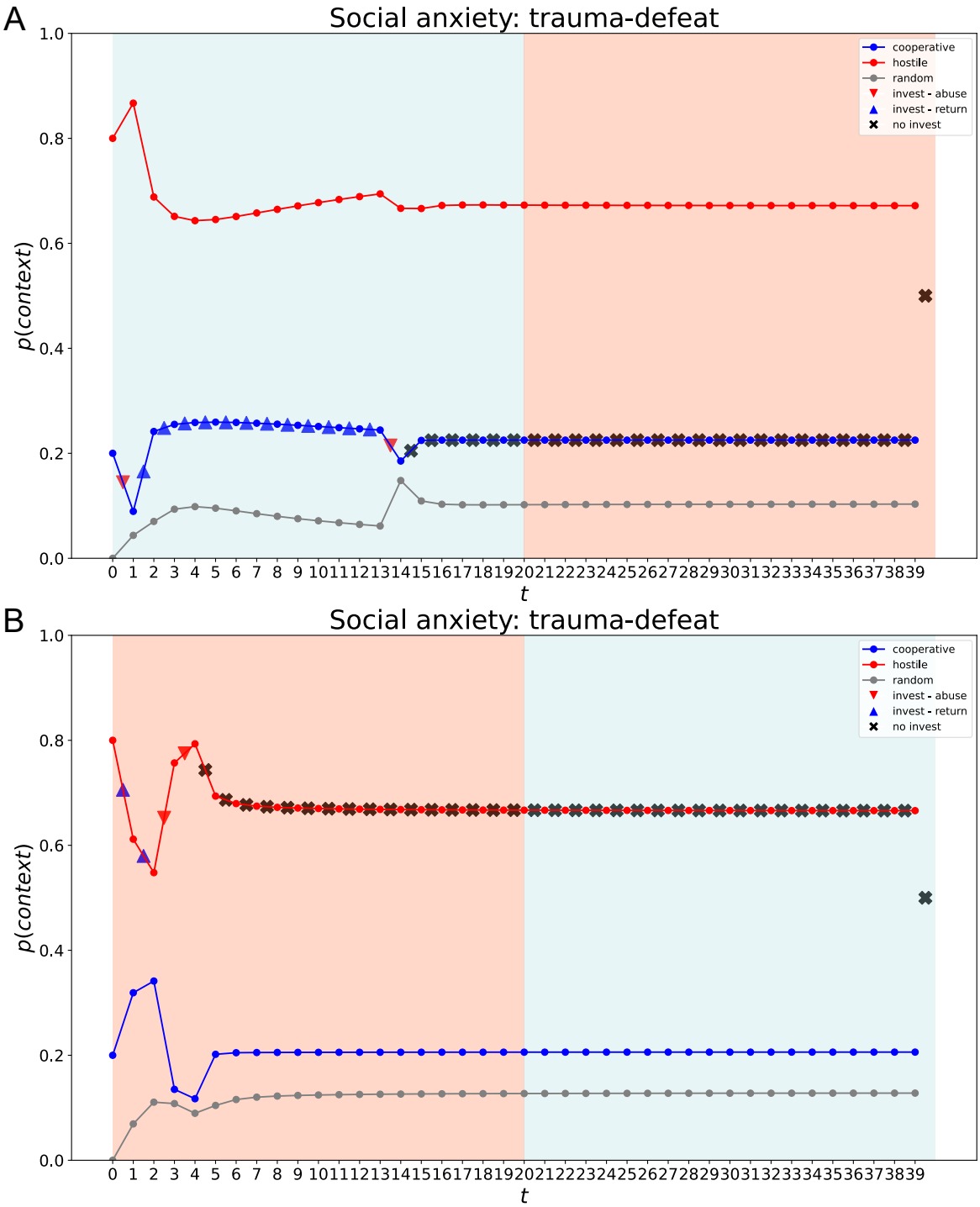


Figure 7. Social anxiety, trauma and defeat.

A the socially anxious (trauma and defeat) agent starts in a cooperative context, followed by a hostile context. The agent holds a relatively strong belief to be in a hostile context, but starts probing the trustee's behavior. Observing a series of t=12 returns of their investment does not change the agent's belief to be in a hostile context. One defect after a long series of returns of investment leads to the agent adopting a rigid avoidant strategy.

B When starting in a hostile context, the agent observes few defects and adopts an avoidant behavioral strategy for the rest of the game.

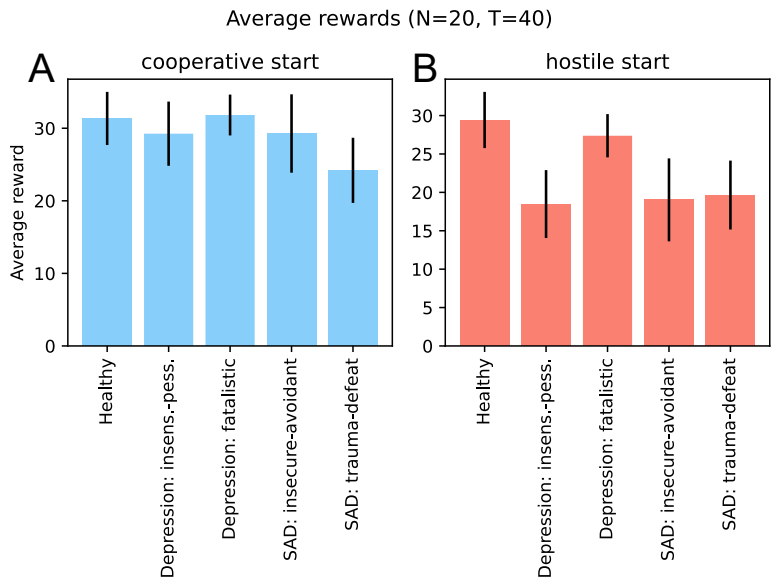


Figure 8. Absolute earned rewards per agent type

A shows average absolute earned rewards (i.e., 50% of the initial investment multiplied by factor 1.5; with standard deviation) per agent type over N=20 simulations of T=40 timesteps, starting with the cooperative context, then switching to the hostile context at t=20. Bars represent standard deviations.

B average earned rewards with standard deviation per agent type over N=20 simulations of T=40 timesteps, starting with the hostile context, then switching to the cooperative context at t=20.

Average reward statistics

We computed the average earned reward, total number of wins and net wins (wins - losses) per agent type in N=20 simulations of T=40 timesteps, where the agents first observed t=20 timesteps of cooperative behavior followed by t=20 timesteps of hostile behavior, or vice versa (**Figure 8**). In a cooperative-to-hostile game, the maximum average reward is earned by the *Type 2 - fatalistic* depressed agent (mean reward: 31.82). Underlying this result is the quick switch to a more avoidant strategy when the context becomes hostile after the first twenty time steps. The least reward is earned by the *Type 2 – trauma and defeat* social phobia agent (mean reward = 24.2) as a result of rigid avoidance behaviors. In a hostile-to-cooperative game, reward is maximized by the healthy agent (mean reward = 29.42). The *Type 1 – insensitive and pessimistic* depressed agent received the lowest reward in this setting (mean reward = 18.47). We further investigate average rewards outcome of the healthy-to-optimistic agent compared to the depressed and socially anxious agents in a longer time series (T=120 with 12 context changes; see supplementary **Figure 33** for details). In a longer series (T=120) with more volatile context dynamics, the healthy agent outearns the depressed- and socially anxious agents (**Figure 35**).

The healthy-to-optimistic agent further observes the highest overall number of *win* outcomes in both the cooperative-to-hostile game (average total wins = 19.5; **Figure 9A**) and in the hostile-to-cooperative

game (average total wins = 16.5; **Figure 9B**). The least absolute number of wins is observed by the *Type 2 – trauma and defeat* social phobia agent in both settings (cooperative-to-hostile: 5.9 average total wins; hostile-to-cooperative: 0.4). Average net wins (i.e., absolute number of wins – absolute number of losses) in a cooperative-to-hostile context are maximized by the *Type 1 – insensitive and pessimistic* depressed agent (average net wins: 8.64) as a function of their lowered threshold to avoid investing with the trustee compared to the healthy agent (average net wins: 1.5). The *Type 1 - depressed* agent however also observes the worst win-to-loss ratio in a hostile-to-cooperative game (average net wins = -5.2). The healthy agent observes the most favorable win-to-loss ratio in the hostile-to-cooperative game (average net wins = -0.65).

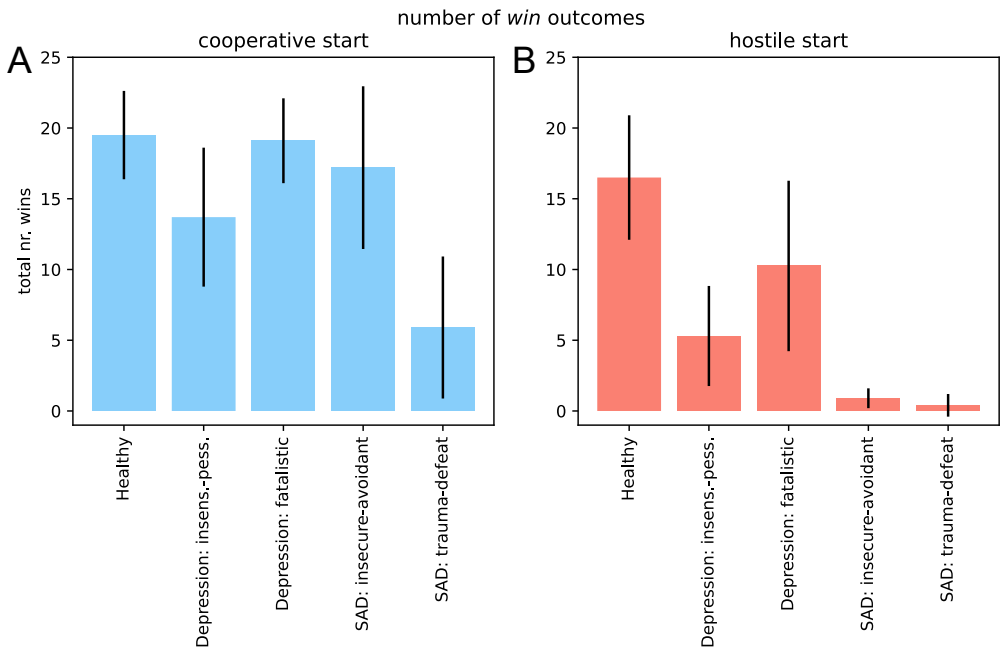


Figure 9. Absolute number of wins per agent type

A. Absolute number of win outcomes per agent type over twenty runs (N=20) in a time series with 40 trials (T=40), starting with a block of t=20 cooperative trials, followed by t=20 hostile trials.

B. Absolute number of win outcomes per agent type over N=20 runs with T=40 trials, starting with a block of t=20 hostile trials, followed by t=20 cooperative trials.

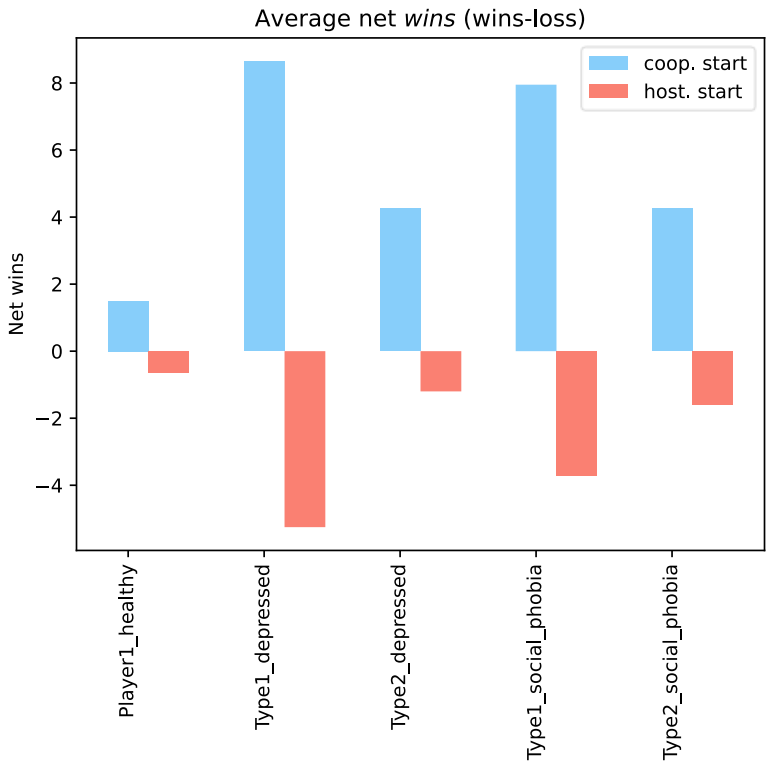


Figure 10. Average net wins per agent type

Average net wins (wins-losses) per agent type in a simulation of T=40 trials over 20 runs, where the blue bars represent a cooperative-to-hostile game, and the red bars represent a hostile-to-cooperative game. Net wins in a cooperative-to-hostile setting are maximized by Type 1 – insensitive and pessimistic depressed agent. Net losses in a hostile-to-cooperative setting are minimized by the healthy-to-optimistic agent.

Model fitting.

Behavioral data. We collected data from N=23 participants with diagnosed depressive disorder, seeking treatment in a local outpatient setting, who played a 60-trial version of the trust game. Participants were randomly assigned to one of three game conditions which differed in the sequence of hidden contexts: a hostile-to-cooperative game (N=7), a cooperative-to-hostile game (N=10; cmp. simulations) and an ambiguous/ random game (N=8). Exemplary data of four participants is shown in **Figure 11**, showing a range of different behavioral strategies in a hostile-to-cooperative game (A-B) and in a cooperative-to-hostile scenario (C-D).

On average, participants earned 39.97€ (standard deviation (SD): 5.84, range: min. 29.5 – max. 46.5) in rewards, of which 50% was paid out to them in €. On average, participants observed 20.58 returns upon investing with the trustee across all conditions (SD: 8.91, range: min. 0 – max. 30) and experienced -0.64 net wins (i.e., number of wins – number of losses; SD: 3.88, range: min. -9, max. 6, see also **Figure 12**). Median BDI score in our sample was $M_{BDI} = 22.0$ (Mean: 24.28, standard deviation: 11.34, range: 7-50; for distribution see supplementary **Figure 36**).

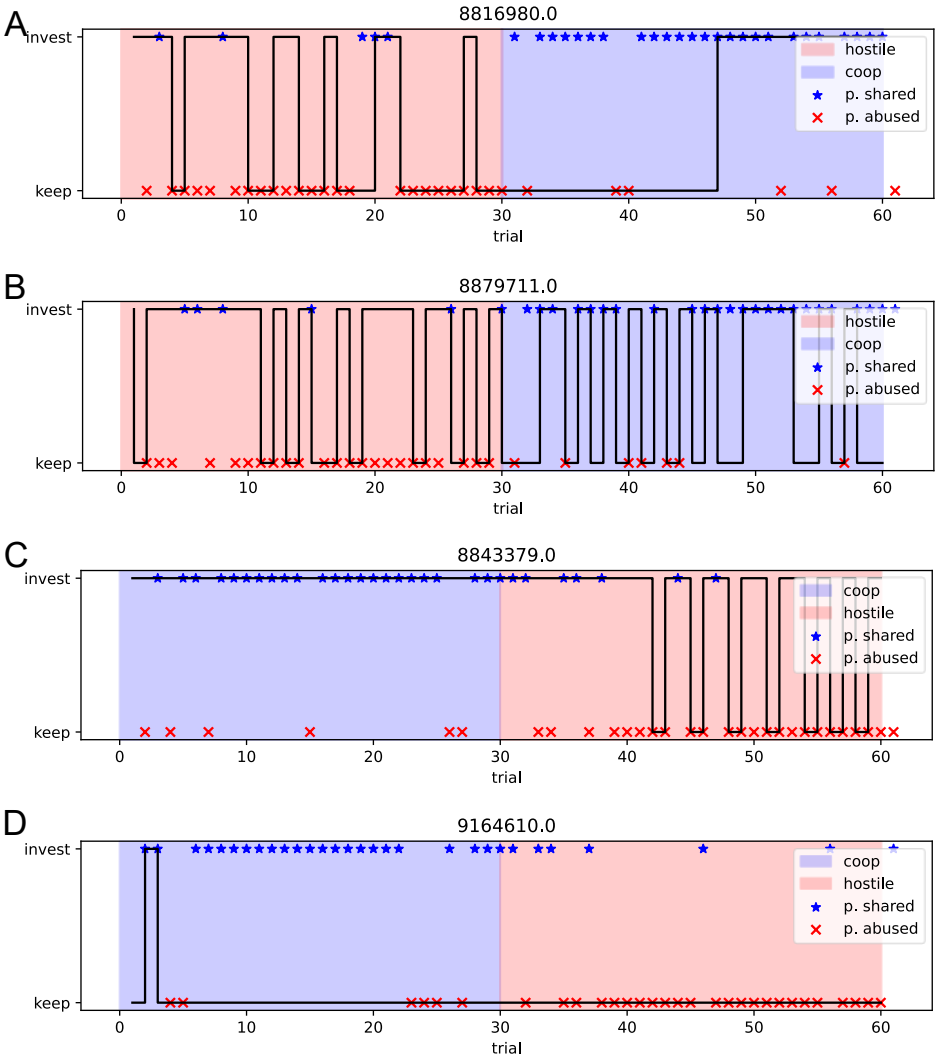


Figure 11. Examples of participants with depressive disorder playing the trust game.

Four example participants playing a 60-trial trust game, either a hostile-to-cooperative game (**A-B**) or a cooperative-to-hostile game (**C-D**). The black step line represents the participant's behavior (0= keep initial budget, 1=invest with trustee). The trustee's response is scattered in the plot (blue star = trustee returned investment, red cross = trustee defect). Note that if the agent does not invest, the trustee's decision to return or defect is not observed by the agent, but plotted here for completeness. The background color indicates the current game context (red= hostile, blue= cooperative).

A. The participant alternates flexibly between investing and keeping their budget in the first phase of the game; however, the participant keeps their budget for a long period during and after the context switches to a cooperative context. **B.** The participant alternates quickly between behavioral responses. **C.** During the cooperative context, the participant invests their budget, whereas after the switch to a hostile context, the participant begins to keep their budget. **D.** After two investments, the participant decided to keep their budget for the rest of the game.

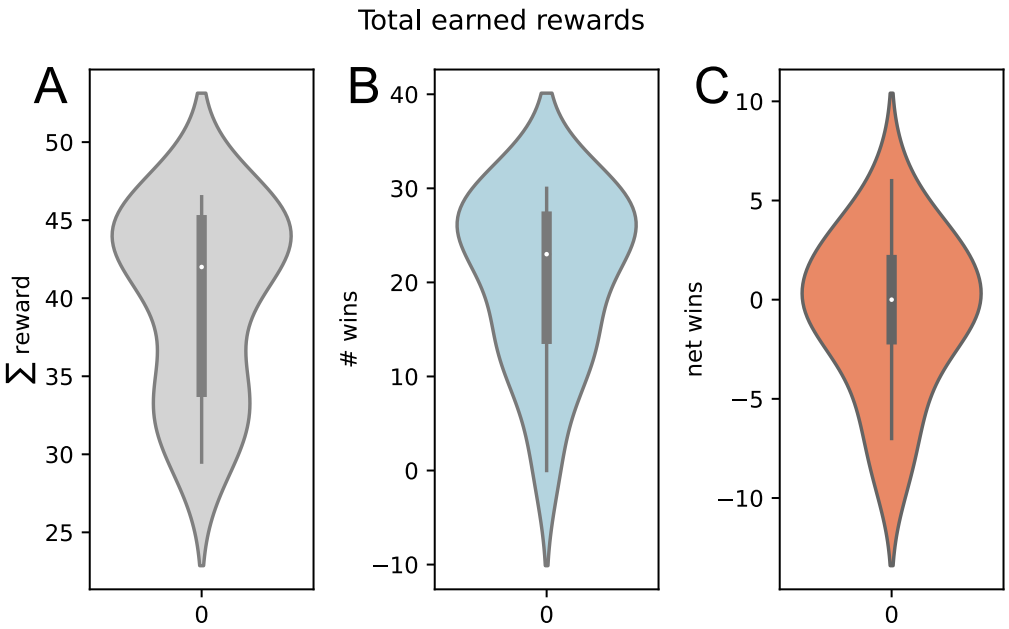


Figure 12. Total rewards earned, number of wins and net wins (wins-losses)

A. Sum of earned rewards per participant (earning 0 for investing and a trustee defect, 0.5 for keeping their budget, and 0.75 for a trustee return on investment).
B. Total number of wins per participant. **C.** Net wins (wins – losses) per participant.

Model fitting. The optimization terminated successfully for all participant datasets. The mean minimized cost was c = 0.283 (SD=0.091, range= min. 0.024 – max. 0.403, see **Figure 13**). The average number of evaluations of the objective functions and its Jacobian and Hessian matrix were 3761.608 (SD=1783.451, range = min. 2054 – max. 8913).

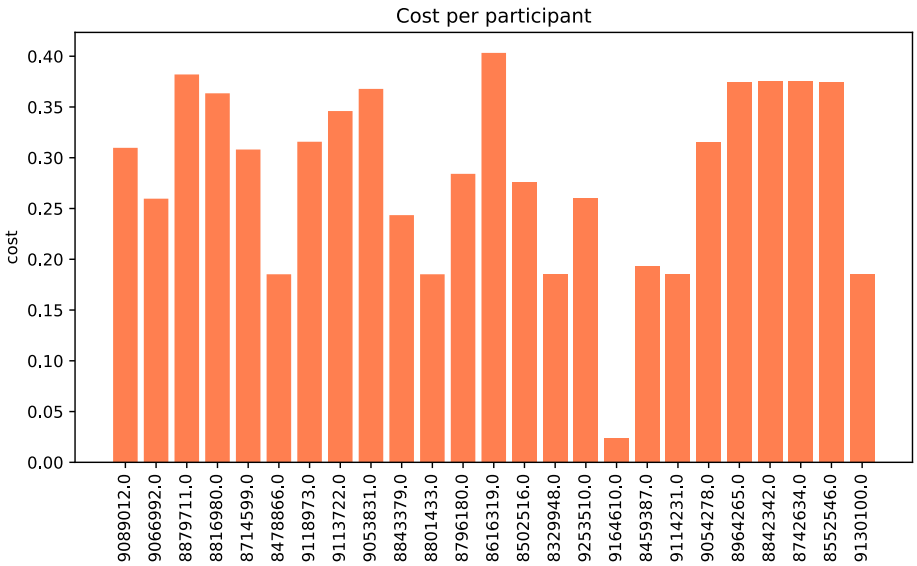


Figure 13. Optimizer cost per participant.

Minimal values for the cost function for each participant across 10 runs of the optimization. The cost function $c = \sum q \cdot \log\left(\frac{q}{a_s}\right) + (1-q) \cdot \log\left(\frac{1-q}{1-a_s}\right)$ was minimized.

For each participant, we obtained 15 parameters that best described their observable behavior in the game. For the observation model A, three parameter values needed to be optimized ($p(\text{win}|\text{cooperative}), p(\text{win}|\text{hostile}), p(\text{win}|\text{random})$). Results are shown in **Figure 14**. Participants' behavior suggests that they correctly inferred a higher probability of observing a return on investment (win) in the cooperative context (average $p(\text{win}|\text{cooperative}) = 0.802$) compared to the hostile ($p(\text{win}|\text{hostile}) = 0.052$) or random contexts ($p(\text{win}|\text{random})$ =0.147). The parameters did not differ significantly between individuals with low- and high symptom severity (**Figure 14**A). Analyzed separately for the three different game conditions (cooperative-to-hostile, hostile-to-cooperative and ambiguous, **Figure 14**B), the highest $p(\text{win}|\text{cooperative})$ can be observed for the cooperative-to-hostile game. $p(\text{win}|\text{random})$ tended to be highest for individuals playing an ambiguous game.

The average optimal transition dynamics B for all participants is shown in **Figure 15**. Participants' behavior was most consistent with the belief that hostile contexts remain stable over time ($p(h|h) = 0.495$, **Figure 15**A), but also cooperative contexts are assumed to be more stable ($p(c|c) = 0.393$). Furthermore, results indicate that on average, participants believed the hostile context would develop

into a random one ($p(r|h) = 0.431$). Considering individuals with low vs. high symptom severity separately (**Figure 15**B), results suggest that individuals with high symptom severity tend to assume the cooperative context to be more stable (High severity: $p(c|c)$=0.403, Low severity: $p(c|c) = 0.384$), and the hostile context to be less stable (High severity: $p(h|h)$=0.481, low severity: $p(h|h) = 0.511$). The behavior of patients with more severe symptom strength was furthermore consistent with a slightly increased belief that a cooperative context will transition into a hostile one (high severity: $p(h|c)$=0.294, low severity: $p(h|c) = 0.268$) compared to participants with less severe symptoms.

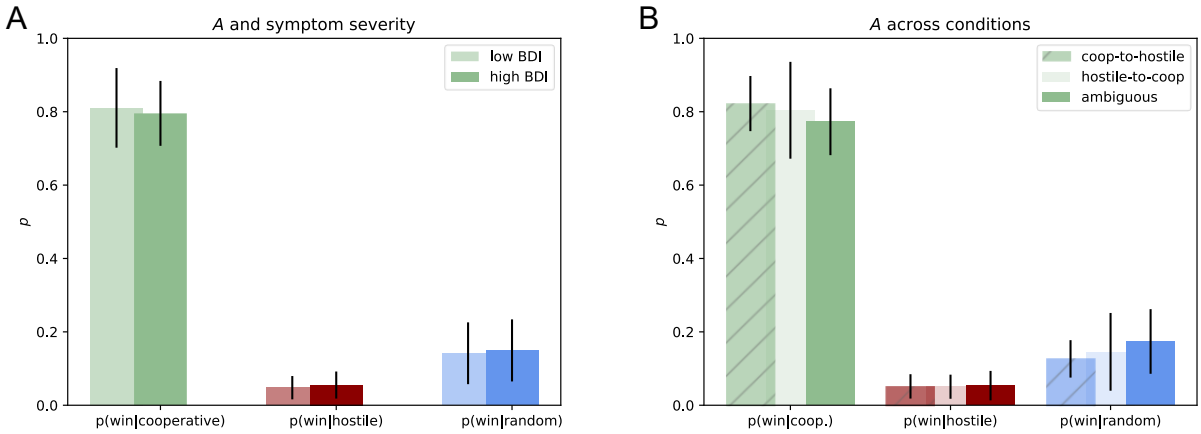


Figure 14. Optimization results for parameter A.

A. Mean parameter values for observation model A, separately for participants with low and high symptom severity (light- and dark bars, respectively). $p(win|cooperative)$ is the probability of observing a return on investment under the cooperative context. $p(win|hostile)$ is the probability of observing a return on investment in the hostile context and finally $p(win|random)$ is the probability of observing a return on investment in the random context.

B. Mean parameter values for observation model A, plotted separately for the three game conditions (cooperative-to-hostile, hostile-to-cooperative and ambiguous/ random).

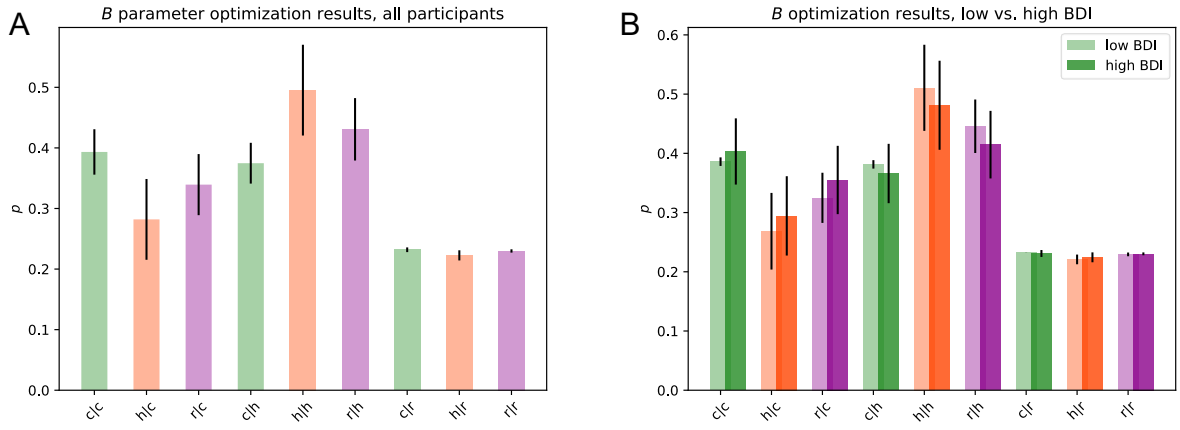


Figure 15. Optimization results for parameter B.

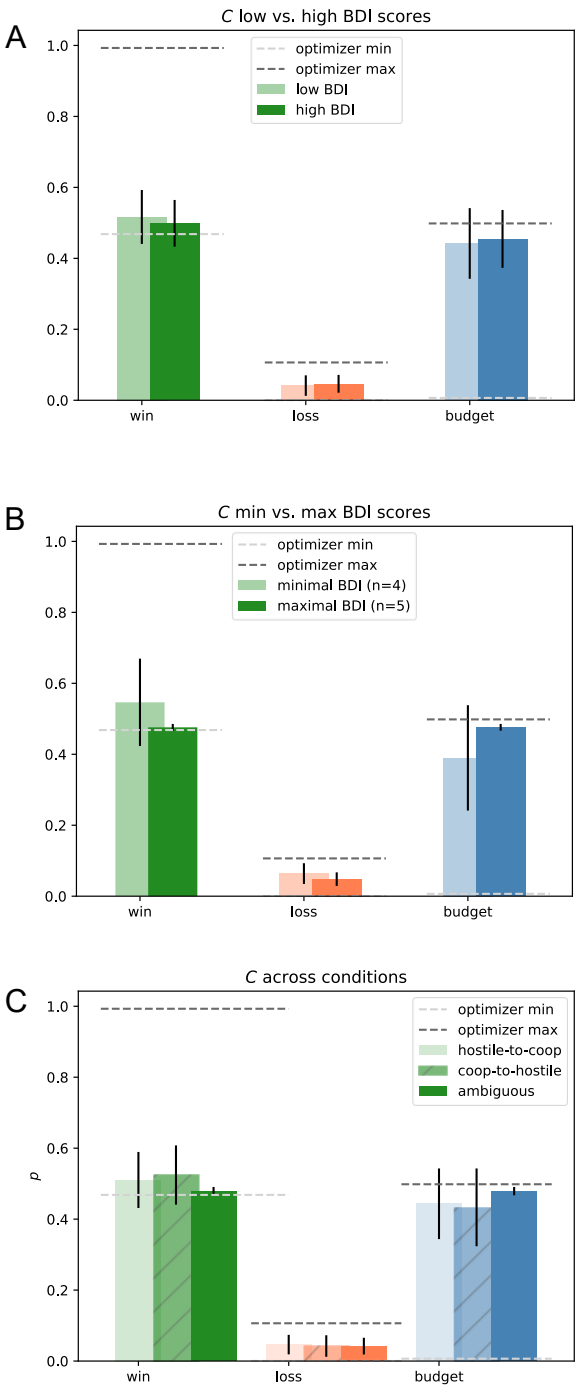
Average parameter values for B, where $c|c$ = cooperative | cooperative transition, $h|c$ = hostile | cooperative transition, $r|c$= random | cooperative transition; $c|h$= cooperative | hostile, $h|h$= hostile |hostile, $r|h$= random | hostile; $c|r$= cooperative | random, $h|r$= hostile | random, $r|r$= random | random transition.

A. Optimization results for B across all participants.

B. Optimization results for B, plotted separately for participants with low (lighter bars) vs. high (darker bars) symptom severity, as measured by BDI.

Optimized prior preferences, or utility, of the different outcomes, governed by C, are shown in **Figure 16**. Differences in preferences for the three outcomes of the game (return on investment, defect and loss, and keep initial budget) between patients with low vs. high symptom severity (as assessed by the BDI) are shown in **Figure 16A**. There is no clear distinction in the mean parameter values between individuals with high- vs. low symptom severity (Low symptom severity: $p(\text{win}) = 0.810, p(\text{loss}) = 0.048, p(\text{budget}) = 0.142$; high symptom severity: $p(\text{win}) = 0.795, p(\text{loss}) = 0.055, p(\text{budget}) = 0.149$). This pattern is strengthened when considering individuals with minimum and maximum BDI scores separately (**Figure 16B**). The effect of the game sequence on C are shown in **Figure 16C**. Individuals in the ambiguous game assign lowest utility to observing a return on investment, and highest to keeping their budget. The cooperative-to-hostile game participants assigned the largest utility to the return on investment (Ambiguous: $p(\text{win}) = 0.479, p(\text{budget}) = 0.479$, hostile-to-cooperative: $p(\text{win}) = 0.510, p(\text{loss}) = 0.046$, cooperative-to-hostile: $p(\text{win}) = 0.524, p(\text{loss}) = 0.042$). For violin distribution plots of C across participants, see supplementary **Figure 37**.

Figure 16. Optimization results for parameter C



A. Preference/ outcome utility for the three possible outcomes (win (green) = return on investment, loss (coral) = trustee defects and agent loses budget, budget (blue) = agent keeps their initial budget), with a median split of participants based on the severity of their depressive symptoms as assessed by the BDI. Individuals with higher symptom severity (darker bars) show slightly reduced preference for the "win" outcomes, and increased preference for keeping their budget, compared to individuals with milder symptoms (lighter bars). Dashed horizontal lines indicate the constraints on the optimization search space.

B. Outcome preference, considered separately for participants with minimal and maximal BDI scores (Score that lies one standard deviation above- or below the sample median).

C. Preference for the three possible outcomes based on the game condition (hostile-to-cooperative game: light-colored bars, cooperative-to-hostile game: striped bars, ambiguous game: dark-colored bars; outcomes are color-coded as above).

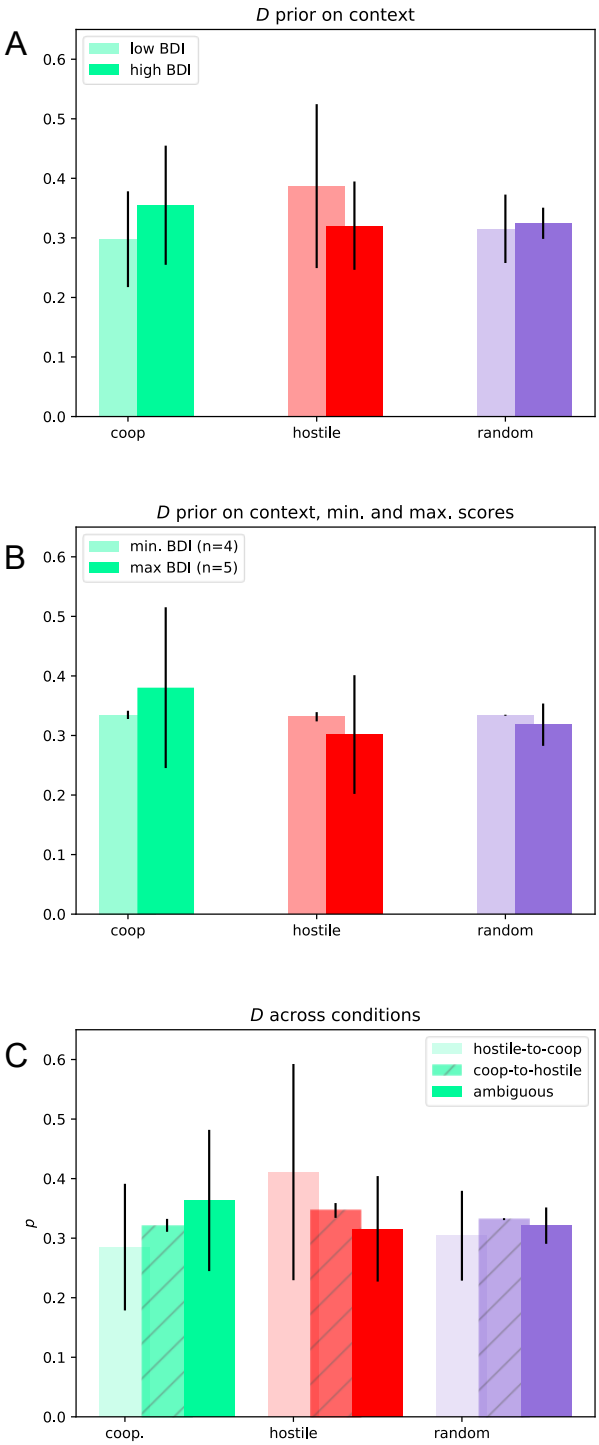
The average optimal prior over context (D) is shown in **Figure 17** (for distribution plots, see supplementary **Figure 38**). Individuals with lower symptom severity show a slightly decreased prior belief to be in a cooperative context, and an increased prior on the hostile context compared to patients with more severe depressive symptoms (low symptom severity: $p(cooperative) = 0.298, p(hostile) = 0.387$; higher symptom severity: $p(\text{cooperative}) = 0.355, p(hostile) = 0.320$, **Figure 17**A). The pattern is similar when considering individuals with minimal and maximum BDI scores separately (**Figure 17B**). Furthermore, in a hostile-to-cooperative game, results suggest an increased prior on the hostile context and a decreased prior belief in a cooperative context compared to the cooperative-to-hostile or ambiguous game (hostile-to-cooperative game: $p(\text{coop.}) = 0.285, p(\text{hostile}) = 0.412, p(\text{random}) = 0.304$; cooperative-to-hostile game: $p(\text{coop.}) = 0.321, p(\text{hostile}) = 0.346, p(\text{random}) = 0.332$; ambiguous game: $p(\text{coop.}) = 0.362, p(\text{hostile}) = 0.316, p(\text{random}) = 0.321$, **Figure 17**C).

Interindividual differences in ε indicate that most participants were strongly reward-driven, with an ε approximating zero (n=20 participants, **Figure 18**). Two participants' behavior was best described by stronger weighting of information over hedonic reward (n=2 participants), while n=3 participants showed a more nuanced weighting between hedonistic and epistemic drives. There were no differences in the distribution of ε between individuals with mild vs. moderate to severe symptoms.

Finally, we inserted the optimized parameters of two patients, representing the low- and high levels of depressive symptomatology, into the generative model and simulated twenty games with these parameters. Results are shown in **Figure 19**. The simulations with parameters derived from the patient with milder depression (patient *md,* subsequently) is shown in **Figure 19A** (BDI score = 11). Compared to the more severely depressed (patient *sd*) patient (BDI score =50, **Figure 19B**), patient *md* forms stronger posterior beliefs to be in a cooperative context in the initial cooperative phase. In contrast, patient *sd* forms stronger beliefs to be in a hostile context in the second phase of the game.

2

Figure 17. Optimization results for parameter D



A. Optimization results for the prior over contexts, D, plotted separately for individuals with low (lighter bars) and high (darker bars) symptom severity.
B. Optimization results for the prior over contexts, plotted separately for individuals with minimal (light bars) and maximum (darker bars) BDI scores.
C. Prior over context D, plotted separately for the game conditions hostile-to-cooperative (lighter bars), cooperative-to-hostile (dashed bars) and ambiguous (darker bars).

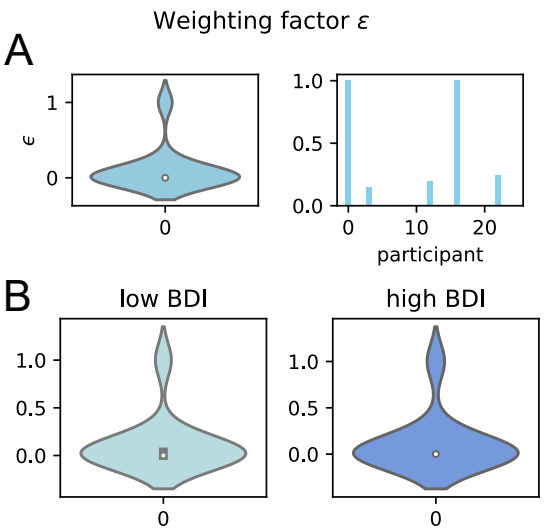


Figure 18. Optimization results for weighting parameter ε.

A. Individual optimized parameter values for ε, the parameter governing the weighting of hedonic, reward-oriented and epistemic, information-oriented drives.

B. Distribution of optimized weighting parameter ε, for low- (light blue) and high (dark blue) symptom severity.

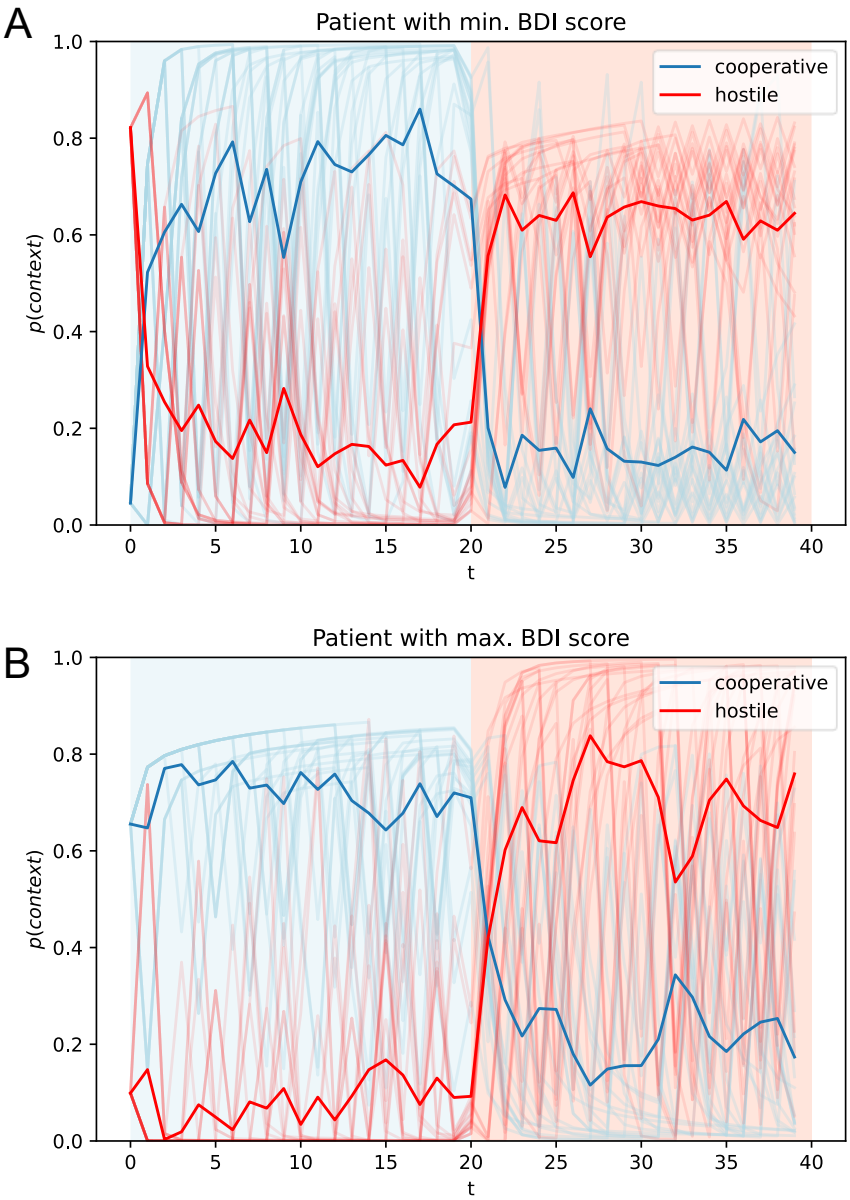


Figure 19. Simulated time series with optimized parameters of patients

A. Simulated time series of a patient with a low BDI score. Plotted are twenty cooperative-to-hostile games (the thin lines represent individual games) and the average inferred posterior belief over contexts (thick line).

B. Simulated time series of a patient with the highest BDI score (twenty cooperative-to-hostile games).

Discussion

In this study, we report a series of simulations in which an artificial agent engages in a classical *trust game* and fit this model to behavioral data from outpatients with depression playing the game. Using Active Inference, the agents infer the interpersonal context from the trustee's behavior. We developed agents in which several biases are combined to simulate depression and social anxiety and finally inverted the model onto behavioral data of patients to retrieve individually optimal parameters of the generative model.

In simulations, we demonstrated how the combination of different parameter perturbations may approximate behavioral strategies of individuals with social anxiety disorder and depression. Depression here resulted from the combination of outcome devaluation and pessimistic prior beliefs (*reward-insensitive and pessimistic*), or rigid beliefs about context dynamics in combination with pessimistic priors (*fatalistic*). Social anxiety disorder resulted from an increased risk aversion in combination with a high prior belief to be in a random, unpredictable interpersonal context (*insecure-avoidant*) or an increased loss avoidance in combination with pessimistic prior- and state transition beliefs (*trauma-and-defeat*).

Pessimistic priors and fatalistic beliefs about the temporal dynamics of hidden states were at the core of the depressed agents' generative models. In line with this, individuals with depression show a variety of pessimistic interpretation biases (Everaert et al., 2017), pessimistic tendencies such as learned helplessness and low self-efficacy (Seligman, 1972) and potentially a lack of interpersonal skills (Segrin, 2000). A lack of reward valuation and pessimistic prior beliefs may also relate to a reduced motivation to engage with their environment in patients (Fernández-Theoduloz et al., 2019). In the trust game, the depressed agents earned significantly less compared to healthy agents during a hostile-to-cooperative series. However, the depressed agents outcompete the healthy agent in a *cooperative-to-hostile* series. Here, the depressed agents switch to an avoidant behavioral strategy soon after the context switch. The healthy agent's optimistic beliefs about their environment and their strong motivation to win a joint reward keeps them from adopting an avoidant strategy, even after observing a series of defects. In the trust game, avoiding the trustee's decision is an intelligent strategy in a hostile context. However, this strategy comes at the expense of potentially missing a switch to a cooperative context, explaining why the depressed agents earn significantly less compared to the healthy agent in the *hostile-to-cooperative* series. Our simulation supports the idea that in some environments, depression may be an adaptive response that protects the individual from exploitation (Prins et al., 2015). However, the optimistic (healthy) agent is best equipped to flexibly navigate dynamically changing contexts over longer time periods, as indicated by a simulation with more volatile context dynamics (appendix **2**). Meanwhile, individuals with depression or anxiety may show less flexible behavior as a consequence of e.g. learned helplessness (Seligman, 1972), aberrant reward processing (Lewinsohn, 1974) or cognitive immunization (Rief & Joormann, 2019). Further, some research suggests that individuals with

depression show a more realistic interpretation of action-outcome contingencies (*depressive realism*, Dunning & Story, 1991; Moore & Fresco, 2012). In line with this, the depressed agents adapt their posterior beliefs over contexts more quickly and accurately and their behavioral strategy switched quicker in hostile contexts, compared to the more optimistic agent in our simulation.

Depressive symptoms often manifest episodically, with patients experiencing alternating phases of severe and relatively mild symptoms. In Active Inference, episodic courses may be a function of the transition model (or B matrix). If individuals believe for example that both hostile and cooperative contexts are relatively stable, then a few negative interpersonal triggers may be sufficient for the agent to infer a stable, hostile state, which can be updated through repeated exposure to positive interpersonal exchanges.

The "socially anxious" agents focused on risk aversion and showed increased avoidance behaviors. Avoidance is a defining feature of social anxiety disorder (Bögels & Mansell, 2004), which could be mediated through the combination of loss aversion and pessimistic prior beliefs in Active Inference. The distinction between the two subtypes of social anxiety disorder lies in their prior beliefs about initial states, the temporal dynamics of these states, and the degree to which they are motivated to achieve a positive outcome (i.e., observing a successful collaboration vs. keeping the initial budget and not taking the risk). The avoidance behavior of the *insecure-avoidant* socially anxious agent is driven by the insecure belief that the environment is random and unpredictable. In contrast, the *trauma-and-defeat* agent avoids collaborating with the trustee because they hold strong beliefs to be in a hostile environment. The two agents are similar with respect to their behavioral strategies and earned rewards, but the differences in model perturbations may be of crucial importance during treatment. It can be conjectured that cognitive restructuring and social competences training may be the most beneficial to insecure-avoidant agents, while, for example, interpersonal trauma may need addressing in trauma-and-defeat-type patients first.

In the Hierarchical Taxonomy of Psychopathology (HiToP, Kotov et al., 2018), social anxiety disorder and depression are part of the internalizing spectrum, but while social anxiety disorder loads on the "fear" subfactor, depression loads on the "distress" subfactor. In terms of computational transdiagnostic processes, this difference may lie in the preference over outcomes, or the C vector. In one of our models for depression (*reward-insensitive and pessimistic*), social anhedonia is modelled via a reduced preference for positive social rewards. Social anxiety disorder, in contrast, is a function of high levels of risk avoidance, where a disproportionately large weight is placed on risk and loss avoidance.

Fitting our model to the behavioral data of patients with diagnosed depressive disorder of varying symptom severity yielded optimal values for all parameters of the generative model A, B, C, D and weighting parameter ε. We considered the resulting generative model parameters separately for individuals with high- and low symptom severity and the game sequence they played. Both symptom severity and game sequence affected the parameters of the generative model in a subtle way. The

observation model A did not differ between individuals with low vs. high symptom severity, indicating that both groups did not differ in their ability to relate a specific outcome to the corresponding context. The transition dynamics B differed very subtly. More severe symptoms were associated with an increased belief in the stability of the cooperative context, and less stability of the hostile context. Furthermore, higher symptom severity was associated with a slightly increased belief of transitioning into a hostile context when the current context was cooperative, and a decreased belief in a transition from a hostile into a cooperative context. Outcome preferences C did not differ between median-split patient groups. When considering individuals with minimum- and maximum symptom severity, a slight tendency emerged, suggesting higher symptom severity is associated with lower utility assigned to the social reward (return on investment), slightly more risk aversion, and higher preference for keeping their budget compared to individuals with milder symptoms. Counter-intuitively, more severe depressive symptoms were associated with a heightened prior belief (D) to be in a cooperative context, and a lower belief to be in a hostile context. Finally, weighting parameter ε was introduced to govern the weighting of epistemic vs. hedonic drives in the generative model. While ε differed inter-individually, this variance was not related to symptom severity.

When equipping the agent-model with the optimized parameters derived from patient data, we could simulate the patient's behavior over several games. We identified idiosyncrasies in the patient's respective decision-making, with mildly depressed patients forming more precise posterior beliefs over the cooperative context, whereas more severely depressed individuals formed higher posterior beliefs over the hostile context. Our approach demonstrates that data-inspired agent models can be useful in simulating patient behavior efficiently for a large number of games and different contexts.

Limitations and future research

The model presented here provides starting points for explaining heterogeneity in social experience and behavior among individuals with mental health problems. For the simulations, the parameter specifications were chosen so that they align with the literature, but are ultimately based on theoretical considerations. By fitting the model to behavioral data of patients with depression, we show that parameter distributions can slightly differ between individuals with low- vs. high symptom severity. Of note, these results need to be interpreted with caution. An important goal of the optimization procedure was to demonstrate its feasibility in the context of relatively sparse behavioral data and complex generative models such as POMDPs. We cannot yet determine if these changes in model parameters are unique to depression or similar to those in, e.g., social anxiety, and whether they can account for substantial variance when comparing clinical with non-clinical samples. More data from diverse diagnostic categories (ICD-10 social anxiety, depression) or symptom spectra (HiTOP internalizing spectrum, subfactors fear vs. distress) is required to draw conclusions about the specificity of parameter changes. A limitation is the relatively small sample size, and the lack of a non-clinical control group in the dataset used for model fitting. Future work should aim at collecting data from both diagnosed- and

undiagnosed individuals to represent the entire spectrum from mental well-being- to disorder. On the other hand, heterogeneity in model parameters emerged despite the lack of a non-clinical control group, and the boundaries between non-clinical sadness in the general population and depressive disorders are fluent (Tebeka et al., 2021).

Of note, finding clear and distinct computational phenotypes within a diagnostic category has proven to be extremely challenging (Karvelis et al., 2023; Robson et al., 2020). The test-retest reliability of the generative model parameters identified in this task is rather unclear. We nonetheless argue that there is value in understanding how changes to the parameters of a generative model influence behavior to inform etiological concepts, instead of basing computational phenotypes on nosological categories. In case transdiagnostic affiliations can be derived from computational models of behavior, there is still very limited research into the reliability and validity of computational markers. A recent meta-analysis suggests that cognitive tasks and computational markers derived from task behavior shows relatively poor psychometric properties (Karvelis et al., 2023). Longitudinal task batteries may be necessary to equip researchers with the data to disentangle measurement errors, practice effects and state- and trait-like changes in behavior. Further empirical validations of our model may need to consider multiple experiments of interpersonal behavior in tasks with varying degrees of similarity.

Conclusion

We simulated complex interpersonal experiences and behaviors in mental disorders as Active Inference of interpersonal context within a neuroeconomic game. We fit our agent model on behavioral data of participants with diagnosed depressive disorders of varying severity. Our model allows for the simulation of complex interpersonal experiences and behaviors, and enables the operationalization of transdiagnostic and disorder-relevant processes, as well as heterogeneity in social functioning across individuals. Fitting the model to behavioral data can provide more mechanistic insights into biased expectations and assumptions in clinical populations, with implications for personalized diagnosis and treatment.

Our simulations yield three fundamental insights: (1) different generative models of the environment can result in similar interpersonal strategies, (2) avoidant interpersonal behaviors are adaptive and functionally sensible in exploitative environments and (3) exploration, which appears to be a crucial characteristic of healthy interpersonal behavior, can lead to losses in some environments; however, on average, it produces the most robust outcome rates in dynamically changing environments. Developing computational phenotypes based on observable behaviors is an important cornerstone of personalized diagnostics and tailored interventions. By fitting our model to the behavioral data of patients, we can obtain approximations to the computations underlying patient's behavioral strategies. Neuroeconomic games offer a relatively naturalistic, yet well-controlled experimental framework. Future research needs to further provide empirical validation of these ideas, ideally within longitudinal, multi-task studies.

Declarations

Ethics approval and consent to participate. Not applicable.

Consent for publication. Not applicable.

Availability of data and materials. The datasets generated and/or analyzed during the current study are available in the research data repository of the Philipps-University of Marburg, https://gitlab.uni-marburg.de/fb04/ag-endres/pymdp_depression.

Competing interests. This work was conducted as part of the cluster initiative "The Adaptive Mind" (TAM), funded by the Hessian Ministry for Science and Arts (Hessisches Ministerium für Wissenschaft und Kunst) which brings together scientists from experimental psychology, clinical psychology, and artificial intelligence to improve the understanding of how the human mind successfully adapts to changing conditions. This includes the role of social expectations in the progression of mental disorders.

Funding. This work was funded by the Hessian Ministry of Higher Education, Research, Science, and the Arts as part of the cluster initiative "The Adaptive Mind".

Authors Contributions. Conceptualization: ALE,LK,WR; Methodology: ALE,JP,DE; Data collection – LK, RZ, AH; Writing - Original Draft: ALE,LK Writing - Review & Editing: all authors.

Acknowledgments. We like to thank all the collaborators of the cluster initiative "The Adaptive Mind" (TAM) for the productive discussions that made this project possible.

Prior versions. A version of this manuscript is available on the PsyArXiv preprint server: https://osf.io/preprints/psyarxiv/8aexf.

All code and preprocessed data is publicly available at https://github.com/Eckertal/pymdp_depression.

The development repository under https://gitlab.uni-marburg.de/fb04/ag-endres/pymdp_depression is accessible with a university guest account.

References

Alden, L. E., & Taylor, C. T. (2004). Interpersonal processes in social phobia. *Clinical Psychology Review*, *24*(7), 857–882. https://doi.org/10.1016/j.cpr.2004.07.006

Alloy, L. B., Olino, T., Freed, R. D., & Nusslock, R. (2016). Role of reward sensitivity and processing in major depressive and bipolar spectrum disorders. *Behavior Therapy*, *47*(5), 600–621.

Arcelus, J., Haslam, M., Farrow, C., & Meyer, C. (2013). The role of interpersonal functioning in the maintenance of eating psychopathology: A systematic review and testable model. *Clinical Psychology Review*, *33*(1), 156–167. https://doi.org/10.1016/j.cpr.2012.10.009

Backs-Dermott, B. J., Dobson, K. S., & Jones, S. L. (2010). An evaluation of an integrated model of relapse in depression. *Journal of Affective Disorders*, *124*(1–2), 60–67. https://doi.org/10.1016/j.jad.2009.11.015

Baker, S. R., & Edelmann, R. J. (2002). Is social phobia related to lack of social skills? Duration of skill-related behaviours and ratings of behavioural adequacy. *British Journal of Clinical Psychology*, *41*(3), 243–257. https://doi.org/10.1348/014466502760379118

Barca, L., & Pezzulo, G. (2020). Keep your interoceptive streams under control: An active inference perspective on anorexia nervosa. *Cognitive, Affective, & Behavioral Neuroscience*, *20*, 427–440.

Barkus, E., & Badcock, J. C. (2019). A transdiagnostic perspective on social anhedonia. *Frontiers in Psychiatry*, *10*, 216.

Barrett, L. F., Quigley, K. S., & Hamilton, P. (2016). An active inference theory of allostasis and interoception in depression. *Philosophical Transactions of the Royal Society B: Biological Sciences*, *371*(1708), 20160011.

Beck, A. T., Steer, R. A., & Carbin, M. G. (1988). Psychometric properties of the Beck Depression Inventory: Twenty-five years of evaluation. *Clinical Psychology Review*, *8*(1), 77–100. https://doi.org/10.1016/0272-7358(88)90050-5

Beeson, C. M. L., Brittain, H., & Vaillancourt, T. (2020). The temporal precedence of peer rejection, rejection sensitivity, depression, and aggression across adolescence. *Child Psychiatry & Human Development*, *51*(5), 781–791. https://doi.org/10.1007/s10578-020-01008-2

Bell, C. M., Ridley, J. A., Overholser, J. C., Young, K., Athey, A., Lehmann, J., & Phillips, K. (2018). The role of perceived burden and social support in suicide and depression. *Suicide and Life-Threatening Behavior*, *48*(1), 87–94. https://doi.org/10.1111/sltb.12327

Benrimoh, D., Parr, T., Vincent, P., Adams, R. A., & Friston, K. (2018). Active Inference and Auditory Hallucinations. *Computational Psychiatry (Cambridge, Mass.)*, *2*, 183–204. https://doi.org/10.1162/cpsy_a_00022

Berg, J., Dickhaut, J., & McCabe, K. (1995). Trust, reciprocity, and social history. *Games and Economic Behavior*, *10*(1), 122–142.

Berg, M., Feldmann, M., Kirchner, L., & Kube, T. (2022). Oversampled and undersolved: Depressive rumination from an active inference perspective. *Neuroscience & Biobehavioral Reviews*, 104873.

Bögels, S. M., & Mansell, W. (2004). Attention processes in the maintenance and treatment of social phobia: Hypervigilance, avoidance and self-focused attention. *Clinical Psychology Review*, *24*(7), 827–856.

Bracha, A., & Brown, D. J. (2012). Affective decision making: A theory of optimism bias. *Games and Economic Behavior*, *75*(1), 67–80.

Brown, H., Adams, R. A., Parees, I., Edwards, M., & Friston, K. (2013). Active inference, sensory attenuation and illusions. *Cognitive Processing*, *14*(4), 411–427. https://doi.org/10.1007/s10339-013-0571-3

Charpentier, C. J., Aylward, J., Roiser, J. P., & Robinson, O. J. (2017). Enhanced risk aversion, but not loss aversion, in unmedicated pathological anxiety. *Biological Psychiatry*, *81*(12), 1014–1022.

Conway, C. C., Forbes, M. K., & South, S. C. (2022). A Hierarchical Taxonomy of Psychopathology (HiTOP) primer for mental health researchers. *Clinical Psychological Science*, *10*(2), 236–258. https://doi.org/10.1177/21677026211017834

Cullen, M., Davey, B., Friston, K. J., & Moran, R. J. (2018). Active inference in OpenAI Gym: A paradigm for computational investigations into psychiatric illness. *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging*, *3*(9), 809–818.

Cuthbert, B. N., & Insel, T. R. (2013). Toward the future of psychiatric diagnosis: The seven pillars of RDoC. *BMC Medicine*, *11*(1), 126. https://doi.org/10.1186/1741-7015-11-126

Da Costa, L., Parr, T., Sajid, N., Veselic, S., Neacsu, V., & Friston, K. (2020). Active inference on discrete state-spaces: A synthesis. *Journal of Mathematical Psychology*, *99*, 102447.

Dalgleish, T., Black, M., Johnston, D., & Bevan, A. (2020). Transdiagnostic approaches to mental health problems: Current status and future directions. *Journal of Consulting and Clinical Psychology*, *88*(3), 179–195. https://doi.org/10.1037/ccp0000482

Dieguez, S., Wagner-Egger, P., & Gauvrit, N. (2015). Nothing happens by accident, or does it? A low prior for randomness does not explain belief in conspiracy theories. *Psychological Science*, *26*(11), 1762–1770.

Dunning, D., & Story, A. L. (1991). Depression, realism, and the overconfidence effect: Are the sadder wiser when predicting future actions and events? *Journal of Personality and Social Psychology*, *61*(4), 521.

Everaert, J., Podina, I. R., & Koster, E. H. (2017). A comprehensive meta-analysis of interpretation biases in depression. *Clinical Psychology Review*, *58*, 33–48.

Ewen, A.-C. I., Rief, W., & Wilhelm, M. (2022). Exploring the path of persisting dysfunctional expectations—Development of the immunization scale IMS. *Frontiers in Psychology*, *13*, 1033078.

Fernández-Theoduloz, G., Paz, V., Nicolaisen-Sobesky, E., Pérez, A., Buunk, A. P., Cabana, Á., & Gradin, V. B. (2019). Social avoidance in depression: A study using a social decision-making task. *Journal of Abnormal Psychology*, *128*(3), 234–244. https://doi.org/10.1037/abn0000415

Fletcher, P. C., & Frith, C. D. (2009). Perceiving is believing: A Bayesian approach to explaining the positive symptoms of schizophrenia. *Nature Reviews Neuroscience*, *10*(1), 48–58.

Friston, K., FitzGerald, T., Rigoli, F., Schwartenbeck, P., & Pezzulo, G. (2016). Active inference and learning. *Neuroscience & Biobehavioral Reviews*, *68*, 862–879.

Galinsky, A. D., Magee, J. C., Gruenfeld, D. H., Whitson, J. A., & Liljenquist, K. A. (2008). Power reduces the press of the situation: Implications for creativity, conformity, and dissonance. *Journal of Personality and Social Psychology*, *95*(6), 1450.

Gamble, B., Moreau, D., Tippett, L. J., & Addis, D. R. (2019). Specificity of future thinking in depression: A meta-analysis. *Perspectives on Psychological Science*, *14*(5), 816–834.

Gao, S., Assink, M., Cipriani, A., & Lin, K. (2017). Associations between rejection sensitivity and mental health outcomes: A meta-analytic review. *Clinical Psychology Review*, *57*, 59–74. https://doi.org/10.1016/j.cpr.2017.08.007

Gerrans, P., & Murray, R. J. (2020). Interoceptive active inference and self-representation in social anxiety disorder (SAD): Exploring the neurocognitive traits of the SAD self. *Neuroscience of Consciousness*, niaa026.

Glimcher, P. W., & Rustichini, A. (2004). Neuroeconomics: The consilience of brain and decision. *Science*, *306*(5695), 447–452.

Gradin, V. B., Pérez, A., MacFarlane, J. A., Cavin, I., Waiter, G., Engelmann, J., Dritschel, B., Pomi, A., Matthews, K., & Steele, J. D. (2015). Abnormal brain responses to social fairness in depression: An fMRI study using the Ultimatum Game. *Psychological Medicine*, *45*(6), 1241–1251.

Haarsma, J., Kok, P., & Browning, M. (2022). The promise of layer-specific neuroimaging for testing predictive coding theories of psychosis. *Schizophrenia Research*, *245*, 68–76.

Haley, W. E., & Strickland, B. R. (1986). Interpersonal betrayal and cooperation: Effects on self-evaluation in depression. *Journal of Personality and Social Psychology*, *50*(2), 386.

Hames, J. L., Hagan, C. R., & Joiner, T. E. (2013). Interpersonal processes in depression. *Annual Review of Clinical Psychology*, *9*, 355–377.

Harris, R. A. (2014). Chronic pain, social withdrawal, and depression. *Journal of Pain Research*, 555–556.

Hartley, C. A., & Phelps, E. A. (2012). Anxiety and decision-making. *Biological Psychiatry*, *72*(2), 113–118.

Heins, C., Millidge, B., Demekas, D., Klein, B., Friston, K., Couzin, I., & Tschantz, A. (2022). pymdp: A Python library for active inference in discrete state spaces. *arXiv Preprint arXiv:2201.03904*.

Hinterbuchinger, B., Kaltenboeck, A., Baumgartner, J. S., Mossaheb, N., & Friedrich, F. (2018). Do patients with different psychiatric disorders show altered social decision-making? A systematic

review of ultimatum game experiments in clinical populations. *Cognitive Neuropsychiatry*, *23*(3), 117–141. https://doi.org/10.1080/13546805.2018.1453791

Hooley, J. M. (2010). Social factors in schizophrenia. *Current Directions in Psychological Science*, *19*(4), 238–242.

Horga, G., & Abi-Dargham, A. (2019). An integrative framework for perceptual disturbances in psychosis. *Nature Reviews Neuroscience*, *20*(12), 763–778.

Horwitz, A. G., Berona, J., Czyz, E. K., Yeguez, C. E., & King, C. A. (2017). Positive and Negative Expectations of Hopelessness as Longitudinal Predictors of Depression, Suicidal Ideation, and Suicidal Behavior in High-Risk Adolescents. *Suicide and Life-Threatening Behavior*, *47*(2), 168–176. https://doi.org/10.1111/sltb.12273

Hudson, J. L., & Rapee, R. M. (2000). The origins of social phobia. *Behavior Modification*, *24*(1), 102–129.

Joormann, J., & Gotlib, I. H. (2006). Is this happiness I see? Biases in the identification of emotional facial expressions in depression and social phobia. *Journal of Abnormal Psychology*, *115*(4), 705–714. https://doi.org/10.1037/0021-843X.115.4.705

Karvelis, P., Paulus, M. P., & Diaconescu, A. O. (2023). Individual differences in computational psychiatry: A review of current challenges. *Neuroscience & Biobehavioral Reviews*, 105137.

Kim, S.-S., Chung, Y., Perry, M. J., Kawachi, I., & Subramanian, S. V. (2012). Association between Interpersonal Trust, Reciprocity, and Depression in South Korea: A Prospective Analysis. *PLOS ONE*, *7*(1), e30602. https://doi.org/10.1371/journal.pone.0030602

King-Casas, B., & Chiu, P. H. (2012). Understanding interpersonal function in psychiatric illness through multiplayer economic games. *Biological Psychiatry*, *72*(2), 119–125. https://doi.org/10.1016/j.biopsych.2012.03.033

Kirchner, L., Schummer, S. E., Krug, H., Kube, T., & Rief, W. (2022). How social rejection expectations and depressive symptoms bi-directionally predict each other – A cross-lagged panel analysis. *Psychology and Psychotherapy: Theory, Research and Practice*, *95*(2), 477–492. https://doi.org/10.1111/papt.12383

Kotov, R., Krueger, R. F., & Watson, D. (2018). A paradigm shift in psychiatric classification: The Hierarchical Taxonomy Of Psychopathology (HiTOP). *World Psychiatry*, *17*(1), 24.

Kube, T. (2023). Biased belief updating in depression. *Clinical Psychology Review*, 102298.

Kupferberg, A., Bicks, L., & Hasler, G. (2016). Social functioning in major depressive disorder. *Neuroscience & Biobehavioral Reviews*, *69*, 313–332.

Lazarus, S. A., Cheavens, J. S., Festa, F., & Rosenthal, M. Z. (2014). Interpersonal functioning in borderline personality disorder: A systematic review of behavioral and laboratory-based assessments. *Clinical Psychology Review*, *34*(3), 193–205.

Lewinsohn, P. M. (1974). A behavioral approach to depression. *Essential Papers on Depression*, 150–172.

Liebke, L., Bungert, M., Thome, J., Hauschild, S., Gescher, D. M., Schmahl, C., Bohus, M., & Lis, S. (2017). Loneliness, social networks, and social functioning in borderline personality disorder. *Personality Disorders: Theory, Research, and Treatment*, *8*(4), 349–356. https://doi.org/10.1037/per0000208

Linson, A., & Friston, K. (2019). Reframing PTSD for computational psychiatry with the active inference framework. *Cognitive Neuropsychiatry*, *24*(5), 347–368.

Liu, R. T., Kleiman, E. M., Nestor, B. A., & Cheek, S. M. (2015). The hopelessness theory of depression: A quarter-century in review. *Clinical Psychology: Science and Practice*, *22*(4), 345–365. https://doi.org/10.1037/h0101732

Lorian, C. N., & Grisham, J. R. (2010). The safety bias: Risk-avoidance and social anxiety pathology. *Behaviour Change*, *27*(1), 29–41.

Lüdtke, T., & Westermann, S. (2023). Negative expectations regarding interpersonal interactions in daily life are associated with subclinical depressive symptoms in a student sample: A prospective experience sampling study. *Motivation and Emotion*, *47*(1), 125–136. https://doi.org/10.1007/s11031-022-09985-8

Maercker, A., & Horn, A. B. (2013). A socio-interpersonal perspective on PTSD: The case for environments and interpersonal processes. *Clinical Psychology & Psychotherapy*, *20*(6), 465–481.

Maier, S. F., & Seligman, M. E. (2016). Learned helplessness at fifty: Insights from neuroscience. *Psychological Review*, *123*(4), 349.

Maisto, D., Barca, L., Van den Bergh, O., & Pezzulo, G. (2021). Perception and misperception of bodily symptoms from an active inference perspective: Modelling the case of panic disorder. *Psychological Review*, *128*(4), 690.

Mauritz, M. W., Goossens, P. J., Draijer, N., & Van Achterberg, T. (2013). Prevalence of interpersonal trauma exposure and trauma-related disorders in severe mental illness. *European Journal of Psychotraumatology*, *4*(1), 19985.

Montague, P. R., Dolan, R. J., Friston, K. J., & Dayan, P. (2012). Computational psychiatry. *Trends in Cognitive Sciences*, *16*(1), 72–80.

Moore, M. T., & Fresco, D. M. (2012). Depressive realism: A meta-analytic review. *Clinical Psychology Review*, *32*(6), 496–509.

Moutoussis, M., Trujillo-Barreto, N. J., El-Deredy, W., Dolan, R. J., & Friston, K. J. (2014). A formal model of interpersonal inference. *Frontiers in Human Neuroscience*, *8*, 160.

Nestor, B. A., Sutherland, S., & Garber, J. (2022). Theory of mind performance in depression: A meta-analysis. *Journal of Affective Disorders*, *303*, 233–244.

Niu, G., Shi, X., Yao, L., Yang, W., Jin, S., & Xu, L. (2022). Social Exclusion and Depression among undergraduate students: The mediating roles of rejection sensitivity and social self-efficacy. *Current Psychology*. https://doi.org/10.1007/s12144-022-03318-1

Ong, D. C., Zaki, J., & Gruber, J. (2017). Increased cooperative behavior across remitted bipolar I disorder and major depression: Insights utilizing a behavioral economic trust game. *Journal of Abnormal Psychology*, *126*(1), 1.

Palmer, C. J., Lawson, R. P., & Hohwy, J. (2017). Bayesian approaches to autism: Towards volatility, action, and behavior. *Psychological Bulletin*, *143*(5), 521.

Panitz, C., Endres, D., Buchholz, M., Khosrowtaj, Z., Sperl, M. F. J., Mueller, E. M., Schubö, A., Schütz, A. C., Teige-Mocigemba, S., & Pinquart, M. (2021). A Revised Framework for the Investigation of Expectation Update Versus Maintenance in the Context of Expectation Violations: The ViolEx 2.0 Model. *Frontiers in Psychology*, *12*. https://www.frontiersin.org/articles/10.3389/fpsyg.2021.726432

Parr, T., Pezzulo, G., & Friston, K. J. (2022). *Active inference: The free energy principle in mind, brain, and behavior*. MIT Press.

Pellicano, E., & Burr, D. (2012). When the world becomes 'too real': A Bayesian explanation of autistic perception. *Trends in Cognitive Sciences*, *16*(10), 504–510.

Pio-Lopez, L., Kuchling, F., Tung, A., Pezzulo, G., & Levin, M. (2022). Active inference, morphogenesis, and computational psychiatry. *Frontiers in Computational Neuroscience*, *16*, 988977.

Poggi, A., Richetin, J., & Preti, E. (2019). Trust and rejection sensitivity in personality disorders. *Current Psychiatry Reports*, *21*, 1–9.

Prins, S. J., Bates, L. M., Keyes, K. M., & Muntaner, C. (2015). Anxious? Depressed? You might be suffering from capitalism: contradictory class locations and the prevalence of depression and anxiety in the USA. *Sociology of Health & Illness*, *37*(8), 1352–1372.

Pulcu, E., Thomas, E. J., Trotter, P. D., McFarquhar, M., Juhász, G., Sahakian, B. J., Deakin, J. F. W., Anderson, I. M., Zahn, R., & Elliott, R. (2015). Social-economical decision making in current and remitted major depression. *Psychological Medicine*, *45*(6), 1301–1313.

Quilty, L. C., Mainland, B. J., McBride, C., & Bagby, R. M. (2013). Interpersonal problems and impacts: Further evidence for the role of interpersonal functioning in treatment outcome in major depressive disorder. *Journal of Affective Disorders*, *150*(2), 393–400. https://doi.org/10.1016/j.jad.2013.04.030

Radke, S., Schäfer, I. C., Müller, B. W., & de Bruijn, E. R. (2013). Do different fairness contexts and facial emotions motivate 'irrational'social decision-making in major depression? An exploratory patient study. *Psychiatry Research*, *210*(2), 438–443.

Rief, W., & Joormann, J. (2019). Revisiting the cognitive model of depression: The role of expectations. *Clinical Psychology in Europe*, *1*(1), 1–19.

Rief, W., Sperl, M. F., Braun-Koch, K., Khosrowtaj, Z., Kirchner, L., Schäfer, L., Schwarting, R. K., Teige-Mocigemba, S., & Panitz, C. (2022). Using expectation violation models to improve the outcome of psychological treatments. *Clinical Psychology Review*, 102212.

Robson, S. E., Repetto, L., Gountouna, V.-E., & Nicodemus, K. K. (2020). A review of neuroeconomic gameplay in psychiatric disorders. *Molecular Psychiatry*, *25*(1), 67–81. https://doi.org/10.1038/s41380-019-0405-5

Scheele, D., Mihov, Y., Schwederski, O., Maier, W., & Hurlemann, R. (2013). A negative emotional and economic judgment bias in major depression. *European Archives of Psychiatry and Clinical Neuroscience*, *263*, 675–683.

Schwartenbeck, P., FitzGerald, T. H., Mathys, C., Dolan, R., & Friston, K. (2015). The dopaminergic midbrain encodes the expected certainty about desired outcomes. *Cerebral Cortex*, *25*(10), 3434–3445.

Schwartenbeck, P., & Friston, K. (2016). Computational phenotyping in psychiatry: A worked example. *Eneuro*, *3*(4).

Scoglio, A. A., Reilly, E. D., Girouard, C., Quigley, K. S., Carnes, S., & Kelly, M. M. (2022). Social functioning in individuals with post-traumatic stress disorder: A systematic review. *Trauma, Violence, & Abuse*, *23*(2), 356–371.

Segrin, C. (2000). Social skills deficits associated with depression. *Clinical Psychology Review*, *20*(3), 379–403. https://doi.org/10.1016/S0272-7358(98)00104-4

Seligman, M. E. (1972). Learned helplessness. *Annual Review of Medicine*, *23*(1), 407–412.

Sharot, T. (2011). The optimism bias. *Current Biology*, *21*(23), R941–R945.

Simon, S., Cain, N. M., Wallner Samstag, L., Meehan, K. B., & Muran, J. C. (2015). Assessing interpersonal subtypes in depression. *Journal of Personality Assessment*, *97*(4), 364–373.

Smith, R., Badcock, P., & Friston, K. J. (2021). Recent advances in the application of predictive coding and active inference models within clinical neuroscience. *Psychiatry and Clinical Neurosciences*, *75*(1), 3–13.

Sripada, C., Angstadt, M., Liberzon, I., McCabe, K., & Phan, K. L. (2013). Aberrant reward center response to partner reputation during a social exchange game in generalized social phobia. *Depression and Anxiety*, *30*(4), 353–361.

Sripada, C. S., Angstadt, M., Banks, S., Nathan, P. J., Liberzon, I., & Phan, K. L. (2009). Functional neuroimaging of mentalizing during the trust game in social anxiety disorder. *Neuroreport*, *20*(11), 984.

Szanto, K., Clark, L., Hallquist, M., Vanyukov, P., Crockett, M., & Dombrovski, A. Y. (2014). The cost of social punishment and high-lethality suicide attempts in the second half of life. *Psychology and Aging*, *29*(1), 84–94. https://doi.org/10.1037/a0035339

Taylor, S., & Wald, J. (2003). Expectations and attributions in social anxiety disorder: Diagnostic distinctions and relationship to general anxiety and depression. *Cognitive Behaviour Therapy*, *32*(4), 166–178.

Tebeka, S., Geoffroy, P. A., Dubertret, C., & Le Strat, Y. (2021). Sadness and the continuum from well-being to depressive disorder: Findings from a representative US population sample. *Journal of Psychiatric Research*, *132*, 50–54. https://doi.org/10.1016/j.jpsychires.2020.10.004

Thimm, J., Holte, A., Brennen, T., & Wang, C. E. (2013). Hope and expectancies for future events in depression. *Frontiers in Psychology*, *4*. https://www.frontiersin.org/articles/10.3389/fpsyg.2013.00470

Tobin, M. C., Drager, K. D., & Richardson, L. F. (2014). A systematic review of social participation for adults with autism spectrum disorders: Support, social functioning, and quality of life. *Research in Autism Spectrum Disorders*, *8*(3), 214–229.

Tokuda, Y., Jimba, M., Yanai, H., Fujii, S., & Inoguchi, T. (2008). Interpersonal trust and quality-of-life: A cross-sectional study in Japan. *PLoS ONE*, *3*(12), e3985. https://doi.org/10.1371/journal.pone.0003985

Tzieropoulos, H. (2013). The Trust Game in neuroscience: A short review. *Social Neuroscience*, *8*(5), 407–416. https://doi.org/10.1080/17470919.2013.832375

Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., Reddy, T., Cournapeau, D., Burovski, E., Peterson, P., Weckesser, W., & Bright, J. (2020). SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods*, *17*(3), 261–272.

Wang, Y., Yang, L. Q., Li, S., & Zhou, Y. (2015). Game theory paradigm: A new tool for investigating social dysfunction in major depressive disorders. *Frontiers in Psychiatry*, *6*(SEP), 128. https://doi.org/10.3389/FPSYT.2015.00128/BIBTEX

Wang, Y., Zhou, Y., Li, S., Wang, P., Wu, G.-W., & Liu, Z.-N. (2014). Impaired social decision making in patients with major depressive disorder. *BMC Psychiatry*, *14*, 1–11.

Wolkenstein, L., Schönenberg, M., Schirm, E., & Hautzinger, M. (2011). I can see what you feel, but I can't deal with it: Impaired theory of mind in depression. *Journal of Affective Disorders*, *132*(1–2), 104–111.

Woods, A., Solomonov, N., Liles, B., Guillod, A., Kales, H. C., & Sirey, J. A. (2021). Perceived Social Support and Interpersonal Functioning as Predictors of Treatment Response Among Depressed Older Adults. *The American Journal of Geriatric Psychiatry*, *29*(8), 843–852. https://doi.org/10.1016/J.JAGP.2020.12.021

Zhang, H., Sun, D., & Lee, T. M. (2012). Impaired social decision making in patients with major depressive disorder. *Brain and Behavior*, *2*(4), 415–423.

Appendix

1. Simulations: Transdiagnostic Processes

In a supplementary series of simulations, the individual parameters of the agent's generative model (**Figure 2**) are perturbed in isolation. With this, we aim to simulate clinically relevant parameter regimes of the inferential process, specifically *observational uncertainty* (A), *fatalistic expectations* (B), *heightened loss aversion* (C), and *pessimistic priors* (D).

Observational uncertainty (A). An inaccuracy in the agent's A matrix corresponds to an inability to "connect the dots" between the trustee's behavior and the hidden state. While a healthy agent confidently expects a positive outcome, or win, in a cooperative context, perturbations in the observation model could lead to e.g., a positive interaction being connected to a hostile context. We hence perturbed A so that (simplified)

$$\underbrace{\begin{pmatrix} .5 \\ .5 \\ .5 \end{pmatrix}}_{win}, \underbrace{\begin{pmatrix} .5 \\ .5 \\ .5 \end{pmatrix}}_{loss} \left.\vphantom{\begin{pmatrix} .5 \\ .5 \\ .5 \end{pmatrix}}\right\} \begin{matrix} coop. \\ hostile \\ random \end{matrix}$$

Now, the agent's uncertainty when mapping observations to hidden states is very high. A cooperative response by their trustee is related to the cooperative context with $p = 0.5$, but also to the hostile context with $p = 0.5$. In clinical psychology, this may relate to perceptual disturbances (Horga & Abi-Dargham, 2019) or "cognitive immunization" (Panitz et al., 2021; Rief et al., 2022), where alternative explanations are found for interpersonal experiences that violate the patient's beliefs about their social surroundings (e.g., "He was nice to me because he thinks that I am a pathetic loser."). It may further relate to a lack of social and emotional skills. In this simulation (*observational uncertainty* (A)), we further set the prior over hidden states to be maximally uninformative (D: $p(context = y) = 0.333$ per context).

Fatalistic expectations (B). For the next simulation, *fatalistic expectations* (B), the generative model is manipulated such that

$$B = \underbrace{\begin{pmatrix} 0.2 \\ 0.05 \\ 0.15 \end{pmatrix}}_{cooperative}, \underbrace{\begin{pmatrix} 0.7 \\ 0.9 \\ 0.15 \end{pmatrix}}_{hostile}, \underbrace{\begin{pmatrix} 0.1 \\ 0.05 \\ 0.7 \end{pmatrix}}_{random} \left.\vphantom{\begin{pmatrix} 0.2 \\ 0.05 \\ 0.15 \end{pmatrix}}\right\} \begin{matrix} coop. \\ host. \\ rand. \end{matrix}$$

Now, the agent believes the cooperative state to be less stable over time ($p(cooperative_t | cooperative_{t-1} = 0.5)$. Instead, the hostile state has increased in stability (p(hostile|hostile) = 0.9). This could translate to an interpersonal pessimism bias, which is frequently observed in patients with depression (Liu et al., 2015; Niu et al., 2022; Panitz et al., 2021). Good things are believed to be futile, whereas negative or uncomfortable situations are assumed to be more stable, as e.g. suggested by research into low self-efficacy or learned helplessness (Maier & Seligman, 2016) or biased expectations in depression (Gamble et al., 2019; Lüdtke & Westermann, 2023). Deviations in

B may further be related to aberrant beliefs about the dynamics of interpersonal situations (Nestor et al., 2022).

Heightened loss aversion (C). The C vector determines the agent's preferred observations by placing a high prior probability over desired outcomes and a low prior probability over undesirable ones. For the simulation *heightened loss aversion* (C), the preferences over outcomes are set to

$$C = softmax\begin{pmatrix} 1 \\ -10 \\ 1.2 \end{pmatrix}\begin{matrix} win \\ loss \\ keep \end{matrix} \rightarrow \begin{pmatrix} 0.45 \\ 0.01 \\ 0.54 \end{pmatrix}$$

An excessive negative weight is placed on losses, compared to a relatively low preference given to the "win" outcome. An agent with such outcome preferences shows inappropriate loss aversion, which may prematurely tip the exploration-exploitation balance towards missing out on valuable epistemic information to avoid a loss. These parameter settings may approximate rejection sensitivity (Gao et al., 2017) and risk aversion (Hartley & Phelps, 2012). The agent was further equipped with uncertain, i.e., equiprobable prior beliefs over hidden states (D).

Pessimistic priors (D). In a final simulation (*pessimistic priors* (D)), we manipulated the D matrix, or prior over hidden states so that

$$D = \begin{pmatrix} 0.1 \\ 0.7 \\ 0.2 \end{pmatrix}\begin{matrix} coop. \\ host. \\ rand. \end{matrix}$$

Here, the agent believes the cooperative context to be highly unlikely, whereas the negative or random contexts are assumed to be more likely. Perturbing the D matrix may formalize transdiagnostic processes such as interpersonal distrust (Poggi et al., 2019) or interpersonal trauma (Mauritz et al., 2013), leading to pessimistic interpersonal prior beliefs. For an overview of all transdiagnostic generative models, see **Table 2**.

Agent name	A_1	B	C	D
Healthy	$\underbrace{\begin{pmatrix}.9\\.15\\.5\end{pmatrix}}_{win}, \underbrace{\begin{pmatrix}.1\\.85\\.5\end{pmatrix}}_{loss}\Bigg\}\begin{matrix}coop.\\hostile\\random\end{matrix}$	$\underbrace{\begin{pmatrix}0.9\\0.3\\0.3\end{pmatrix}}_{coop.}, \underbrace{\begin{pmatrix}0.02\\0.6\\0.3\end{pmatrix}}_{hostile}, \underbrace{\begin{pmatrix}0.08\\0.1\\0.4\end{pmatrix}}_{rand.}\Bigg\}\begin{matrix}coop.\\host.\\rand.\end{matrix}$	$sm\begin{pmatrix}2.5\\-2.2\\1\end{pmatrix}$	$\begin{pmatrix}0.6\\0.35\\0.05\end{pmatrix}$
Observational uncertainty (A)	$\underbrace{\begin{pmatrix}\mathbf{.5}\\\mathbf{.5}\\\mathbf{.5}\end{pmatrix}}_{win}, \underbrace{\begin{pmatrix}\mathbf{.5}\\\mathbf{.5}\\\mathbf{.5}\end{pmatrix}}_{loss}\Bigg\}\begin{matrix}coop.\\hostile\\random\end{matrix}$	$\underbrace{\begin{pmatrix}0.9\\0.32\\0.5\end{pmatrix}}_{coop.}, \underbrace{\begin{pmatrix}0.02\\0.6\\0.3\end{pmatrix}}_{hostile}, \underbrace{\begin{pmatrix}0.08\\0.08\\0.2\end{pmatrix}}_{rand.}\Bigg\}\begin{matrix}coop.\\host.\\rand.\end{matrix}$	$sm\begin{pmatrix}2.5\\-2.2\\1.\end{pmatrix}$	$\begin{pmatrix}0.33\\0.33\\0.33\end{pmatrix}$
Fatalistic expectations (B)	$\underbrace{\begin{pmatrix}.9\\.15\\.5\end{pmatrix}}_{win}, \underbrace{\begin{pmatrix}.1\\.85\\.5\end{pmatrix}}_{loss}\Bigg\}\begin{matrix}coop.\\hostile\\random\end{matrix}$	$\underbrace{\begin{pmatrix}\mathbf{0.2}\\\mathbf{0.05}\\\mathbf{0.15}\end{pmatrix}}_{coop.}, \underbrace{\begin{pmatrix}\mathbf{0.7}\\\mathbf{0.9}\\\mathbf{0.15}\end{pmatrix}}_{hostile}, \underbrace{\begin{pmatrix}\mathbf{0.1}\\\mathbf{0.05}\\\mathbf{0.7}\end{pmatrix}}_{rand.}\Bigg\}\begin{matrix}coop.\\host.\\rand.\end{matrix}$	$sm\begin{pmatrix}2.5\\-2.2\\1.\end{pmatrix}$	$\begin{pmatrix}0.33\\0.33\\0.33\end{pmatrix}$
Heightened loss aversion (C)	$\underbrace{\begin{pmatrix}.9\\.15\\.5\end{pmatrix}}_{win}, \underbrace{\begin{pmatrix}.1\\.85\\.5\end{pmatrix}}_{loss}\Bigg\}\begin{matrix}coop.\\hostile\\random\end{matrix}$	$\underbrace{\begin{pmatrix}0.9\\0.3\\0.3\end{pmatrix}}_{coop.}, \underbrace{\begin{pmatrix}0.02\\0.6\\0.3\end{pmatrix}}_{hostile}, \underbrace{\begin{pmatrix}0.08\\0.1\\0.4\end{pmatrix}}_{rand.}\Bigg\}\begin{matrix}coop.\\host.\\rand.\end{matrix}$	$sm\begin{pmatrix}2.5\\\mathbf{-5.}\\1.\end{pmatrix}$	$\begin{pmatrix}0.33\\0.33\\0.33\end{pmatrix}$
Pessimistic priors (D)	$\underbrace{\begin{pmatrix}.9\\.15\\.5\end{pmatrix}}_{win}, \underbrace{\begin{pmatrix}.1\\.85\\.5\end{pmatrix}}_{loss}\Bigg\}\begin{matrix}coop.\\hostile\\random\end{matrix}$	$\underbrace{\begin{pmatrix}0.9\\0.3\\0.3\end{pmatrix}}_{coop.}, \underbrace{\begin{pmatrix}0.02\\0.6\\0.3\end{pmatrix}}_{hostile}, \underbrace{\begin{pmatrix}0.08\\0.1\\0.4\end{pmatrix}}_{rand.}\Bigg\}\begin{matrix}coop.\\host.\\rand.\end{matrix}$	$sm\begin{pmatrix}2.5\\-2.2\\1.\end{pmatrix}$	$\begin{pmatrix}\mathbf{0.1}\\\mathbf{0.7}\\\mathbf{0.2}\end{pmatrix}$

Table 2. Overview of parameter settings for the simulation of transdiagnostic processes.

In **bold** font are all parameters that were changed compared to the "healthy" agent (top row, for comparison).

1.1 Results: parameter-based simulations.

Observational uncertainty **(A).** When introducing an uninformative distribution as the A parameter of the generative model as a proxy for high inferential uncertainty, the agent is unable to form precise posterior beliefs about any specific context (**Figure 20A**; averaged: **Figure 24**). Their posterior beliefs deviate only slightly from the uninformative prior as a consequence of the agent's inability to connect a specific observation to a context. Any slight deviations from the prior can be attributed to the remaining parameters of the generative model (B, C, D) which are equivalent to the healthy-to-optimistic agent. The switch to a more hostile context after t=20 leads to a slight decrease in P(context=cooperative), and an increase in p(context=random), but no change in the agent's observable behavior. When starting in a hostile context (**Figure 20B**), the agent's beliefs to be in a hostile context similarly do not deviate much from the uninformative prior (max. $p(context = hostile) = 0.33$). Shortly before the context switches, the agent shows rigid avoidance behavior despite their imprecise posterior beliefs. Because of this, the agent does not register the context switch from epoch 20 onwards.

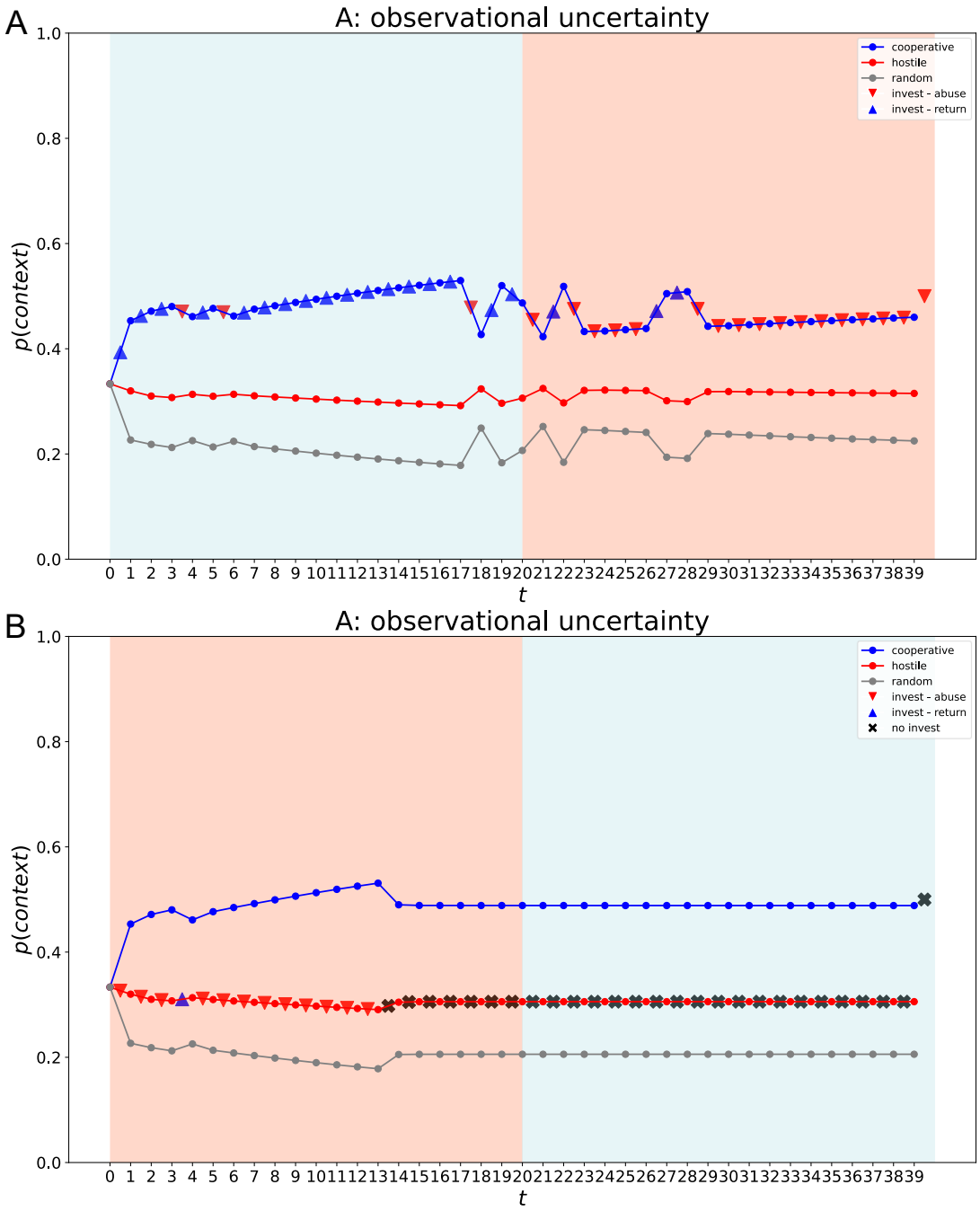


Figure 20. Observational uncertainty (A parameter)

A. An agent with a biased *A* parameter forms only imprecise posterior beliefs about being in any specific context, compared to an agent with a more accurate generative model (comp. Fig. **2**). Negative experiences have a relatively smaller weight to alter the posterior belief compared to a healthy agent. The context switch is barely reflected in the agent's posterior beliefs about context.

B. In a hostile context, the agent's posterior belief to be in a hostile context does not deviate much from the uninformative prior p(context=hostile)=0.33. However, the agent here avoids interacting with the trustee after 13 timesteps.

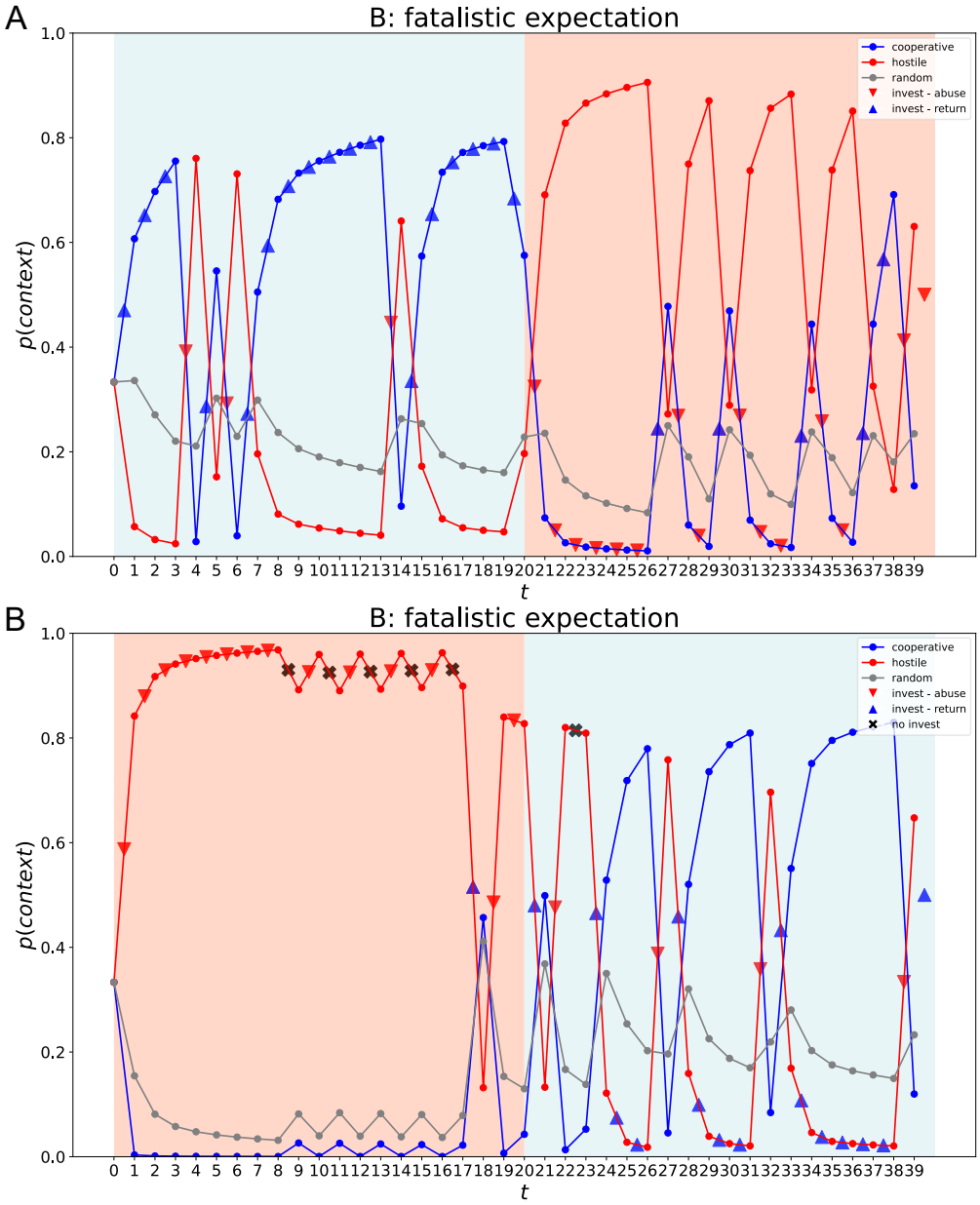


Figure 21. Fatalistic expectations (B parameter) simulations.

A. In a cooperative context, the agent with fatalistic expectations (biased *B* parameter) shows a relatively strong posterior belief to be in a hostile context after observing some interspersed defects. The switch to the hostile context after *t*=20 timesteps is reflected in the agent's posterior beliefs, but this belief does not lead to an avoidant behavioral strategy.

B. In the negative context, precise posteriors are formed over the hostile context. While the agent refrains from investing their budget with the trustee after observing a series of defects, they continue to probe the trustee's behavior. The switch to a more cooperative context leads to alternating spikes in the beliefs to be in a cooperative- or hostile context.

***Fatalistic expectations* (B).** Results from an agent with fatalistic expectations, i.e., a perturbed B parameter, are shown in **Figure 21A,B** (averages in **Figure 25**). When starting in a cooperative context (**Figure 21A**), the agent's posterior belief to be in a cooperative context form slowly compared to the healthy-to-optimistic agent. Individual defects by the trustee impact the agent's beliefs much more strongly than longer series of cooperative observations (e.g. defect in t=4, t3 p(context=cooperative)=.78; t4 p(context=cooperative)=.04). The agent is more uncertain about the context compared to the healthy agent, even after observing a longer series of cooperative actions (max. $p(context = cooperative) = 0.81$). When the context switches, this is reflected in the agent's posterior beliefs (max. $p(context = hostile) = 0.83$). However, this belief is not reflected in a more avoidant behavioral strategy. This lack of behavioral adaptation can be attributed to the intact C parameter, which assigns a high utility to observing the "win" outcome. When first seeing a hostile context (**Figure 21B**), the agent accurately believes to be in a hostile context (max. $p(context = hostile) = 0.94$), and starts avoiding the transaction with the trustee. Nevertheless, they continue to probe the trustee's behavior, which is why the context switch at t=20 is registered. However, the agent subsequently is unable to form precise posterior beliefs about any particular context, with max. $p(context = cooperative) = 0.81$.

***Heightened loss aversion* (C). Figure 22A-B** illustrate results from an agent with a biased preference over outcomes compared to the healthy agent, the transdiagnostic process of *heightened loss aversion* (averaged results **Figure 26**). When starting in a cooperative context, the agent accurately infers its nature and continues investing with the trustee. After the context switches in t=20, observing the trustee defecting leads to large updates in the agent's posterior beliefs and the agent begins to avoid interacting with the trustee. In a hostile-first game (**Figure 22B**), the agent quickly infers to be in a hostile context and flexibly avoids interacting with the trustee. After the context switches, the agent observes one defect and adopts a rigid behavioral strategy that lasts until the end of the game.

***Pessimistic priors* (D).** An agent with pessimistic priors, i.e., a biased D parameter, is shown in **Figure 23(**averaged results, see appendix **Figure 27**). Upon observing a series of returns of their investment, the agent quickly forms an accurate belief to be in a cooperative context (max. $p(context = cooperative) = 0.94$, **Figure 23A**). The context switch at t=20 towards a hostile context is reflected in the agent's beliefs (max. $p(context = hostile) = 0.90$), but this does not lead to avoidance behaviors. In the hostile-to-cooperative game (**Figure 23B**), precise beliefs about being in a hostile context are formed ($p(context = hostile) = 0.90$) and the agent flexibly avoids interacting with the trustee. However, the agent's pessimistic priors are updated and eventually overridden after observing more returns on their investment after the context switch. Since the transition parameter B is equivalent to the healthy agent's parameter, the Markov chain of latent states would converge to the "healthy" steady state even without observations.

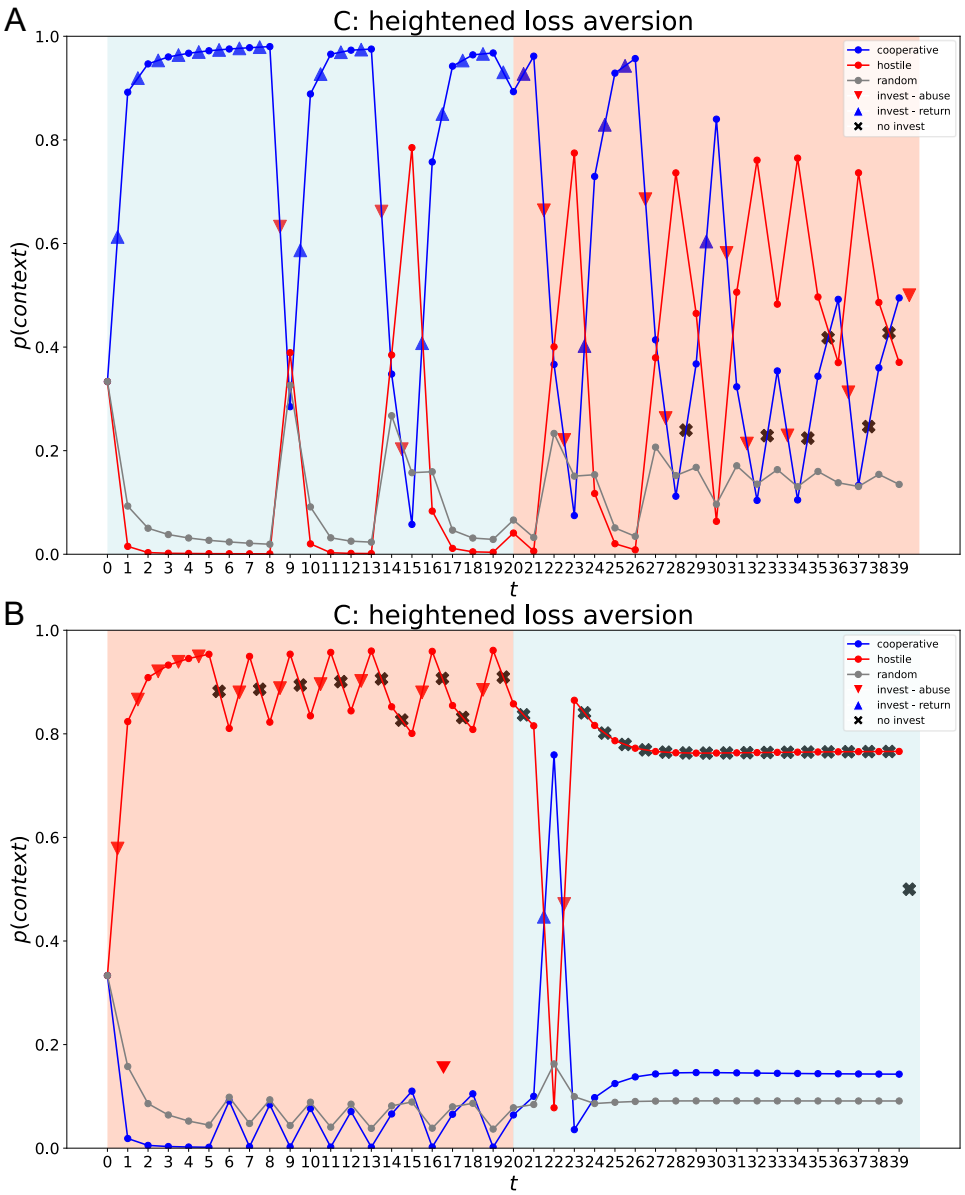


Figure 22. Heightened loss aversion (C parameter)

A The agent with a simulated heightened loss aversion (C) forms precise beliefs about being in a cooperative context at first, but the agent's belief is strongly impacted by observing few defects interspersed within a majority of returns on investment (e.g. t=8: p(context|cooperative)=0.97, after one defect in t=9: p(context|cooperative)=0.28). The agent's belief to be in a cooperative context changes slowly after the context switch at t=20, with some avoidant behaviors emerging from t=29 onwards.

B. When starting in a hostile context, the agent forms a precise belief to be in a hostile context and avoids the transaction with the trustee while continuing to probe the trustee's behavior. After the context switches, one defect in t=23 is observed, after which the agent avoids the trustee until the end of the game.

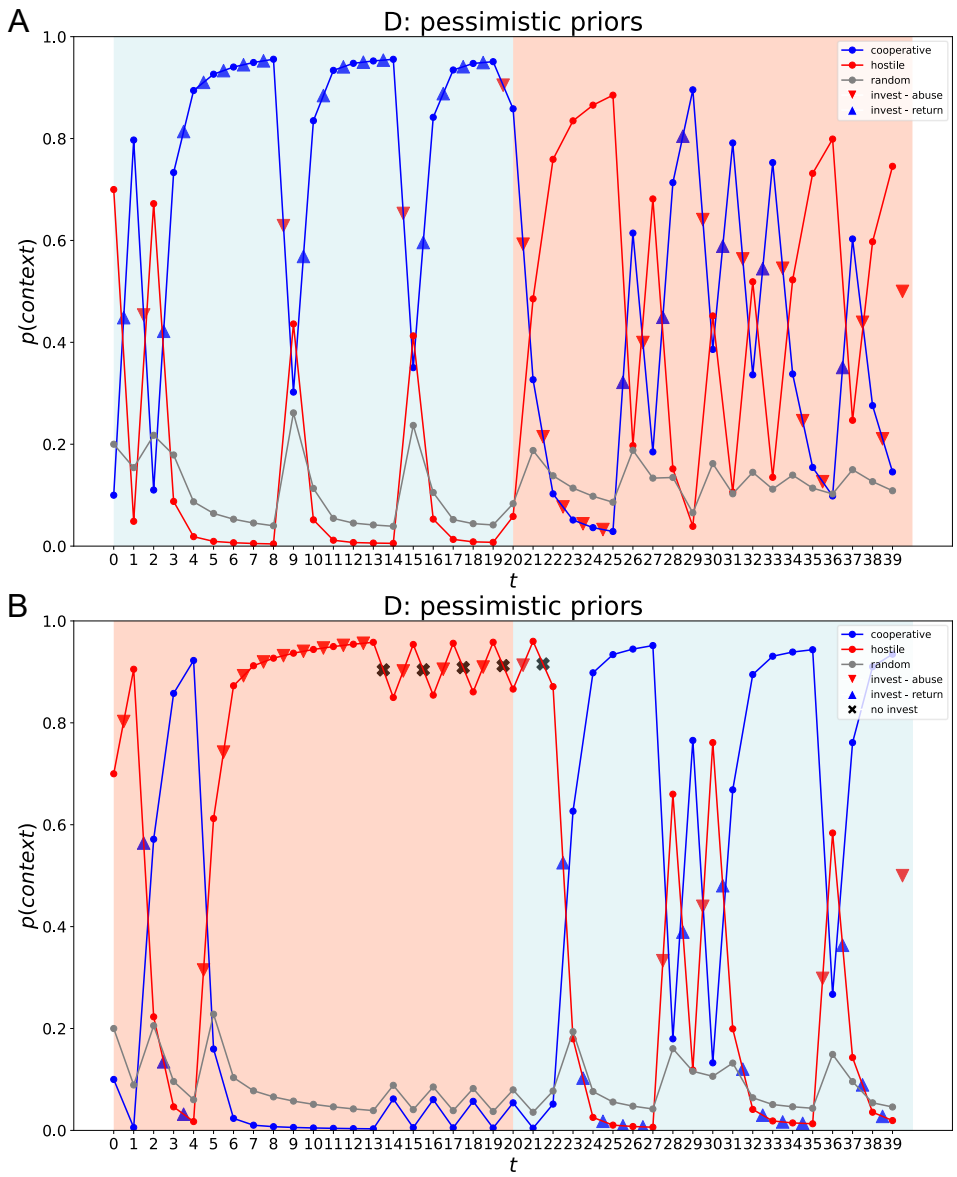


Figure 23. Pessimistic priors (D parameter)

A *Agent* pessimistic priors *is characterized by a heightened a-priori belief at t=0 to be in a hostile context*. When interacting with the environment, their posterior beliefs are updated in accordance with the observations and close to the generative process. After t=20, the hostile context is not clearly identified by the agent and no avoidance is observed.

B Starting in the hostile context, the nature of the context is reflected in the agent's posterior beliefs and the agent begins to avoid the transaction with the trustee while continuing to probe the trustee's behavior. Upon the context switch, the cooperative context is registered by the agent and their posterior beliefs adjust accordingly.

2. Average posterior beliefs

An additional analysis was performed to determine the average posterior beliefs over contexts per agent type. The time series (T=40, context switch after t=20) was run 20 times for each agent type. Results are shown in the subsequent plots (**Figures 15-23**). The plots show average posterior beliefs over the cooperative (blue) and hostile (red) contexts. The thick blue and red lines represent mean posterior beliefs, averaged over 20 runs of the simulation (pale lines represent one game). For convenience, the posterior belief to be in a random state is not shown.

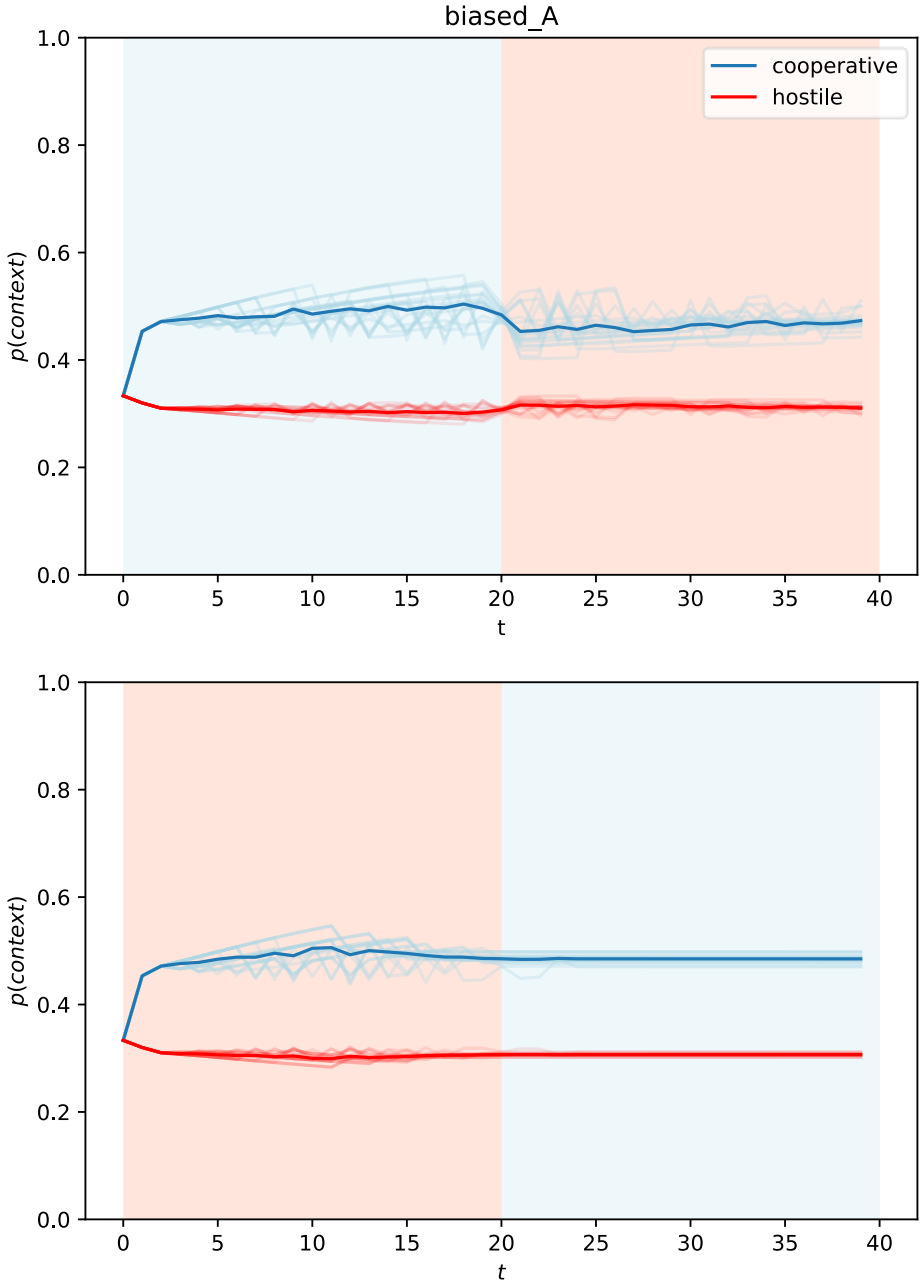


Figure 24. Average time series for agent with observational uncertainty (A perturbation)

A. Beliefs of the "perceptual uncertainty (biased *A*)" agent in the cooperative-to-hostile game. The posterior beliefs do not deviate much from the uninformed prior across all instantiations of the game.

B. Perceptual uncertainty agent in the hostile-to-cooperative game. Similar to the cooperative-to-hostile game, the agent's posterior beliefs do not deviate much from the uninformed prior for the length of the game.

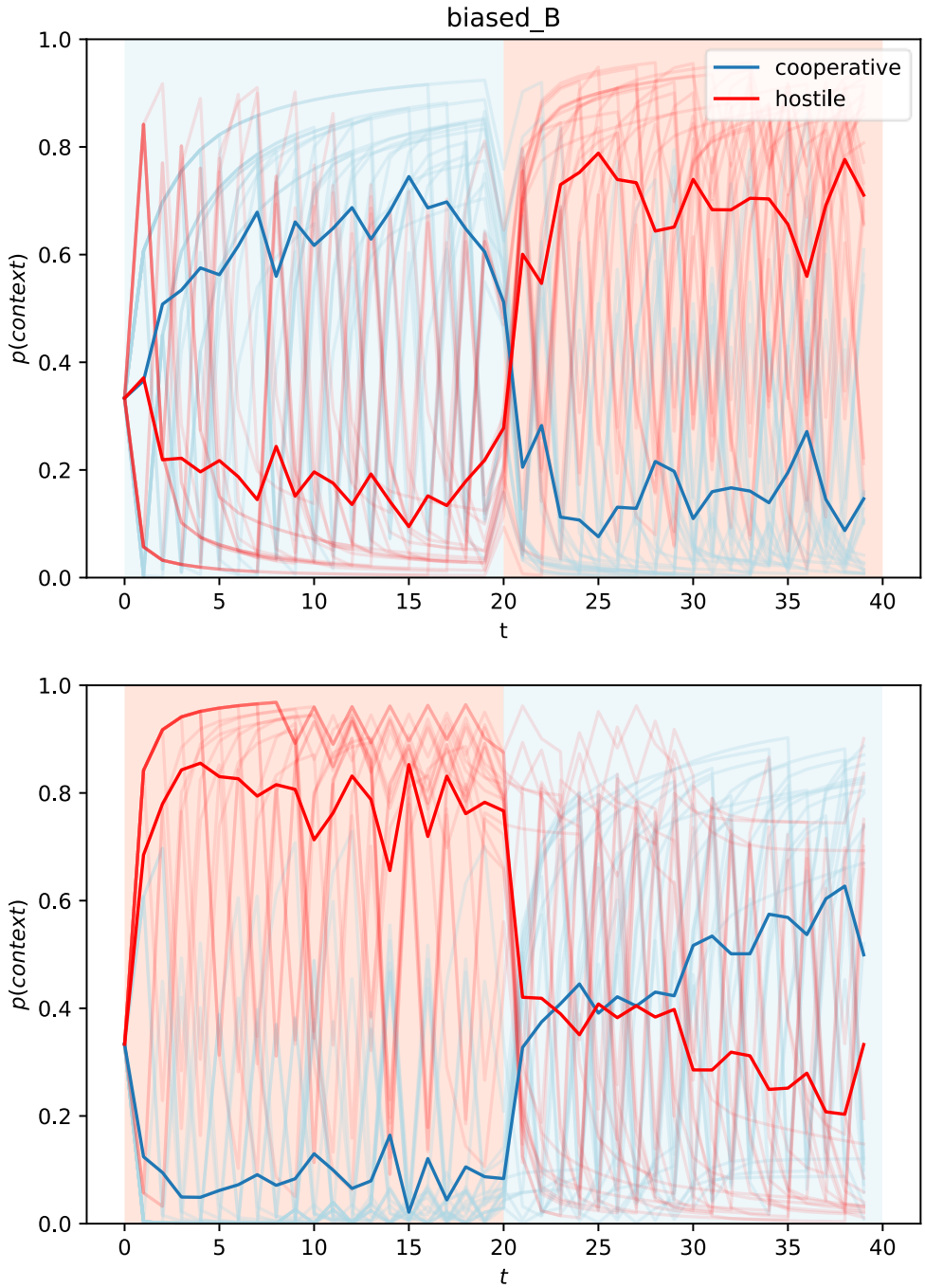


Figure 25. Average time series for agent with fatalistic expectations (biased B matrix)

A. Posterior beliefs of the agent type with fatalistic expectations (biased B) in the cooperative-to-hostile game. The posterior belief to be in a cooperative context accumulates slower compared to the healthy agent. After the context switch, the agent forms more accurate beliefs about the context.

B. In the hostile-to-cooperative game, the agent quickly arrives at a precise belief to be in a hostile context. While the cooperative context is recognized by the agent in most instantiations of the game, the posterior belief to be in a cooperative context does not reach the same precision as the belief about the hostile context in the first twenty trials.

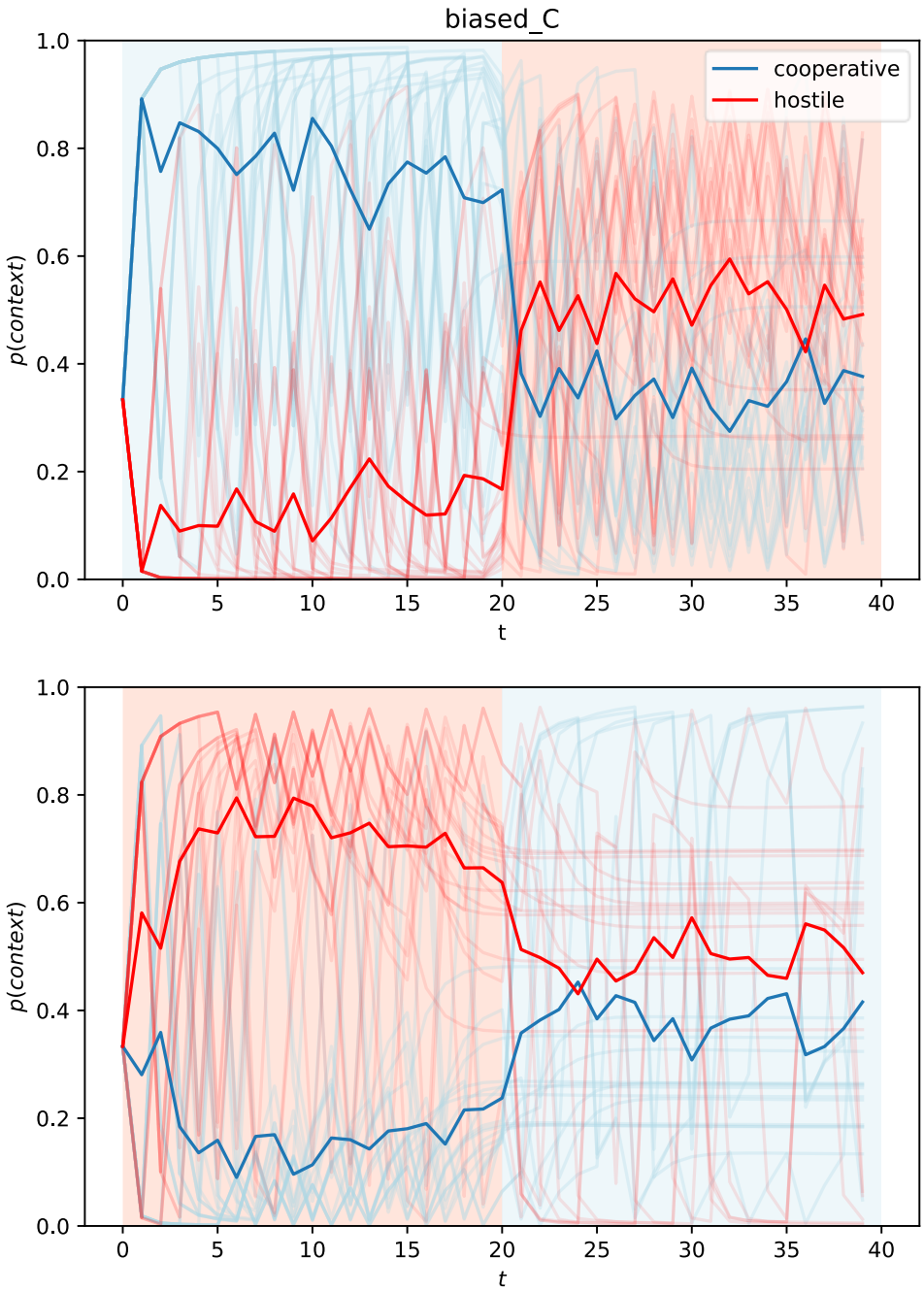


Figure 26. Average time series for agent with loss aversion (C perturbation)

A. In the cooperative-to-hostile game, the agent's belief to be in a cooperative context rises within the first twenty trials. After the context switch, the agent is more certain to be in a hostile context.

B. During the hostile-to-cooperative game, the agent believes to be in a hostile context. After the context switch, the agent does not form a precise belief to be in a cooperative context – many instantiations of the game result in rigid avoidance behavior (horizontal pale lines), where the trustee behavior is no longer sampled.

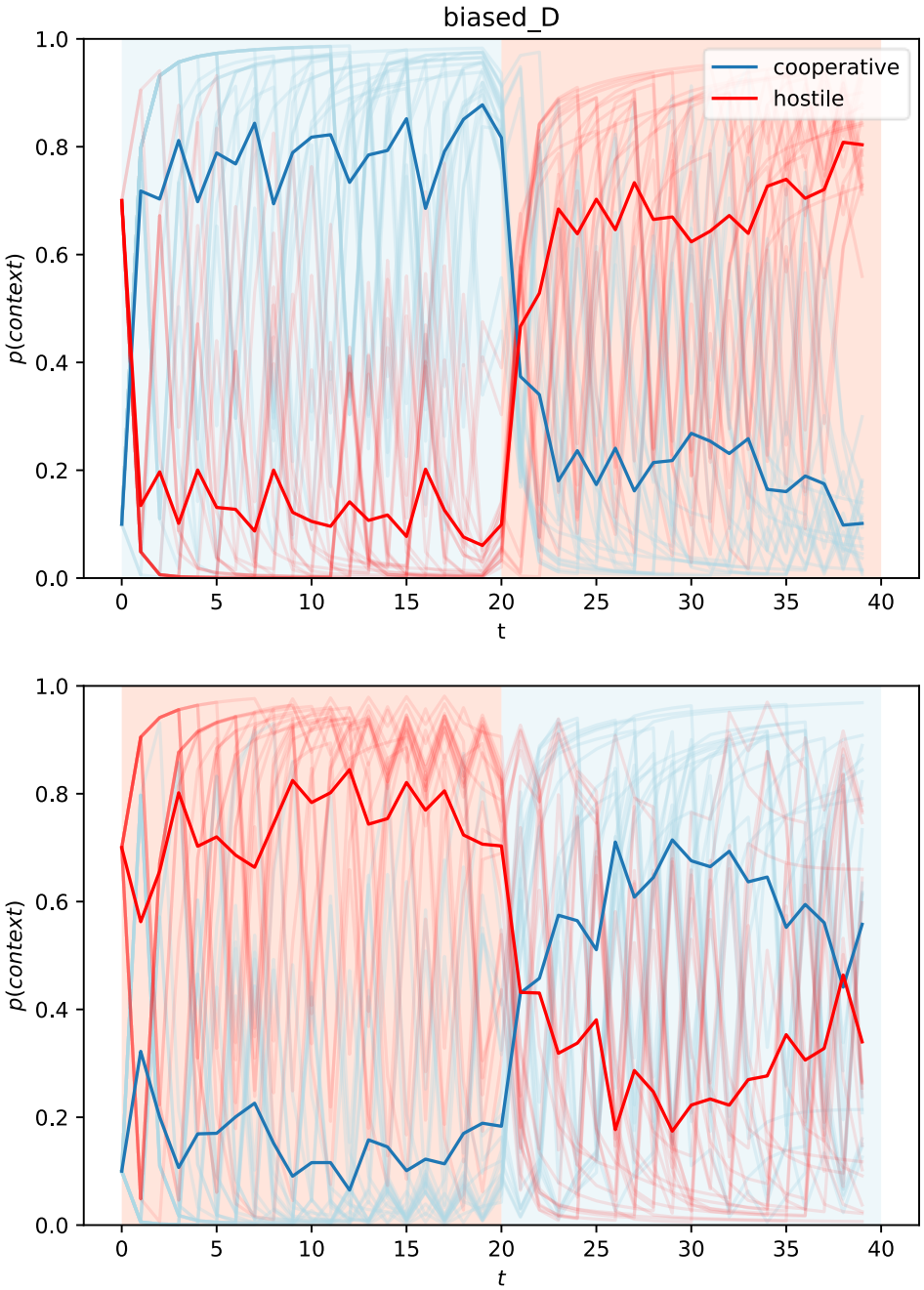


Figure 27. Average time series with pessimistic priors (D perturbation)

A. The agent with pessimistic priors (biased *D*) plays the cooperative-to-hostile game. The agent recognizes both the cooperative and the hostile context.

B. In the hostile-to-cooperative game, the agent's posterior beliefs about the context reflect the true context dynamics, albeit the agent shows less precise beliefs about the cooperative context than the healthy agent.

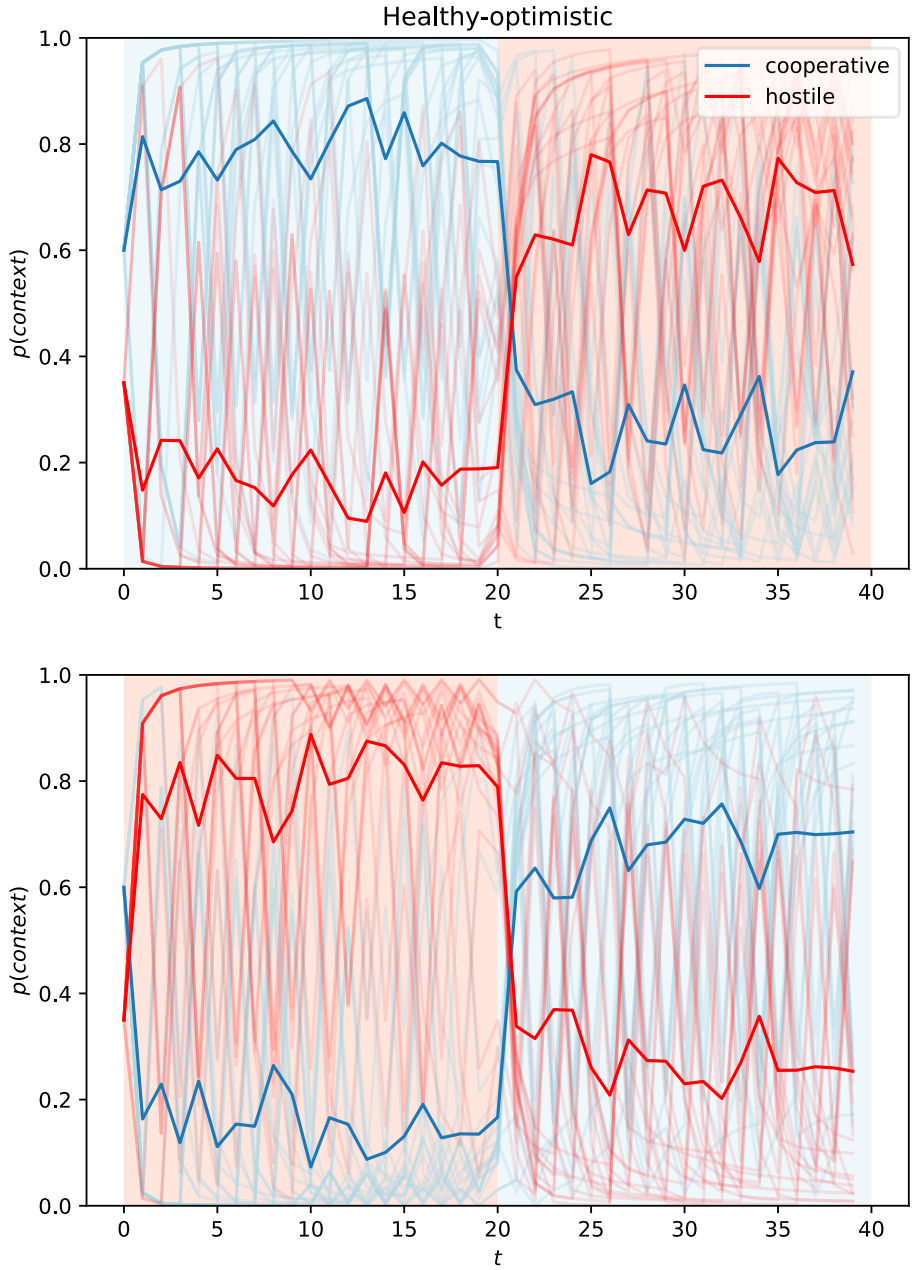


Figure 28. Average time series healthy agent.

A. The healthy-to-optimistic agent succeeds in recognizing both the cooperative and the hostile context after the switch in a cooperative-to-hostile game.

B. In the hostile-to-cooperative game, the healthy-to-optimistic agent forms precise posterior beliefs to be in a hostile context at first, which are updated in line with the context switch to a more cooperative context after t=20.

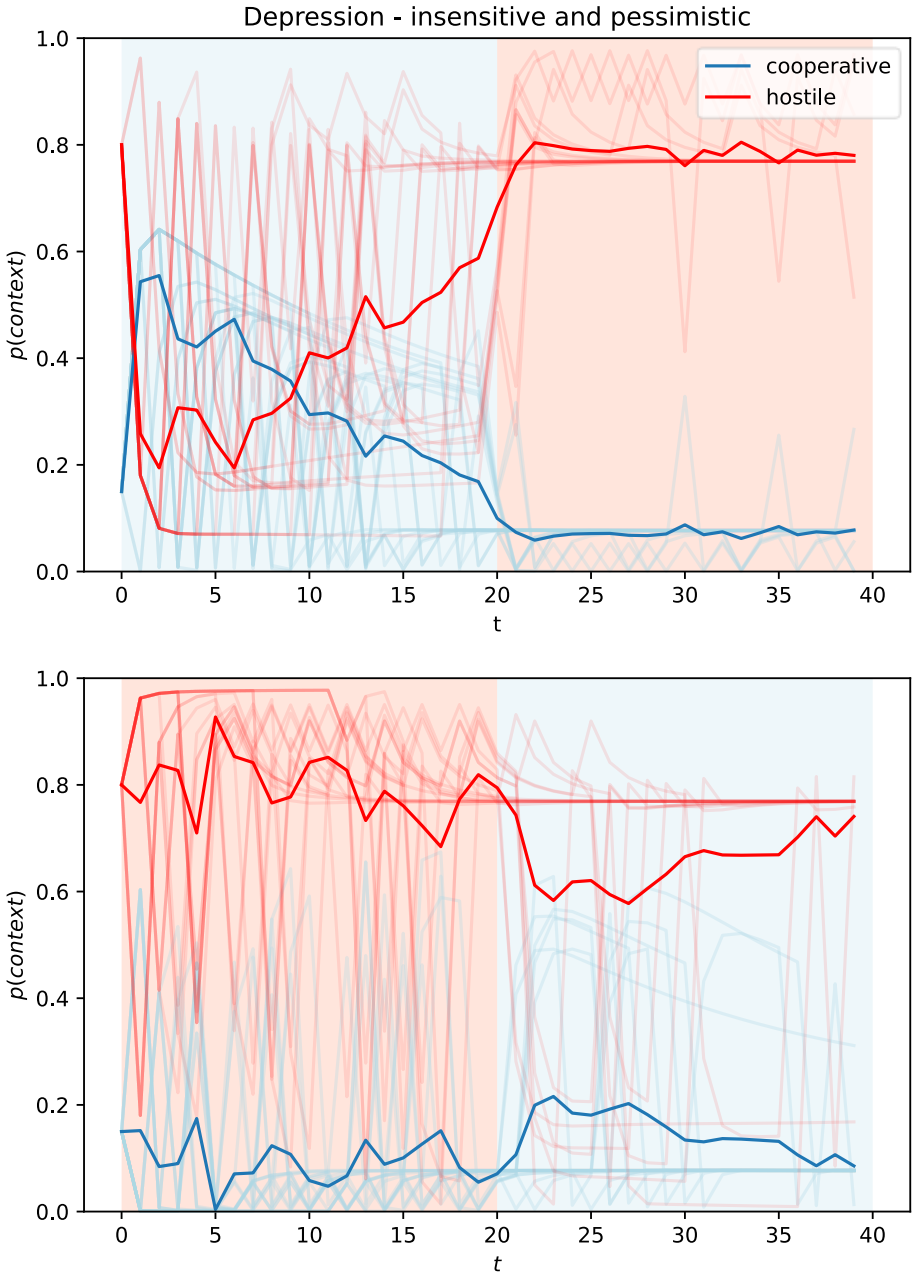


Figure 29. Average time series for depressed agent (reward-insensitive-pessimistic)

A. The reward-insensitive, pessimistic agent is exposed to the cooperative-to-hostile game. The agent's belief to be a in a cooperative context is weakened and develops towards believing to be in a hostile context during the cooperative phase in almost all instantiations of the game. During the hostile phase, the agent shows fixed avoidance behaviors and quits sampling the trustee's behavior.

B. In the hostile-to-cooperative game, the agent forms precise posterior beliefs about the nature of the context in the hostile phase of the game. In most instantiations, this agent shows rigid avoidance behavior (horizontal line). The average belief to be in a cooperative context after the context switch never exceeds p=0.3.

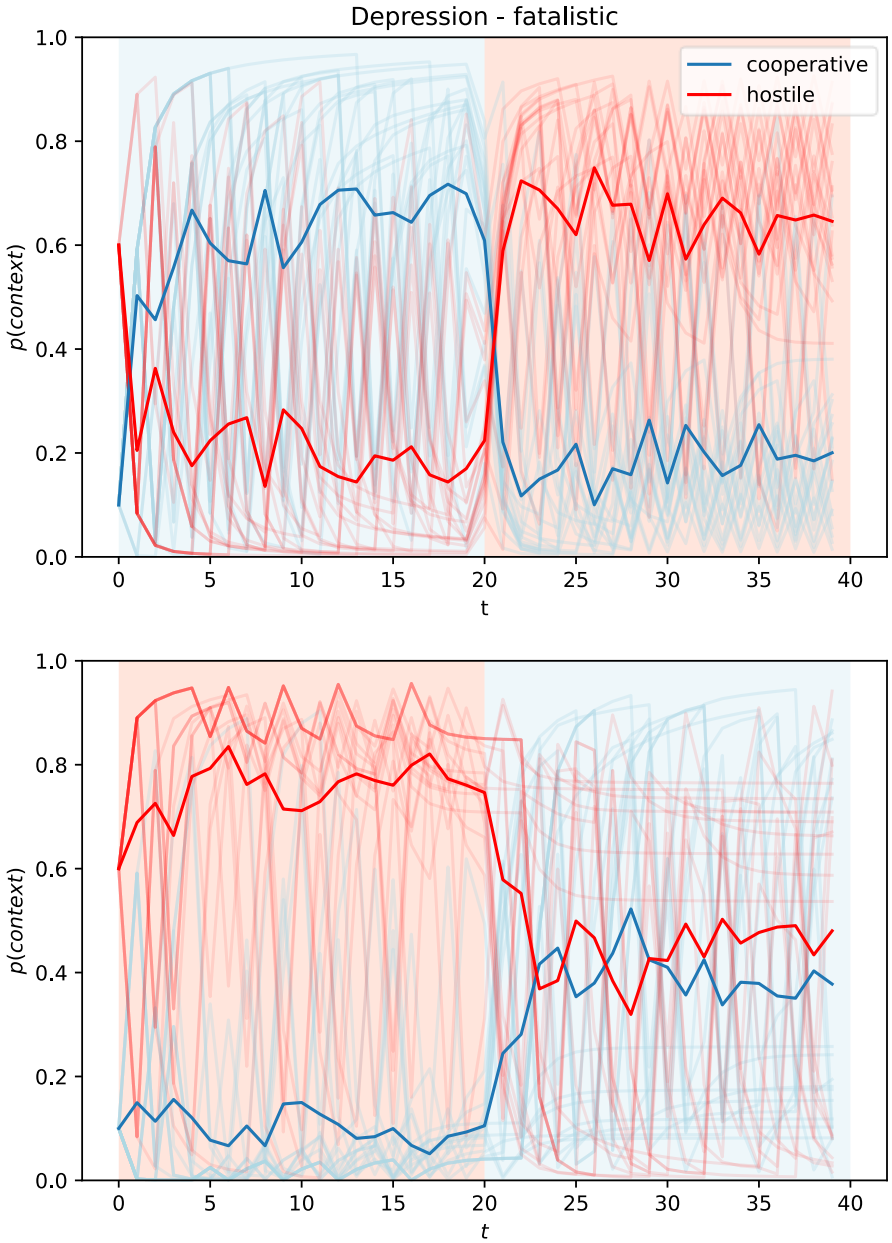


Figure 30. Average time series, depressed agent with fatalistic expectations

A. During a cooperative-to-hostile game, the depressed (fatalistic, type 2) agent does recognize the true context dynamics, albeit the belief to be in a cooperative context is weakened compared to the healthy-to-optimistic agent. The switch to the hostile context after t=20 is reflected in the agent's posterior beliefs.

B. In the hostile-to-cooperative game, the agent accurately infers the hostile context during the first phase of the game. In many instantiations, this agent shows rigid avoidance behaviors (horizontal pale lines), but in some cases, the agent successfully infers the context switch. However, the mean p(context=cooperative) never exceeds p=0.5.

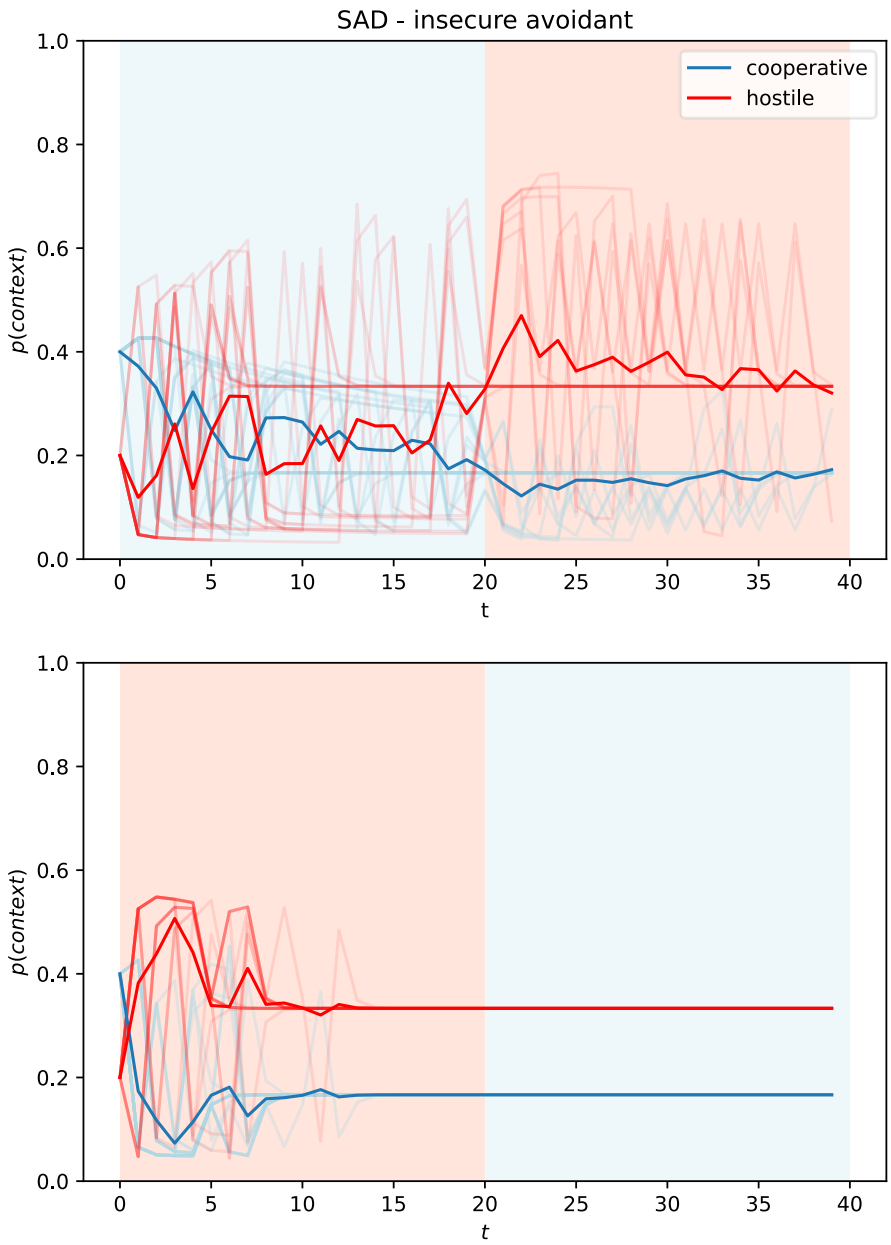


Figure 31. Average time series social phobia, insecure-avoidant.

A. The social-phobia agent (type 1, insecure-avoidant) does not reliably infer a cooperative context during the first cooperative phase of the game. In many instantiations of the game, the agent shows rigid avoidance behaviors. After the context switches to hostile, the agent forms more precise posterior beliefs about being in a hostile context.

B. In a hostile-to-cooperative game, the agent shows brief exploration behavior, before stopping to interact with the agent in all instantiations. The context switch to cooperative is not recognized in any agent instantiation.

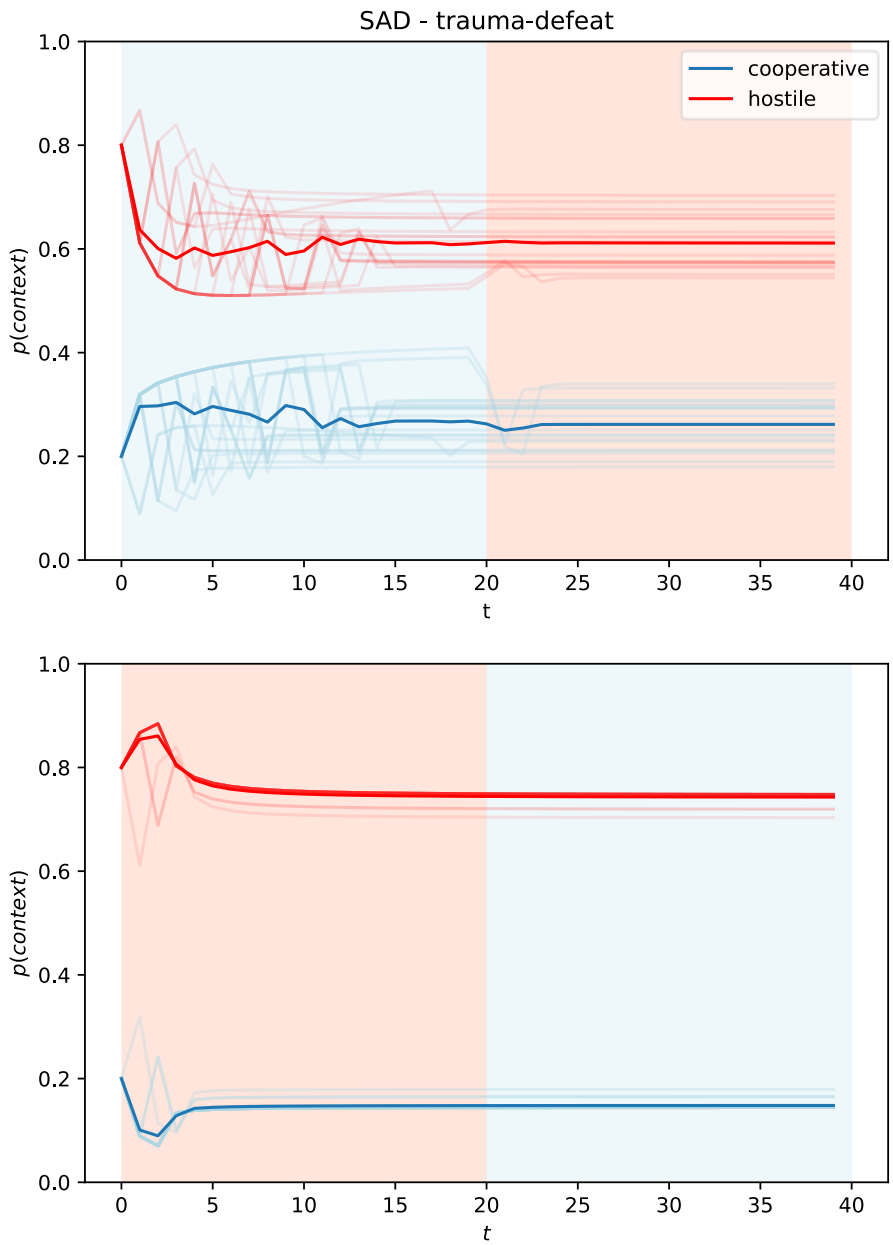


Figure 32. Average time series, social phobia - trauma and defeat.

A. The trauma-and-defeat (type 2 social phobia) agent in a cooperative-to-hostile game. This agent deviates only slightly from the biased prior settings in almost all instantiations. In almost all games, the agent shows rigidity in their behavioral avoidance of interacting with the trustee and stops exploring their behavior during the cooperative phase of the game.

B. In the hostile-to-cooperative game, the agent barely deviates from their prior and explores interactions with the trustee for a maximum of 5 trials before adopting a rigid avoidance strategy.

Discussion: transdiagnostic processes

In the first agent, we targeted the observation model (A, *observational uncertainty*), by decreasing the certainty with which observations are connected to a specific context. As a consequence, the agent struggled making precise context inferences. Biases to the observation model could be related to "cognitive immunization" techniques, where a friendly interaction is connected to a hostile context (Kube, 2023; Panitz et al., 2021; Rief & Joormann, 2019). This thought pattern is frequently observed in patients with depression or social anxiety (Everaert et al., 2017; Rief et al., 2022). The inability to draw "correct" conclusions from a given observation may also be the consequence of a lack of socio-emotional skills, a common impairment in clinical populations (Nestor et al., 2022; Segrin, 2000; Wolkenstein et al., 2011). Many patients further connect positive interpersonal experiences to randomness rather than true benevolence (Ewen et al., 2022). In conclusion, perturbations to the agent's A matrix have the consequence that observations do not have a systematic or sensible effect on state inference.

Next, we perturbed the agent's transition parameter (B), introducing *fatalistic beliefs.* The transition matrix comprises beliefs about the evolution of contexts over time and the agent's action-dependent state transitions. Believing that a hostile context will likely develop into a cooperative one is a computational approximation to the optimism bias in healthy adults (Bracha & Brown, 2012; Sharot, 2011). In contrast, believing that hostile contexts are the norm while cooperative contexts are futile is in line with e.g., hopelessness (Liu et al., 2015), learned helplessness (Seligman, 1972), low self-efficacy (Maier & Seligman, 2016; Niu et al., 2022), or interpersonal distrust (Kim et al., 2012). Empirical data suggests that individuals with depression and social anxiety show deficits in social cognition, mentalization and theory of mind (Baker & Edelmann, 2002; Nestor et al., 2022), which may be a consequence of inaccurate beliefs about the dynamics of interpersonal situations – for example, believing that interpersonal situations will inevitably end in hostility (Lüdtke & Westermann, 2023; Segrin, 2000). Supporting the relevance of expectations, recent data suggest that negative expectations drive subclinical depression to a larger degree than actual negative experiences (Lüdtke & Westermann, 2023). Furthermore, biases in the transition model B may be an appropriate operationalization for altered future thinking in depression. A recent meta-analysis suggests a link between depression and the difficulty to imagine specific positive events in the future, which may reduce motivation to engage in any activities, especially given that the ability to imagine negative future events is unaltered in depression (Gamble et al., 2019).

Gaining a joint win is the preferred outcome in healthy individuals. To simulate psychopathological behavior, we manipulated the C parameter such that the agent puts an increased weight on avoiding losses instead of maximizing wins. Consequently, the agent explores the trustee's behavior less, and starts avoiding any transactions with the trustee quicker compared to the healthy-to-optimistic agent, which may be a useful method to simulate social anhedonia (Barkus & Badcock, 2019). Several studies

suggest aberrant reward valuation and - responsiveness in depression (Alloy et al., 2016). On the other hand, perturbations in the C parameter could be a proxy for rejection sensitivity (Gao et al., 2017) and risk aversion (Hartley & Phelps, 2012), by placing an overly strong weight on avoiding the "loss" outcome.

Finally, we manipulated the agent's prior beliefs over contexts (D, *pessimistic priors*). Particularly, the agent's generative model comprised the belief that a hostile context is more likely compared to a cooperative context a-priori. Individuals with depression expect to experience higher rates of negative events, and lower rates of positive events (Thimm et al., 2013). Specifically, studies suggest that a lack of positive expectations may be more central to depression than the presence of negative expectations, which is in line with a reduced prior on positive experiences (Horwitz et al., 2017). In psychological terms, perturbations to the D parameter may be related to interpersonal distrust (Poggi et al., 2019) or trauma (Mauritz et al., 2013). In the simulation, a biased prior belief was overridden by the agent's observations because the other parameters of the generative model (i.e., observation model, transition matrix, preference over outcomes) were intact. Pessimistic prior beliefs may effectively model experience-dependent negative beliefs that normalize in more favorable environments. Deviations in the D vector may nevertheless be of relevance for clinical phenotypes, especially in combination with perturbations to other parameters of the generative model.

A healthy-to-optimistic agent is most flexible in a longer time series.

An additional supplementary analysis was performed to determine which agent would earn more absolute rewards in a longer simulation with more volatile contexts. The healthy agent outearns all other agents (reward-insensitive and fatalistic depression, and both social phobia agents, see **Figure 8** in main text). More detailed analyses of exemplary time series reveal that the healthy-to-optimistic agent remains capable of detecting context changes throughout the longer time series, unlike the agents with depression and social phobia. This allows the healthy/ optimistic agent to flexibly maximize rewards, also in a situation with very volatile context dynamics. Exemplary developments of three agent's posterior beliefs during the long time series are shown in **Figure 33**. A healthy-to-optimistic agent updates beliefs to be in a hostile environment quickest.

Cooperative-to-hostile games reward loss avoidance; hostile-to-cooperative scenarios reward flexibility.

We computed total earnings in a game with T=40 trials, averaged over 20 games per agent type. Results are shown in **Figure 34**. In a cooperative-to-hostile game, the agent with increased loss avoidance, but an otherwise intact generative model earns the highest absolute rewards. This may be attributed to their quicker adopting of an avoidant behavioral strategy (see also **Figure 22**) compared to other agents. In contrast, in a hostile-to-cooperative game, the healthy agent maximizes total earned rewards due to their unique ability to update a precise posterior belief to be in a hostile context compared to all other agent types.

In a longer time series with more volatile context dynamics, a healthy agent maximizes total earned rewards compared to both agents with depression (type 1 depression: insensitive and pessimistic, type 2 depression: fatalistic) and social phobia (type 1 social phobia: insecure and avoidant, type 2 social phobia: trauma-and-defeat). This can similarly be attributed to the agent's continued exploration despite having formed a precise posterior belief to be in a hostile context (see also **Figure 33**).

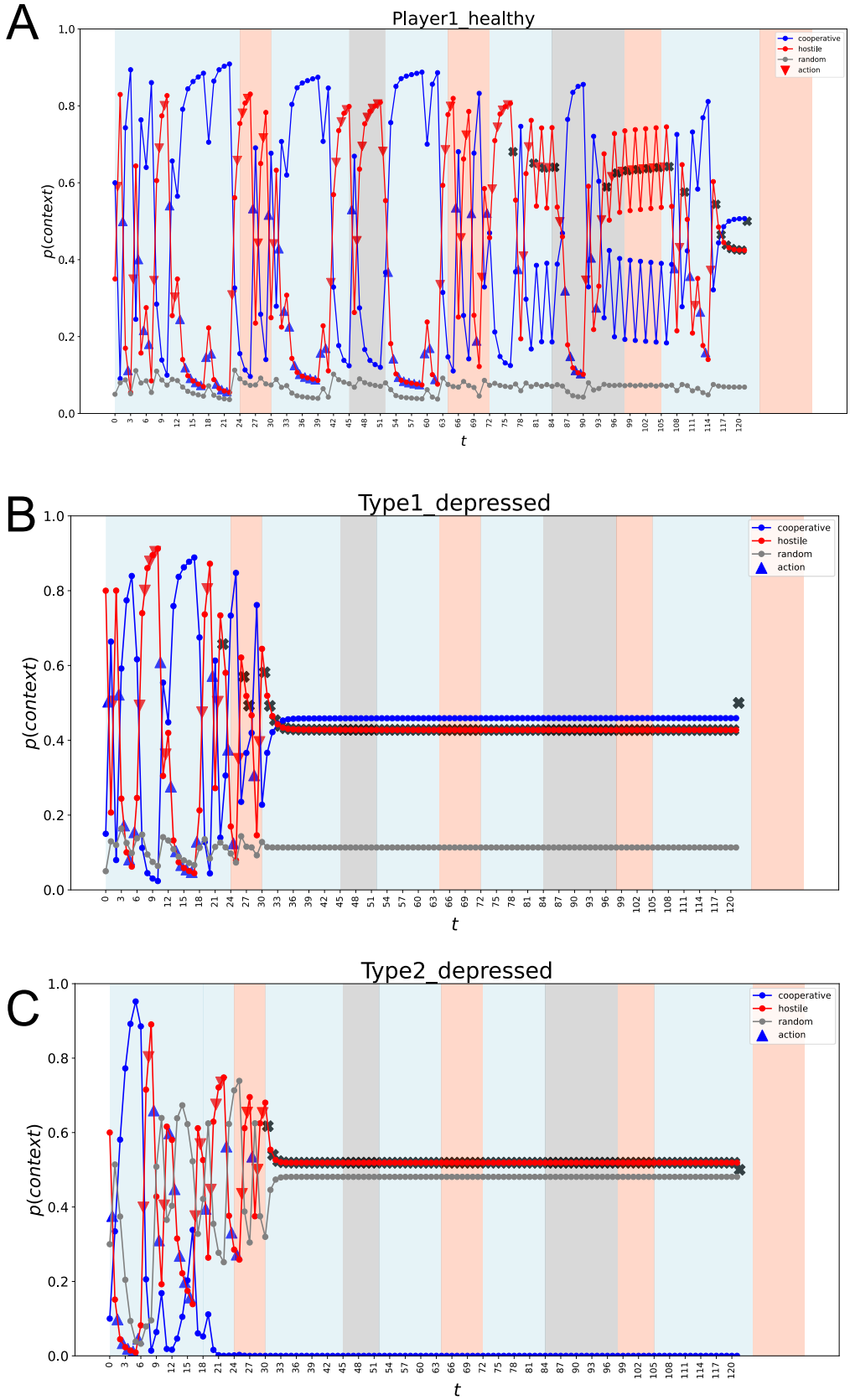


Figure 33. Long time series with volatile contexts (healthy, depression).

Posterior beliefs of three agents healthy (**A**), depression "reward-insensitive and pessimistic", (**B**) and depression "fatalistic expectations", (**C**)

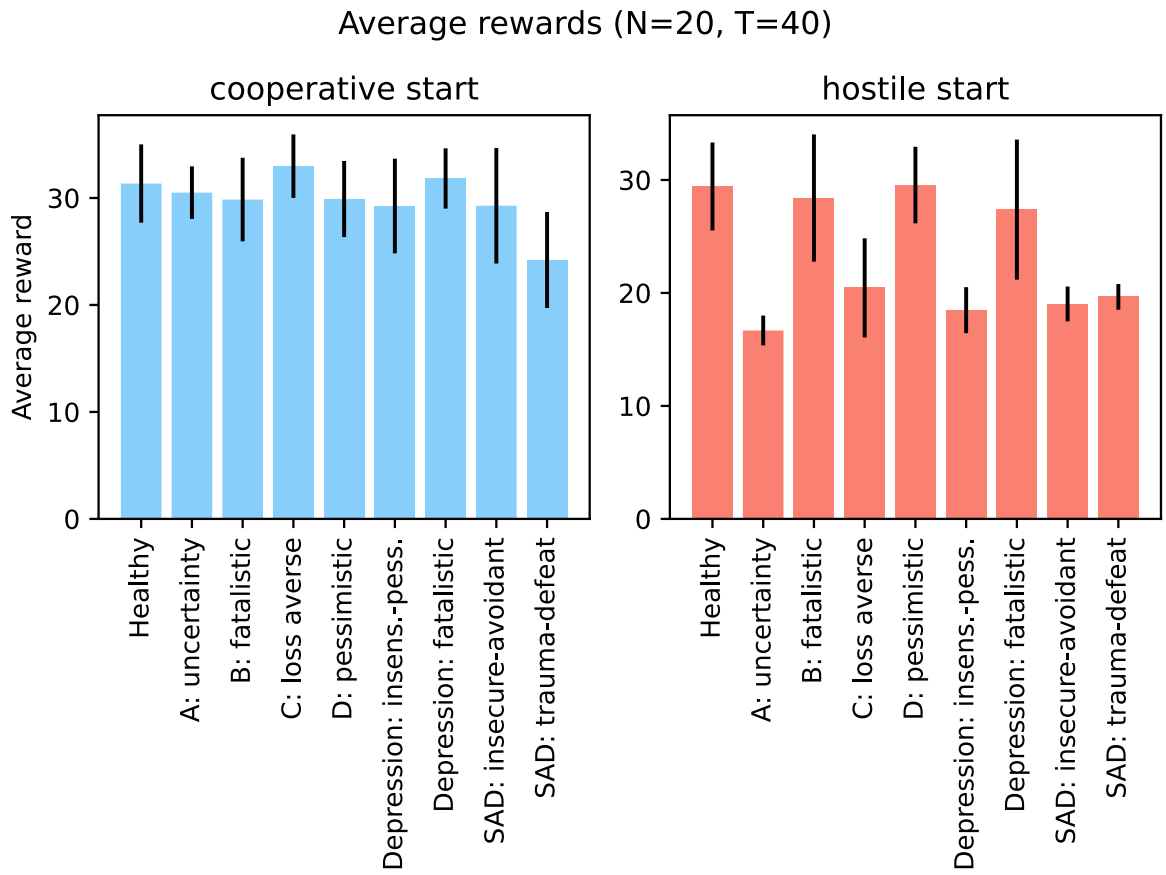


Figure 34. Average earned rewards across all agent types

Across all agent types (disorder-specific and transdiagnostic agents), the more loss averse agent (biased C parameter) outearns all others in a cooperative-to-hostile game. In a hostile-to-cooperative game, the healthy-to-optimistic agent outearns all other agent types. Figure shows average values over 20 simulations incl. standard deviation (error bars).

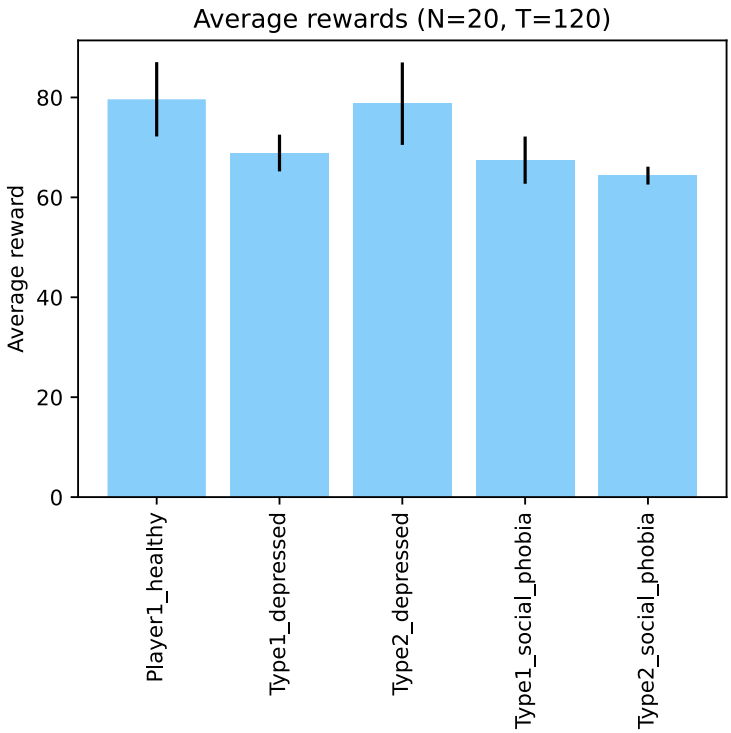


Figure 35. Average earned rewards per agent type, long series (T=120)

The figure shows the average earned rewards per agent type in a longer simulation (T=120) with a total of 12 context switches. The healthy (accurate-to-optimistic) agent outperforms the agents simulating depression as well as both agents with social phobia (earnings healthy-to-optimistic agent: 79.625; type 1 depressed (insensitive and pessimistic: 68.875; type 2 depressed (fatalistic): 78.75; type 1 social phobia (insecure avoidant): 67.45; type 2 social phobia (trauma-and-defeat): 64.35).

Participant data

Distribution of BDI scores.

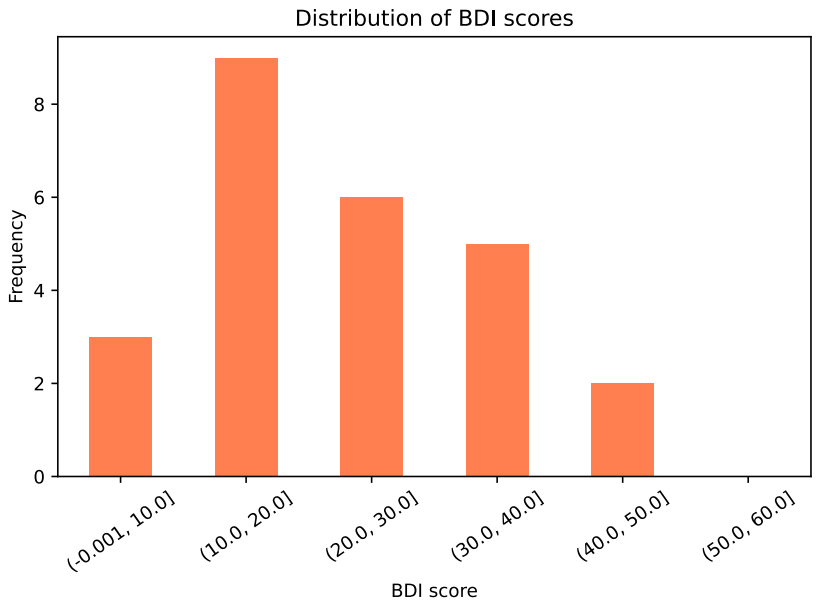


Figure 36. BDI scores of patient sample.

The figure shows binned frequencies of BDI scores in the sample of outpatients with diagnosed depressive disorders.

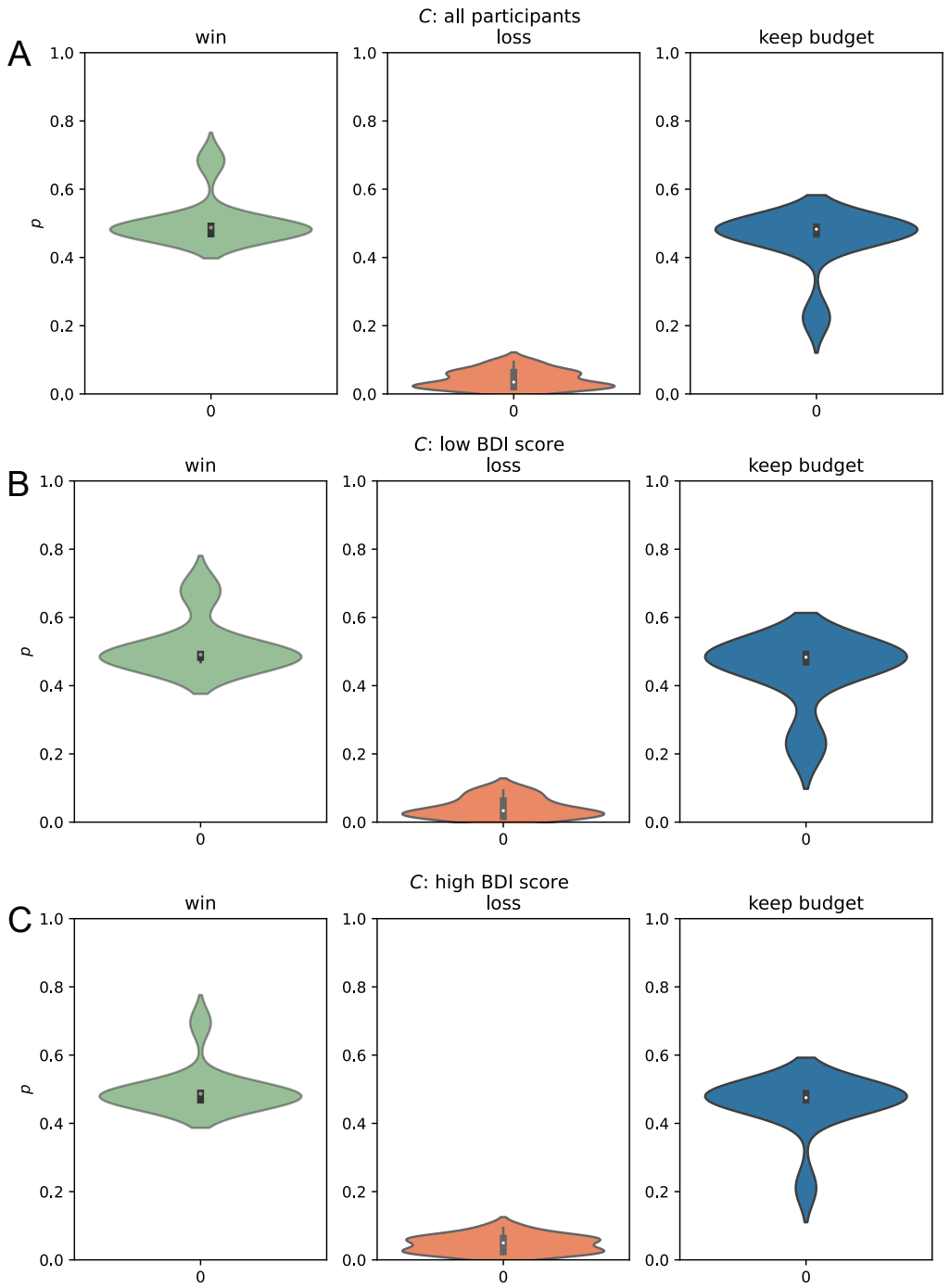


Figure 37. Distribution plots for parameter C.

A. Distribution of parameter C across all participants.

B. Distribution of C in participants who reported below-median depressive symptom severity.

C. Distribution of C in participants who reported above median depressive symptom severity.

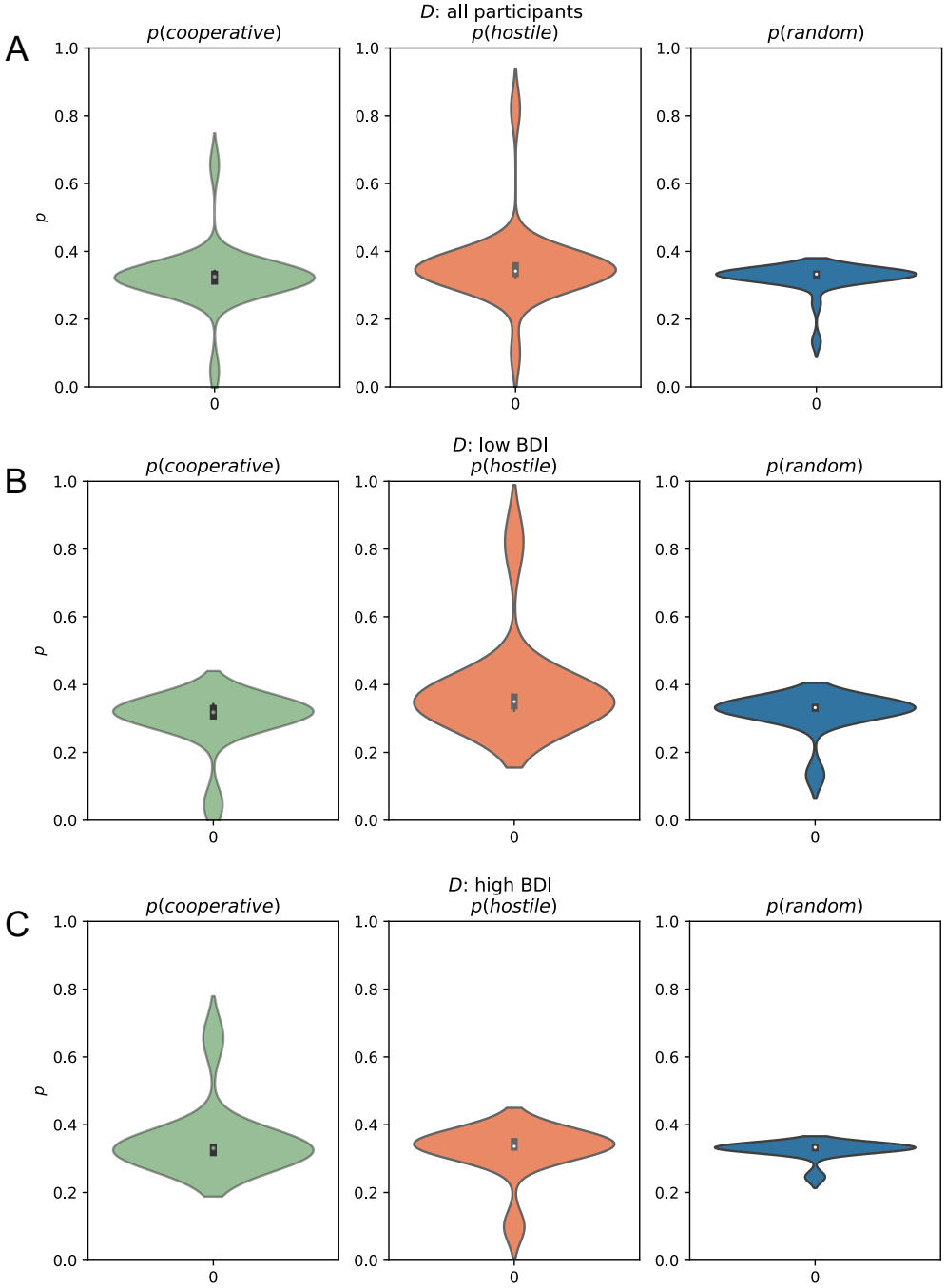


Figure 38. Optimization results for parameter D (distributions).

A. Parameter values *D* (prior over context) for cooperative (green), hostile (red) and random (blue) context priors across all participants.

B. Prior over contexts (*D*) for cooperative, hostile and random context of participants with low symptom severity.

C. *D* for cooperative, hostile and random context for participants with high symptom severity.